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Spin-selective transport in semiconductor spintronics and single-defect quantum technology

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Spin-Selective Transport in Semiconductor Spintronics and Single-Defect Quantum Technology

by

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of the requirements for the
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Abstract

This thesis represents a theoretical exploration of the consequences of coupling spin-correlations with spin coherent phenomena. These correlations can arise from intrinsic properties like spin-orbit interaction or extrinsic elements such as spin-polarized contacts non-equilibrium systems.

In conjugated organics, the weak spin-orbit interaction acts to mix spin and orbital states. This mixing links the spin dynamics to spatial orientation. The probability of a spin flipping orientation, *e.g.* spin-up to spin-down with respect to a global quantization axis, is now dependent on the relative orientation of the two sites. We predict that consideration of the steric environment can result in an order of magnitude increase in spin lifetimes limited by the spin-orbit interaction. In the theoretical limit where the unhybridized orbital at each site is parallel to the global quantization axis the spin-orbit induced relaxation is completely quenched.

In systems where the spin correlation is provided by polarized contacts, incoherent transport through a localized spin center generates dynamical effects in the dc magnetoresistance. Using the stochastic Liouville formalism we calculate clearly-defined resonances in the dc magnetoresistance when the applied magnetic field matches the exchange interaction with a nearby spin. At these resonances spin precession in the applied magnetic field is canceled by spin evolution in the exchange field, preserving a dynamic bottleneck for spin transport through the dopant. Similar features emerge when the dopant spin is coupled to nearby nuclei through the hyperfine interaction. These features provide a precise means of measuring exchange or hyperfine couplings between localized spins near a surface using spin-polarized scanning tunneling microscopy, without any ac electric or magnetic fields, even when the exchange or hyperfine energy is orders of magnitude smaller than the thermal energy.

Public Abstract

Typically when one thinks of information and technology they reflect on the tremendous progress made in controlling the flow of electric current - a quantity proportional to the transport of electronic charge. The conservation of charge makes it a reliable quantity for encoding information and provides a sound foundation for technological devices. However, the electron's charge is not the only useful conserved quantity. The intrinsic angular momentum or "spin" of the electron provides an additional handle for calculation and information storage. Unlike charge, the spin is a quantum mechanical quantity and therefore allows one to imagine devices that skirt the limitations imposed by classical systems, *e.g.* through the inherent ambiguity in the state of a spin, known as *coherence*, that allows it to exist as more than one physical outcome at once, until measured. While useful, this coherent state is easily destroyed through interactions between the spin and its environment. As a result, probing the delicate evolution of these spin states is a serious challenge, especially at the single-spin level. In addition, any engineering that prolongs the lifetime of the spin is highly desirable.

This thesis focuses on methods of probing and even controlling the coherent dynamics of spin through consideration of how they evolve in the presence of forbidden transitions. These forbidden transitions result from linking the spin state to additional characteristics like spatial orientation or the number of available final states during transport. Through this linking, the transport of certain spin states is not possible. As a first example, the electron spin is linked to spatial orientation through the influence of its motion around a nucleus. If transport is the result of discrete hops from one site to another, the relative orientation of neighboring sites can determine the lifetime of a given spin's polarization. We predict that modification to the relative molecular orientation in organic conjugated semiconductors leads to ten-fold increases in the spin lifetime.

In addition to linking the spin to spatial orientation, one can link the spin of an isolated defect in a semiconductor to available states in a contact by building the contact from ferromagnetic materials. Contacts of this type will preferentially admit a certain spin orientation restricting current flow through the defect when the resident spin does not match this orien-

tation. The coherent evolution of the spin will rotate the orientation periodically, opening transport channels for states initially in an orientation with limited transport. We predict, in this case, that modification of the coherent evolution rate of the defect spin will lead to changes in the current measured through the defect.

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Chapter 1

The march toward atomic-scale technology

1.1 Converging on the quantum world

The study of spin-based electronics and quantum technology is experiencing a groundswell of interest from groups focused on a broad range of applications. This is due, at least in part, to the convergence in ideas from the electronics community and the spin-based technology community. The discovery of Giant Magnetoresistance in the late 1980's and early 1990's marks a node in the tree of scientific progress where researchers began viewing spin as a useful degree of freedom. As this branch began to grow, a seemingly disparate effort was continuing in the electronics industry to relentlessly pursue the miniaturization of the transistor - a ubiquitous component in electronic devices referred to as the "nerve cell" of the information age by one of its founders, William Shockley. This push has made relevant a different set of rules than those governing classical technology - the quirky and often counter intuitive rules of the quantum world. It is here, at the nexus between modern electronics and control of the spin degree of freedom, that most likely will mark the genesis of the next technological era.

1.1.1 Electron tubes

The revolution sparked by the transistor has its roots in 1904 with the invention of the two element electron tube, or Fleming valve. In the late 1800's John Fleming was conducting research on a little understood observation made by Thomas Edison. In his efforts to invent an incandescent light bulb, Edison noted that electric current flows through vacuum from an incandescent filament to a cooler metallic plate. At the time, this process was seen as an

impediment to Edison and not well understood. This thermal liberation of electrons is now known as thermionic emission or the Edison effect and set in motion a series of discoveries and inventions that allowed for the age of electronics.

The Fleming valve is the first useful application of the vacuum tube. The utility of the device relies on the ability to allow current to flow in one direction and suppresses current in the opposite direction - what is commonly known as a thermionic diode. The Fleming valve was used to rectify AC signals for use in power supplies. An extra terminal was added in 1906 by Lee de Forest in a device he called the “Audion”. The Audion was the predecessor to the transistor, allowing for the amplification of weak electronic signals. The third electrode was a grid and a small electrical signal applied to the grid could control a large signal from filament to the metal plate. The primary use of the Audion was in amplifying radio receivers.

1.1.2 The transistor

In 1947, a group of researchers - William Shockley, Walter Brattain, and John Bardeen - at Bell labs created an electrical amplification device made from a semiconducting material, germanium. The device was called a “transistor” due to the fact that it was a resistive or semiconducting device that amplified signals transferred through it. The transistor would go on to revolutionize the world of electronics. The principle of the device is similar to that of the Audion, except there were no vacuum tubes, grids, plates, or heating elements required. The innovation allowed for a reduction in power consumption and an increase in speed.

At the heart of a transistor is the junction of two different types of semiconductors, one p-type and one n-type. P-type semiconductors are doped with atoms containing less valence electrons than the host creating an absence of an electron, called a “hole”. This hole acts as a positive charge and rather than electrons moving to fill a hole the hole is thought to move and carry current, hence the “p” for “positive” carriers. Conversely, n-type semiconductors have negatively charged carriers - electrons. They are doped with atoms containing additional electrons relative to the host. At the p-type/n-type interface a depletion region (the word “depletion” referring to the lack of charge carriers) forms due to the diffusion of electrons

from the n-type material to the holes of the p-type material. This limits the flow of current through the device. The depletion region is controllable through a voltage applied to a third contact called the “gate”. In modern data storage and manipulation protocols, the transistor acts as the switch, turning on and off current at high speed.

1.1.3 Pushing the limits of classical technology

Over the past seven decades the scale of transistors has dramatically decreased. The first integrated circuit contained only four transistors at roughly 100 microns in length. As an example of the progress made in this area, a modern processor like the Apple A13 found in the iPhone 11 Pro contains 8.5 billion transistors within an area just over 98 mm². This means a single transistor occupies roughly 100 nm² or a length of 10 nm! This is still twice as long as the minimum gate lengths predicted for silicon based transistors. Once the gate lengths drop below 5 nm quantum tunneling between source and drain lead to a critical loss of electrostatic control in the device. Effort has been made to avoid this issue by using materials other than silicon, such as MoS₂ [1], where the gate length was reduced to 1 nm. Rather than fending off the threat of quantum behavior it is possible to leverage it and broaden the range of possibilities.

Over the course of this chapter the content focuses on developments in utilizing the spin degree of freedom, starting from an introduction to spin and its mechanics and culminating in an overview of progress in the field of spintronics and solotronics - a area of research focused on the advantage of a single spin. The remainder of the thesis is focused on the theoretical investigation of manipulating and probing spin through correlated incoherent transport.

1.2 Introduction to spin

In the late 1800’s, Pieter Zeeman observed that the spectral lines of atoms split in the presence of external magnetic fields. It wasn’t until the development of quantum mechanics in the 1920’s that this phenomena was linked to an internal degree of freedom of the electron itself. In 1925, Wolfgang Pauli reasoned that the various atomic spectra would make sense

if the atomic states were described by, in addition to the azimuthal and magnetic quantum numbers, a fourth quantum number *and* that only one electron occupied each state - Pauli's *exclusion principle*. That same year, two Dutch physics students, George Uhlenbeck and Samuel Goudsmit, linked this additional quantum number to an intrinsic degree of freedom of the electron. While this achievement is viewed as a step forward in the old quantum theory, also known as the Bohr-Sommerfeld theory, it was predicted in the relativistic quantum mechanical theory of Paul Dirac later that decade. It is now understood that angular momentum comes from two sources: orbital motion and “spin”. The latter is associated with this internal degrees of freedom wholly separate from rotations in real space and lacks any responsible classical analog. This lack of direct intuition, however, should not limit the physical utility of the concept.

1.2.1 Spinors, probabilities, and expectation values

In quantum mechanics, the state of an entity is described by a state-vector. Spin is described, in a non-relativistic theory, by a two-component vector called a *spinor*. The spin value along the three spatial axes in Euclidean space are calculated in quantum mechanics through linear operators in the two-dimensional Hilbert space. Since these projections correspond to observable quantities, the eigenvalue of the operators must be real and therefore Hermitian. Any Hermitian operator in 2-dimensional Hilbert space can be written in terms of the identity and the three Pauli matrices,

$$\hat{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \hat{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \hat{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.1)$$

The operator corresponding to the total spin is proportional to the vector of Pauli matrices,

$$\mathbf{S} = \frac{\hbar}{2} \begin{pmatrix} \hat{\sigma}_1 \\ \hat{\sigma}_2 \\ \hat{\sigma}_3 \end{pmatrix}, \quad (1.2)$$

where $\hbar$ is the reduced Planck's constant. To calculate the states of maximum spin along a given axis we look for the eigenvectors of the corresponding spin operator. The Pauli

matrices in Eq(1.1) are said to be written in the representation of $\hat{\sigma}_3$. For convenience lets take this axis to be the z-axis with $\hat{\sigma}_1$ corresponding to the x-axis and $\hat{\sigma}_2$ corresponding to the y-axis. The eigenstates of the operator $\hat{S}_z = \hbar\hat{\sigma}_z/2$ are by inspection,

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.3)$$

and the eigenstates of the other two spin operators can be written as coherent superpositions of these states i.e. all state-vectors can be written as a linear combination of eigenstates of any observable within the same space. The general expression for an electron spinor in the $\hat{S}_z$ representation is found through calculation of the eigenstates of $\mathbf{S} \cdot \mathbf{n}$, where $\mathbf{n} = \{\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta\}$ is an arbitrary unit vector in spherical coordinates, θ is the polar angle, and ϕ is the azimuthal angle. The solutions are

$$\begin{aligned} |\mathbf{n}; \uparrow\rangle &= \cos\frac{\theta}{2}|\uparrow\rangle + e^{i\phi}\sin\frac{\theta}{2}|\downarrow\rangle \\ |\mathbf{n}; \downarrow\rangle &= \sin\frac{\theta}{2}|\uparrow\rangle + e^{i\phi}\cos\frac{\theta}{2}|\downarrow\rangle. \end{aligned} \quad (1.4)$$

Due to Born's statistical interpretation of the wave function, the spin of an electron is inherently indeterminate in general. An arbitrary spin-up state from Eq(1.4) has a probability to measure $\hbar/2$ along z of

$$P_{\uparrow z} = |\langle\uparrow|\mathbf{n}; \uparrow\rangle|^2 = \cos^2\frac{\theta}{2} \quad (1.5)$$

and the probability of measuring $-\hbar/2$ is

$$P_{\downarrow z} = |\langle\downarrow|\mathbf{n}; \uparrow\rangle|^2 = \sin^2\frac{\theta}{2}. \quad (1.6)$$

Additionally, an important quantity to calculate is the average value of an observable. In quantum mechanics this is called the *expectation value* and once again, for the spin-up state in Eq(1.4) the expectation value is

$$\langle\hat{S}_z\rangle = \langle\mathbf{n}; \uparrow|\hat{S}_z|\mathbf{n}; \uparrow\rangle = \frac{\hbar}{2}\left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right) = \hbar\left(\cos^2\frac{\theta}{2} - \frac{1}{2}\right). \quad (1.7)$$

1.2.2 The Hamiltonian

A description of the Zeeman effect can now be given in terms of the interaction of the spin magnetic moment with the external field. As in classical physics, a magnetic moment

experiences a torque in the presence of a magnetic field. The magnetic moment of an individual electron is no different. When analyzed in the basis of spin eigenstates of the operator parallel to the applied field, the interaction with a static magnetic field can be shown to lift the degeneracy between spin-up and spin-down. The spin magnetic moment is

$$\boldsymbol{\mu}_s = -g_0 \frac{\mu_B}{\hbar} \mathbf{S}, \quad (1.8)$$

where $g_0 = 2.0023$ is the Landé g-factor, μ_B is the Bohr magneton, and $\mathbf{S}$ is the spin operator. As one would expect, the energy of this moment in a static magnetic field, $\mathbf{B}$, is

$$\begin{aligned} E &= -\boldsymbol{\mu}_s \cdot \mathbf{B} \\ &= \frac{g_0 \mu_B}{2} \mathbf{B} \cdot \boldsymbol{\sigma}, \end{aligned} \quad (1.9)$$

where $\boldsymbol{\sigma} = \{\sigma_x, \sigma_y, \sigma_z\}^T$ is the vector of Pauli matrices. For an electron the spin is $S=1/2$ and the effect of this term in the Hamiltonian is to lift the degeneracy of the spin states by breaking spatial inversion symmetry. The difference in energy between the spin “up” and spin “down” states is

$$\begin{aligned} \Delta E &= E_{\uparrow} - E_{\downarrow} \\ &= g_0 \mu_B |\mathbf{B}|. \end{aligned} \quad (1.10)$$

This energy splitting is referred to as the *Zeeman energy*.

1.3 Evolution of quantum states

1.3.1 Coherent evolution

Time evolution of a state is determined by the Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle, \quad (1.11)$$

where $\hat{H}$ is the Hamiltonian. The coherent evolution of the density matrix stems from the Schrödinger equation and can be written as

$$i\hbar \frac{\partial}{\partial t} \rho = [\hat{H}, \rho]. \quad (1.12)$$

This form, known as the *Liouville-von Neumann equation*, shows that the coherent evolution of the density matrix, ρ , is determined by the commutator of the Hamiltonian with the density matrix. For a review of the density matrix see Appendix B.

1.3.2 Influence of a dc magnetic field

In the presence of a dc magnetic field we discussed that the spin states are no longer degenerate. This interaction can lead to coherent evolution.

Consider the general expression for a spin-1/2 in the basis of $\hat{S}_z$ eigenstates given in Eq(1.4). Solving the Schrödinger equation for the Hamiltonian of a static field along the quantization axis,

$$\hat{H} = g\mu_B B_z \hat{S}_z, \quad (1.13)$$

gives a time dependent phase shift proportional to the Zeeman energy:

$$|\hat{n}; \uparrow(t)\rangle = \cos \frac{\theta}{2} e^{-iE_{\uparrow}t/\hbar} |\uparrow\rangle + \sin \frac{\theta}{2} e^{i(\phi - E_{\downarrow}t/\hbar)} |\downarrow\rangle. \quad (1.14)$$

For a state along the field axis, this time dependent phase is physically irrelevant. States with a finite polar angle relative to the field axis will maintain the same projection onto the axis, but the azimuthal angle will change with time. Physically this manifests as precession about the field axis and is called *Larmor precession*.

1.3.3 Adding an ac component: magnetic resonance

An important example of these dynamics is electron spin resonance (ESR). As an illustration of the coherent evolution of an electron spin let's consider the dynamics associated with dc and ac magnetic fields. A localized spin-1/2 electron in the presence of a constant magnetic field along the z -axis will no longer have degenerate spin states as shown above. The splitting between spin-up and spin-down, Eq(1.10) can be rewritten as

$$\hbar\omega_p = g\mu_B B_z, \quad (1.15)$$

where g is the gyromagnetic factor, μ_B is the Bohr magneton, and ω_p is the precession frequency. Given the following spin state

$$\phi(t) = \begin{pmatrix} c_0(t) \\ c_1(t) \end{pmatrix} \quad (1.16)$$

written in the z -basis, applying a rotating field in the xy -plane

$$\begin{aligned} B_x &= B_0 \cos \omega_p t, \\ B_y &= B_0 \sin \omega_p t. \end{aligned} \tag{1.17}$$

Solving the time-dependent Schrödinger equation for the spin in the presence of the static field B_z and the rotating field in the xy -plane gives the following normalized solution for the time dependent spinor,

$$\phi(t) = \begin{pmatrix} \cos(\Omega t) \exp(-i\omega_p t/2) \\ -i \sin(\Omega t) \exp(i\omega_p t/2) \end{pmatrix}, \tag{1.18}$$

where $\Omega = g\mu_B B_0/2\hbar$. Solving now for the spin expectation value along each of the orthonormal axes yields

$$\begin{aligned} \langle S_z \rangle &= \frac{\hbar}{2} \cos(2\Omega t) \\ \langle S_y \rangle &= -\frac{\hbar}{2} \sin(2\Omega t) \cos(\omega_p t) \\ \langle S_x \rangle &= -\frac{\hbar}{2} \sin(2\Omega t) \sin(\omega_p t). \end{aligned} \tag{1.19}$$

From these expressions it is seen that for a spin initially along the z -axis after applying a pulse for $t = \pi/4\Omega$ the expectation value along the z -axis vanishes because the spinor has evolved into the xy -plane. A pulse of this type is referred to as a $\pi/2$ pulse because of the polar rotation of $\pi/2$ radians on the Bloch sphere. Similar reasoning applies for a π pulse which flips a spin-up to a spin-down or vice versa.

1.3.4 Exchange

Electronic interaction also results from the symmetry of the many body wave function. Since electrons are fermions, the total wave function must be antisymmetric with respect to exchange. The spatial portion of the wave function can be symmetric,

$$|\Phi(\mathbf{r}_1, \mathbf{r}_2)\rangle_s = \frac{1}{\sqrt{2}}(\phi_1(\mathbf{r}_1)\phi_2(\mathbf{r}_2) + \phi_1(\mathbf{r}_2)\phi_2(\mathbf{r}_1)) \tag{1.20}$$

or antisymmetric

$$|\Phi(\mathbf{r}_1, \mathbf{r}_2)\rangle_a = \frac{1}{\sqrt{2}}(\phi_1(\mathbf{r}_1)\phi_2(\mathbf{r}_2) - \phi_1(\mathbf{r}_2)\phi_2(\mathbf{r}_1)). \tag{1.21}$$

Similarly, the spin states can be symmetric

$$\begin{aligned}
\chi_{T_+}(1, 2) &= \chi_{\uparrow}(1)\chi_{\uparrow}(2) \\
\chi_{T_-}(1, 2) &= \chi_{\downarrow}(1)\chi_{\downarrow}(2) \\
\chi_{T_0}(1, 2) &= \frac{1}{\sqrt{2}}[\chi_{\uparrow}(1)\chi_{\downarrow}(2) + \chi_{\downarrow}(1)\chi_{\uparrow}(2)]
\end{aligned} \tag{1.22}$$

or antisymmetric

$$\chi_S(1, 2) = \frac{1}{\sqrt{2}}[\chi_{\uparrow}(1)\chi_{\downarrow}(2) - \chi_{\downarrow}(1)\chi_{\uparrow}(2)]. \tag{1.23}$$

There are three possible symmetric spin states, hence they are referred to as the spin “triplets”. The sole antisymmetric pairing is called the spin “singlet”.

The total wave function then must be the product of an antisymmetric spatial state with a spin triplet, or a symmetric spatial state with a spin singlet. Although the distinction between these two possible states is due to the fermionic nature of the particles involved, there are implications on the energy structure of the spin manifold. Exchange, therefore, plays a role in the spin-dynamics of localized centers as discussed in chapter 3. Due to the correlation with the spin states, the energy difference between the triplet and singlet can be expressed as a spin Hamiltonian,

$$H_{\text{ex}} = J\mathbf{S}_1 \cdot \mathbf{S}_2, \tag{1.24}$$

where $\mathbf{S}_i$ is the vector of spin operators for spin i .

1.3.5 Hyperfine interaction

In addition to interactions with external fields, electronic spins can also interact with the field of nuclei. In many applications, the electronic wave function extends over many host atoms and each atom contributes to an effective nuclear magnetic field, $\mathbf{B}_{\text{nuc}}$, that is randomly oriented in space. In quantum dot systems the number of nuclei that the electron probability distribution interacts with can be on the order of $\sim 10^6$. In this case, the relevant hyperfine Hamiltonian is the Fermi contact term,

$$H_{\text{hf}} = \frac{8\pi}{3} \frac{\mu_0}{4\pi} g_0 \mu_B \sum_i \hbar \gamma_{N,i} \mathbf{I}_i \otimes \mathbf{S} \delta(\mathbf{r} - \mathbf{R}_i), \tag{1.25}$$

where $\mathbf{R}_i$ is the nuclear position, $\mathbf{I}_i$ is the nuclear spin operator, $\mathbf{r}$ is the electron position, $\mathbf{S}$ is the electron spin operator, γ_N is the gyromagnetic ratio of the nucleus, μ_B is Bohr's magneton, and μ_0 is the permeability of vacuum. In cases where the spin and orbital states of the wave function are not correlated *i.e.*, the wave function can be written as a product of orbital and spin states, an effective spin Hamiltonian can be written to represent Eq(1.25),

$$H_{\text{hf}} = \sum_i A_i \mathbf{I}_i \otimes \mathbf{S} \quad (1.26)$$

with

$$A_i = \frac{8\pi}{3} \frac{\mu_0}{4\pi} g_0 \mu_B \hbar \gamma_{N,i} |\psi(\mathbf{R}_i)|^2 \quad (1.27)$$

as a quantity that varies in space according to the final factor.

To gain insight into the type of dynamics this Hamiltonian imparts to the electron, the product of operators can be recast in terms of raising and lowering operators,

$$\mathbf{I}_i \otimes \mathbf{S} = \frac{1}{2}(S_+ \otimes I_- + S_- \otimes I_+) + S_z \otimes I_z, \quad (1.28)$$

where $S_{\pm} = S_x \pm iS_y$ and $I_{\pm} = I_x \pm iI_y$. The first term acts to flip the electron spin while inversely flipping the spin of the nucleus and the second acts as a splitting similar to the Zeeman effect.

In this case where the electron is interacting with many spins, a classical *Overhauser* field can model the system,

$$\mathbf{B}_N = \frac{1}{g^* \mu_B} \left\langle \sum_i A_i \mathbf{I}_i \right\rangle. \quad (1.29)$$

The spatial variation of this effective field causes a change in the relative phase between two electrons. The effective spin Hamiltonian is

$$H = \frac{1}{2} g^* \mu_B (\mathbf{B}_{N1} \cdot \boldsymbol{\sigma}_1 + \mathbf{B}_{N2} \cdot \boldsymbol{\sigma}_2), \quad (1.30)$$

where $\mathbf{B}_{Ni}$ is the effective nuclear field at site i and $\boldsymbol{\sigma}_i$ is the vector of Pauli matrices for the

spin at site i . Writing the Hamiltonian in matrix form in the basis $\{|S\rangle, |T_0\rangle, |T_+\rangle, |T_-\rangle\}$,

$$\begin{pmatrix} 0 & \Delta_z & \frac{-\Delta_x - i\Delta_y}{\sqrt{2}} & \frac{\Delta_x - i\Delta_y}{\sqrt{2}} \\ \Delta_z & 0 & \frac{\Sigma_x + i\Sigma_y}{\sqrt{2}} & \frac{\Sigma_x - i\Sigma_y}{\sqrt{2}} \\ \frac{-\Delta_x + i\Delta_y}{\sqrt{2}} & \frac{\Sigma_x - i\Sigma_y}{\sqrt{2}} & \Sigma_z & 0 \\ \frac{\Delta_x + i\Delta_y}{\sqrt{2}} & \frac{\Sigma_x + i\Sigma_y}{\sqrt{2}} & 0 & -\Sigma_z \end{pmatrix} \quad (1.31)$$

where $\Delta_i = B_{1i} - B_{2i}$ and $\Sigma_i = B_{1i} + B_{2i}$, shows that the difference in fields between the two sites contributes to coupling between the singlet and all three triplet states. In the presence of an external field the coupling between the singlet and T_+ and the singlet and T_- become irrelevant due to Zeeman splitting, but the coupling between the singlet and T_0 remains since they both do not couple to the field. The hyperfine field then combines S and T_0 into bonding and antibonding combinations forming the states $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$.

1.3.6 Non-unitary evolution

Spin relaxation times have been measured in quantum dots (QD) using read-out schemes. Information about the relaxation time is gleaned by varying the wait-time after an electron is injected [2, 3]. For successful read-out this time t_w should be longer than the characteristic tunneling time but shorter than the spin relaxation time. As t_w increases, the number of swaps with the reservoir decreases because the excited state of the QD has time to relax before being read. The data can be fit to an exponential form for the probability of measuring an excited spin state as a function of wait-time:

$$p_{\text{ex}} = \alpha + C \exp(-t_w/T_1), \quad (1.32)$$

where the parameter α accounts for instances where the detector current exceeds the threshold despite the spin being in the ground state due to thermally activated tunneling or electrical noise.

The Lindblad equation

In section 1.3.1 the coherent evolution of the density matrix was briefly discussed and the Liouville-von Neumann equation was presented. The coherent processes are often important,

but never the whole picture. A more realistic interpretation involves coupling the system of interest to the environment - a reference state about which a certain amount of ignorance is assumed. In general, the combination of the environment and the system evolve coherently. In practice, tracking the degrees of freedom of these bath states is impractical and often unnecessary. Rather than describe the evolution of the entire composite system, it is possible in many cases to trace over the bath states and evolve the reduced density matrix according to a dynamical map. In the case of a static bath, the evolution only takes place in the system and the Markovian approximation holds. Here the general form for an equation describing evolution of the system is non-unitary and often referred to as the Lindblad master equation,

$$\frac{d}{dt}\rho_S(t) = -i[H, \rho_S(t)] + \mathcal{D}(\rho_S(t)), \quad (1.33)$$

where

$$\mathcal{D}(\rho_S) \equiv \sum_k \gamma_k \left(A_k \rho_S A_k^\dagger - \frac{1}{2} A_k^\dagger A_k \rho_S - \frac{1}{2} \rho_S A_k^\dagger A_k \right). \quad (1.34)$$

In this expression, the operators A_k and $A_k^\dagger$ are called the Lindblad operators and they represent the system's interaction with the environment, γ_k is the rate associated with the interaction, and ρ_S is the reduced density matrix for the system. The term denoted by $\mathcal{D}(\rho_S)$ is called the *dissipator* of the system.

For a system consisting of a single spin-1/2 particle the Lindblad operators are

$$A_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, A_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, A_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (1.35)$$

with

$$\gamma_1 = \frac{1}{2T_1}, \gamma_2 = \frac{1}{2T_1}, \gamma_3 = \frac{1}{2T_2} - \frac{1}{4T_1}. \quad (1.36)$$

Here the Lindblad terms encompass both spin-lattice and spin-spin relaxation with relaxation times T_1 and T_2 respectively.

1.3.7 Application to sensing and information

Basic sensing protocol

Apart from new concepts in traditional devices, spintronics brings about the opportunity to develop non-classical schemes for information processing and sensing. The latter involves relying on the quantum properties of a system - discrete energy levels or quantum coherence - to measure a physical quantity such as electric or magnetic fields. An excellent review of this topic and the model for a good portion of the following treatment is provided by Degen *et al.* [4]. The procedure for sensing begins with initialization. The initialization is necessary to define the unitary transformation that evolves the sensor into the desired sensing state. Once in this state, the sensor will evolve according to the Hamiltonian of the environment for a period of time. At this point in general the state is in a superposition of observable readout states. Projective readout (as opposed to POVM techniques) is probabilistic due to the quantum nature of the superposed final state. This means that in a two-dimensional Hilbert space spanned by the readout states $\{|0\rangle, |1\rangle\}$ there will be a probability p for measuring state $|0\rangle$ and probability $(1 - p)$ for measuring state $|1\rangle$. Therefore, this process will need to be averaged either over many sequential runs or over many identical non-interacting systems. In some cases the resulting averaged probability distribution is enough to provide useful information about the Hamiltonian being probed. In general though, it would be necessary to modify a controllable parameter like the sensing period or the unitary transformation after initialization to obtain the set of probabilities $\{p_k\}$. This is a simple yet sufficient procedure for sensing and is by no means the most efficient.

As an example of how the above sensing procedure would work it can be applied to the Ramsey measurement. The quantum sensor is first initialized into the state $|0\rangle$ before a $\pi/2$ pulse is applied:

$$\hat{U}_{\pi/2} = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \quad (1.37)$$

written in the $\{|0\rangle, |1\rangle\}$ basis. The pulse rotates the state into the coherent superposition

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle). \quad (1.38)$$

The state then is free to evolve under the influence of the Hamiltonian $\hat{H}_0$ which generates a relative phase $\phi = \omega_0 t$, where ω_0 is proportional to the energy splitting between $|0\rangle$ and $|1\rangle$,

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{-i\omega_0 t}|1\rangle). \quad (1.39)$$

By applying a second $\pi/2$ pulse the state is converted back to the measurable state,

$$|\alpha\rangle = \frac{1}{2}(1 + e^{-i\omega_0 t})|0\rangle + \frac{1}{2}(1 - e^{-i\omega_0 t})|1\rangle. \quad (1.40)$$

Reading out the final state gives a transition probability of

$$\begin{aligned} p &= 1 - |\langle 0|\alpha\rangle|^2 \\ &= \frac{1}{2}[1 - \cos(\omega_0 t)]. \end{aligned} \quad (1.41)$$

Plotting the transition probability as a function of time provides an oscillatory signature called Ramsey fringes that provide a direct measurement of the energy splitting between $|0\rangle$ and $|1\rangle$.

Noise and the minimum detectable signal

In all practical applications of a sensing protocol there will be error in measuring the transition probability p . For instance, in the processes outlined above the readout of the quantum state is probabilistic by nature and the only way to determine p is to repeat the measurement N times. Of course since N must be a finite integer the value of p must be estimated. The distribution of measurements of this type is binomial and the variance of the binomial distribution is what leads to a $1/\sqrt{N}$ dependence on the noise. This type of noise is referred to as *quantum projection noise*. While this noise is a result of making a finite number of measurements there exists a more intrinsic form of noise related to the unintentional coupling to the environment.

Noise resulting from relaxation and decoherence results because of the random transitions and phases that develop if the sensing time is comparable to the relaxation time or the

decoherence time respectively. This noise is given by an exponentially decreasing probability to measure p as time increases,

$$\delta p_{\text{obs}}(t) = \delta p(t)e^{-\chi(t)}, \quad (1.42)$$

where $\delta p(t)$ is the probability without decoherence and $\chi(t)$ is typically found to obey a power law

$$\chi(t) = (\Gamma t)^a. \quad (1.43)$$

Here Γ is the relaxation or decoherence rate and a is usually a number between one and three.

Similarly, errors in initialization or manipulation will result in a reduction of the observed probability. This reduction comes in as a constant factor,

$$\delta p_{\text{obs}} = \beta \delta p, \quad (1.44)$$

where β is positive and less than 1.

The quantum projection noise is introduced to the system as the state is read out and as mentioned above, it originates from the quantum nature of the system. Readout can also introduce classical noise. If the classical contribution is small compared to the quantum contribution then one talks about “single shot readout” where the discrete possible quantum outcomes broaden and have some overlap in the tails of the distributions. In this case a threshold value is set and results are associated with the discrete quantum outcomes. This introduces error due to the reassignment of values based on the threshold. On the other hand, when the classical noise outweighs the quantum projective noise only a single peak appears and p is estimated by the average of the distribution. The total readout uncertainty is then the sum of the quantum and classical contributions.

The signal-to-noise ratio for quantum sensing is given by the ratio of the noise introduced through decoherence and relaxation to the value of the readout noise. Since the change in probability δp is related to the change in signal δV through $\delta p = \delta V^q |\partial_V^q p(t)|$ it can be shown that the signal-to-noise ratio is

$$\text{SNR} = \delta V^q |\partial_V^q p(t)| e^{-\chi(t)} 2C(t_m) \frac{\sqrt{T}}{\sqrt{t + t_m}}, \quad (1.45)$$

where T is the total time available for the measurement, t_m is the time it takes to initialize, manipulate, and readout the sensor, and $C(t_m)$ is a function of t_m because the efficiency typically improves for longer readout times.

The minimum detectable signal is defined as the signal that yields unit SNR for an integration time of one second ($T = 1$ s),

$$v_{\min}^q \propto \frac{e^{\chi(t)} \sqrt{t + t_m}}{2C(t_m) \gamma^q t^q}, \quad (1.46)$$

where the expression $\delta V^q |\partial_V^q p(t)| \propto (\gamma t \delta V)^q$ was used. This expression suggests that the sensing time should be as long as possible while not introducing large exponential increase in sensitivity from the decoherence and relaxation. The optimum time should be somewhere near the characteristic time for relaxation and/or decoherence $t \approx T_\chi$. Additionally the sensitivity can be made even smaller by optimizing t_m as well as using advanced quantum schemes.

Manipulation of double quantum dot spin systems

In the double-dot system spin effects like Pauli blocking must be considered in addition to charge effects like the Coulomb blockade. In what follows we will consider primarily two quantum dots coupled in series. While there are reasons one might wish to couple dots in parallel [5], the spin dependent transport phenomena of dots in series are of most interest to the work presented in later chapters. The main spin-based transport interaction is spin (or Pauli) blocking. For simplicity consider two dots in series with a maximum occupation of two. Applying a bias between source and drain V_{sd} causes a current to flow from the left dot to the right dot. Additionally, the potential at each dot can be tuned by individual gate voltages V_L and V_R on the left and right dots respectively. For a sufficiently high potential barrier between the two dots the relative spin orientation does not matter, but when the potential is low enough to allow tunneling between the two dots the relative spin orientation plays a critical role. Due to the antisymmetric nature of electronic wave functions the antisymmetric spin state (the singlet state) corresponds to the symmetric orbital state. The symmetric orbital state is represented as the product of two single-spin states $\psi_0(x_1)\psi_0(x_2)$

corresponding to the ground state of the quantum dot. In order for a spin at the left dot to hop to the right dot with a identical spin the spin state would be symmetric and the orbital state would then have to be antisymmetric. This necessitates the introduction of an additional orbital state $\psi_1(x)$, which would by definition sit higher in energy than the ground state ψ_0 , to form the antisymmetric pairing $\psi_0(x_1)\psi_1(x_2) - \psi_0(x_2)\psi_1(x_1)$. When this spin-triplet state sits at a higher energy than the initial state of the singly occupied left dot no current will flow into the right dot causing a *spin blockade*.

The concept of the Pauli blockade can be combined with ESR techniques by applying an oscillating magnetic field B_{ac} perpendicular to a constant applied field B_z . This can be realized by flowing current through a conducting strip covering the dot system. The conductive strip is isolated from the underlying gate fingers by a dielectric. In addition to the field B_{ac} there is an intrinsic magnetic field associated with the nuclei of the host. In the case where the electron spins interact with many nuclei the nuclear field can be assumed to generate an effective magnetic field pointing in an arbitrary direction and, since the nuclear environment at the left dot is different than that at the right dot, a field gradient is established. This nuclear field mixes $|T_0(1,1)\rangle$ and $|S(1,1)\rangle$ (where $|\Psi(n_L, n_R)\rangle$ is the state Ψ with occupation n_L at the left dot and occupation n_R at the right dot), but leaves the spin-parallel states untouched due to the relatively large Zeeman splitting in the z -direction. With the inclusion of the hyperfine interaction blockades formed by the $|T_0(1,1)\rangle$ state are relieved and with the application of ESR (assuming only the left dot is resonant with B_{ac}) the spin-parallel states $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ can be rotated to $|\downarrow\uparrow\rangle$ and $|\uparrow\downarrow\rangle$ respectively then to $|S(1,1)\rangle$ via the hyperfine field.

Coherent manipulation of a single spin requires lowering the potential at the left dot below the double occupation energy of the right dot while the spin in the left dot is being manipulated. This is the so-called *Coulomb blockade* regime since the electrostatic repulsion between two electrons at the same dot makes the configuration energetically unfavorable.

The stripline method for implementing ESR is discussed in this section for pedagogical reasons. In practice this approach is a bit ham-fisted in the sense that it is difficult to

scale. The conductive striplines are bulky and the magnetic fields required are too large to be practical when considering the many coupled dots required for commercial application. Alternative approaches have relied on electrical manipulation via ac gate pulses and the spin-orbit interaction [6–8].

1.4 States in transit: from bands to hops

1.4.1 States in periodic environments

All discussion up to this point has been limited to one or two electrons in the absence of a host environment. First we consider an ideal crystalline host under the nearly free electron model, then we take the opposite approach by assuming electrons are closely bound to each site in the crystal in the tight-binding model.

Nearly free electron model

In introducing the solution to the electron wave function in the bulk of a crystal an analysis of the Schrödinger equation with a periodic potential is required,

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right] \psi(x) = E\psi(x) \quad (1.47)$$

where

$$V(x) = V(x + ma). \quad (1.48)$$

In this expression for the potential a is the lattice constant and m is a positive integer. Since the Fourier transform of the periodic potential only contains plane waves with wavenumbers $h_n = n2\pi/a$ the potential can be expanded as

$$V(x) = \sum_{n=-\infty}^{+\infty} V(h_n) e^{ih_n x}. \quad (1.49)$$

Following the treatment in any textbook on solid state physics leads to the formulation of the Bloch equations,

$$\psi_k(x) = e^{ikx} u_k(x) \quad (1.50)$$

with

$$u_k(x + a) = u_k(x). \quad (1.51)$$

This well known solution to the single electron Schrödinger equation with a periodic potential describes an itinerant electron modified by a function with the same periodicity of the lattice.

1.4.2 Band transport from localized states

Tight-binding model

The tight-binding model provides a familiar example of coherent transport in solid state systems. A sufficient condition for coherent transport is to show that a definite phase relationship exists between various probability amplitudes in the system. The description of the tight-binding approach rests on a few assumptions: (i) the Hamiltonian near a lattice site is well defined by the Hamiltonian of a single atom $\mathcal{H}_{\text{at}}$, (ii) the wave function $\phi_a(\mathbf{r})$, defined by $\mathcal{H}_{\text{at}}\phi_j(\mathbf{r}) = E_j\phi_j(\mathbf{r})$, is strongly localized at lattice site j , and (iii) the atomic wave function is a decent approximation to a stationary state of the crystal as is $\phi_j(\mathbf{r} + \mathbf{T})$, where $\mathbf{T}$ is the translation operator of the lattice. For simplicity a one-dimensional lattice of spacing a is considered along with a single non-degenerate energy level at each site. Due to the translational symmetry of the system the crystal Hamiltonian has identical elements along the diagonal and the hopping integrals between sites are identical,

$$\langle \phi_a(x - t_n) | H | \phi_a(x - t_n) \rangle = E_0 \quad (1.52)$$

and

$$\langle \phi_a(x - t_n) | H | \phi_a(x - t_{n\pm 1}) \rangle = \gamma. \quad (1.53)$$

Relying on assumption (ii) the hopping integral over distances larger than a are neglected.

The set of localized functions $\{\phi_a(x - t_n)\}$ do not satisfy Bloch's theorem, but delocalized functions that *do* satisfy Bloch's theorem can be constructed by taking linear superpositions of the localized states,

$$\Phi(k, x) = \frac{1}{\sqrt{N}} \sum_{\mathbf{n}} e^{ikt_n} \phi_a(x - t_n), \quad (1.54)$$

where N is the number of unit cells in the crystal. Starting from the origin one can now show, from the relation

$$\Phi(k, x + t_m) = \frac{e^{ikt_m}}{\sqrt{N}} \sum_n e^{ik(t_n - t_m)} \phi_a(x - t_n + t_m) = e^{ikt_m} \Phi(k, x), \quad (1.55)$$

that the probability amplitude for finding an itinerant electron on site n $\langle \Phi(k, x + t_n) | \Phi(k, 0) \rangle = e^{ikt_m}$ has a definite phase relationship.

The energy dispersion relation for this system is

$$E(k) = E_0 + 2\gamma \cos ka; \quad (1.56)$$

which shows that even for the highly simplified case the N degenerate isolated states are smeared into a band of width $4|\gamma|$.

1.4.3 Influence of impurities

The introduction of non-native elements in a crystal alter the dynamics of carriers. The impurity can either provide additional electrons to the system or subtract electrons. In the former case the impurity is said to be a *donor* and in the latter an *acceptor*. If the ionization energy of a donor is small relative to the energy gap it is a shallow (or hydrogenic) level. For these types of impurities the structure and chemical nature of the atom itself has minimal influence on the dynamic properties of carriers in the crystal. The reason for this is because the small ionization potential weakly localizes a carrier in the vicinity of the atom and in typical cases the carrier can extend over many lattice constants and as a result the impurity is well approximated by the modified potential of a point charge,

$$U(r) = \frac{e^2}{\kappa r}, \quad (1.57)$$

where e is the electronic charge, r is the distance to the center of the impurity, and κ is the dielectric permittivity of the lattice. These shallow impurities curb the delocalized nature of Bloch states - solutions to the ideal lattice. The wave function of a shallow impurity can be calculated within an effective mass theory.

Effective mass theory

The aim of effective mass theory is to reduce a problem from an $(N+1)$ -body problem to a one-body problem by treating the single body as moving in a field equal to the difference between the actual potential and the perfect lattice potential. The two approximations made are that: (1) the defect potential given by Eq(1.57) is varying slow enough to ignore Fourier components outside the first Brillouin zone, and (2) the envelop function, which is the eigenfunction of the defect Hamiltonian, is sufficiently diffuse. The definition of sufficiently diffuse will be made clear soon.

To start, we begin with the effective mass equations in the single electron approximation. The band function Hamiltonian is denoted H_0 such that

$$H_0\psi_n(\mathbf{k}, \mathbf{r}) = E_n(\mathbf{k})\psi(\mathbf{k}, \mathbf{r}) \quad (1.58)$$

where $\psi_n(\mathbf{k}, \mathbf{r})$ are Bloch functions like those described in section 1.4.1. The defect potential $U(\mathbf{r})$ can be added to the Hamiltonian of the perfect lattice and the objective becomes to find the solutions to the equation

$$(H_0 + U)\Psi = E\Psi, \quad (1.59)$$

where Ψ is the function of the defect electron.

Since the Bloch functions form a basis and the defect potential acts as a perturbation, it makes sense to expand the defect wave function in terms of the band states,

$$\Psi(\mathbf{r}) = \sum_{n', \mathbf{k}'} \langle \psi_{n', \mathbf{k}'} | \Psi \rangle \psi_{n', \mathbf{k}'}. \quad (1.60)$$

Plugging Eq(1.60) into Eq(1.59) and multiplying by $\langle \psi_{n, \mathbf{k}} |$ gives

$$(E_n(\mathbf{k}) - E) \langle \psi_{n, \mathbf{k}} | \Psi \rangle + \sum_{n', \mathbf{k}'} \langle \psi_{n, \mathbf{k}} | U | \psi_{n', \mathbf{k}'} \rangle \langle \psi_{n', \mathbf{k}'} | \Psi \rangle. \quad (1.61)$$

Assuming that $\langle \psi_{n', \mathbf{k}'} | \Psi \rangle$ is relevant only near the center of the Brillouin ($u_{n, \mathbf{k}} \approx u_{n, 0}$ and $u_{n', \mathbf{k}'} \approx u_{n', 0}$) zone we can break the integral of U into the product of two integrals - one

that varies rapidly and one that varies slowly:

$$\begin{aligned}\langle\psi_{n,\mathbf{k}}|U|\psi_{n',\mathbf{k}'}\rangle &= \int_{\text{unit cell}} u_{n,0}^* u_{n',0} d\mathbf{r} \frac{1}{V_0} \int e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}} U(\mathbf{r}) d\mathbf{r} \\ &= \frac{\delta_{n,n'}}{V_0} \int e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}} U(\mathbf{r}) d\mathbf{r}.\end{aligned}\tag{1.62}$$

Substituting Eq(1.62) into Eq(1.61) gives a closed form equation for the coefficients $\langle\psi_{n,\mathbf{k}}|\Psi\rangle$,

$$(E_n(\mathbf{k}) - E)\langle\psi_{n,\mathbf{k}}|\Psi\rangle + \sum_{\mathbf{k}'} U(\mathbf{k}, \mathbf{k}') \langle\psi_{n,\mathbf{k}'}|\Psi\rangle,\tag{1.63}$$

with

$$U(\mathbf{k}, \mathbf{k}') = \frac{1}{V_0} \int e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}} U(r) d\mathbf{r}.\tag{1.64}$$

Using Eq(1.57) as the defect potential $U(r)$ results in a solution for the defect wave function that resembles a factor with the periodicity of the lattice and an envelope function resembling the radial contribution of the hydrogen atom,

$$\Psi(\mathbf{r}) = \frac{1}{\sqrt{\pi a^3}} u_{n,0}(\mathbf{r}) e^{-r/a},\tag{1.65}$$

in the simple case of a single band extremum at the center of the Brillouin zone. Here a is the localization radius and $u_{n,0}$ is a function with the same periodicity as the lattice for band n and wave vector $k = 0$. From Eq(1.65) the impurity wave function is a Bloch function at the bottom of the band modified by a hydrogenic exponential decay.

Looking back, we have discussed the wave function for the case of nearly free electrons in a periodic lattice, tightly bound electrons occupying degenerate sites, and shallow impurities in an ideal lattice. Next we address transitions between sites that are no longer degenerate in energy.

1.4.4 Transitions between non-degenerate states

The tight-binding model introduced in the previous section illustrates how coherent transport results from periodic potentials even despite using a basis of localized atomic states. The model, however, assumes that all sites are resonant *i.e.* there is no energetic disorder. One question to ask is how do transitions occur when this condition is relaxed? Analysis of

this type is self motivated by the fact that any real material will have variations in the site energy due to fluctuations in the local fields. This does not detract from the importance of the ideal crystal analysis as it may be the case that fluctuations are irrelevant for the types of calculations of interest. Never the less, this situation ought to be addressed for its importance in magnetic resonance applications and disorder materials such as π -conjugated polymers, the latter being a topic discussed at length in the next chapter.

Oscillation between two discrete states

From time-dependent perturbation theory, one calculates the transition probability $P_{if}(t)$ as a function of the perturbation operator (see appendix C) and here the transitions due to a sinusoidal or constant perturbation is addressed. Using a perturbation of the form

$$\hat{W}(t) = \hat{W} \sin \omega t \quad (1.66)$$

the state vector is calculated from the general expression

$$b_n^{(1)}(t) = \frac{1}{i\hbar} \int_0^t e^{i\omega_{ni}t'} \hat{W}_{ni}(t') dt', \quad (1.67)$$

to yield

$$b_n^{(1)}(t) = \frac{\hat{W}_{ni}}{2i\hbar} \left[\frac{1 - e^{i(\omega_{ni} + \omega)t}}{\omega_{ni} + \omega} - \frac{1 - e^{i(\omega_{ni} - \omega)t}}{\omega_{ni} - \omega} \right], \quad (1.68)$$

with the transition probability given by $P_{if}(t; \omega) = \lambda^2 |b_f^{(1)}(t)|^2$. For a resonant absorption of energy $\hbar\omega \approx \hbar\omega_{fi}$ only the second term in Eq(1.68) is relevant with transition probability

$$P_{if}(t; \omega) = \frac{|W_{fi}|^2}{4\hbar^2} \left(\frac{\sin[(\omega_{fi} - \omega)t/2]}{(\omega_{fi} - \omega)/2} \right)^2. \quad (1.69)$$

For a constant period of time t , the transition probability is a maximum for a frequency near ω_{fi} and for constant radiation frequency, there is an oscillating transition between the two discrete levels. When considering transitions between states in a material there exists a characteristic time called the *correlation time*, τ_c beyond which coherent behavior is lost. Typically for disordered materials like organic semiconductors this time is very short. The finite correlation time leads to broadening in the energy levels and suppresses the oscillatory

behavior. Transition probabilities in this regime are described by the famous Fermi's golden rule,

$$P_{if} = \frac{2\pi}{\hbar} |\langle \phi_f | W | \phi_i \rangle|^2 \rho(E_f), \quad (1.70)$$

with $\rho(E_f)$ the density of states at the energy E_f of the final states.

After this discussion on transition probabilities between non-degenerate states we next discuss a tight-binding model that incorporates energetic disorder.

1.4.5 The Anderson model

Inhomogeneities in the lattice - whether resulting from compositional/substitutional disorder or positional/structural disorder - act to localize the electronic wave function. This localization can come about from variations in the local potential (Anderson localization) or from disorder in the spatial configuration of centers (Lifshitz localization). In general the electron wave function can be written as a superposition of states centered at sites throughout the lattice. Delocalized electrons are described by an infinite superposition of site functions each with roughly equal weight. Localized electrons are described by a superposition where the weighting falls off exponentially with separation from a certain site. In 1958 Anderson studied the absence of diffusion in a model with disorder. The Anderson model is a tight-binding model with a single state per lattice site and a Hamiltonian of the following form

$$H = \sum_j \epsilon_j a_j^\dagger a_j + \sum_{j,m \neq 0} I(m) a_j^\dagger a_{j+m}, \quad (1.71)$$

where $I(m)$ is a constant overlap integral between nearest neighbor sites j and $j+m$ (0 otherwise), and the disorder is introduced through the site energies ϵ_j which are assumed random and uncorrelated. The site energy distribution is taken to be uniform over some interval W . For the case where all site energies are the same, sharp band edges form in the density of states due to the translational symmetry of the lattice - the dispersion relation has the familiar form $E(k) = 2t \cos(ka)$, and the eigenstates are Bloch functions. Disorder acts to smooth the edges of the density of states into tails and solutions become localized with exponential falloff.

Anderson argued that when the energy interval W is sufficiently broad relative to the overlap energy $I(m)$ all states are localized. As the ratio W/I decreases past a critical value extended states appear in the middle of the band and mobility edges form separating the delocalized states in the middle of the band from localized states in the tails. Depending on the ratio W/I transport can fall into one of two regimes: multiple trapping or multiple hopping.

Multiple trapping regime

Systems that exhibit weak disorder experience conduction through trap and release events at low carrier density. In this regime there are two classes of carriers: (i) mobile carriers and (ii) immobile carriers. The mobile carriers exhibit band-like transport with random scattering events while the immobile carriers are localized until released by a phonon. The two classes are separated by the mobility edge and mobile carriers are typically less than $k_B T$ above the mobility edge. Since the released electron is delocalized the trap landscape can be modeled assuming dynamic disorder.

Multiple hopping regime

Strong disorder results in carriers that are predominantly localized with no band-like transport. In this case all hops are seen as phonon assisted and the site disorder is static. This has implications on the spin dynamics of carriers since there is a finite chance of hopping back and forth between two sites with static disorder. Conduction within the multiple hopping regime has been successfully modeled as current through a network of random resistors by Miller and Abrahams in 1960 [9].

1.5 Calculating currents

1.5.1 Miller-Abrahams rates

When the charge carriers of a system are localized, conduction takes place through hopping transport. An important model for transport between two localized sites is that developed by

Miller and Abrahams [9]. This model is intended to calculate the probability of an inelastic transition between two donor states as well as the transition rate. Under the influence of a weak electric field the transition rate will favor a particular direction set by the field and a current will flow. This inter-donor current corresponds to a resistance and so current through a dopant network can be modeled as a current through a resistor network. The intersite resistance has the form

$$R_{ij} = \gamma_{ij}^0 \exp\left(\frac{2r_{ij}}{a} + \frac{\epsilon_{ij}}{k_B T}\right), \quad (1.72)$$

where r_{ij} is the spatial separation between site i and j , a is the localization radius of the carrier wave function, ϵ_{ij} is the energy difference between site i and j , T is the temperature, k_B is the Boltzmann constant, and γ_{ij}^0 is the hopping rate at zero spatial separation and equal site energy.

1.5.2 Master equation approach

A more versatile approach to calculating current through, in the simplest case, a single state uses the machinery of the density matrix. As discussed in earlier sections, the coherent evolution of a state is determined through the Liouville-von Neumann equation and on-site relaxation can be included through Lindblad operators. Coherent transport would involve adding a hopping term to the Hamiltonian, but calculations of this sort are typically best dealt with through non-equilibrium Green's functions. Conversely, incoherent transport is handled with ease using the master equation approach. Let's consider a system consisting of a single site that can either be vacant or occupied by a single spin-1/2 carrier. In this scenario, the space of the system is a Fock space composed of the 1x1 empty state ρ_0 and the 2x2 occupied state $\rho_{1/2}$,

$$\rho = \rho_0 \oplus \rho_{1/2}. \quad (1.73)$$

The resulting matrix is a 3x3. To describe the transitions of charge onto and off of the site we can use the following set of Lindblad operators,

$$\begin{aligned} A_1 &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, A_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ A_3 &= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, A_4 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \end{aligned} \quad (1.74)$$

with

$$\gamma_1 = \gamma_2 = \gamma_{\text{empty}}, \gamma_3 = \gamma_4 = \gamma_{\text{fill}}. \quad (1.75)$$

Substituting Eq(1.74) and Eq(1.75) into the Lindblad master equation Eq(1.33) and solving for the steady state, $d\rho/dt = 0$, gives a solution for the density matrix. As a result of the conservation of charge, the current into the site is equal to the current out of the site. The current into the site is given by

$$i_{\text{in}} = e\gamma_{\text{fill}}\rho_{11}, \quad (1.76)$$

and the current out of the site is

$$i_{\text{out}} = \frac{e}{2}\gamma_{\text{empty}}(\rho_{22} + \rho_{33}), \quad (1.77)$$

the former amounting to a trace over the portion of Fock space representing an unoccupied site and the latter amounting to a trace over the occupied portion of the Fock space. The analytic result is

$$i = i_{\text{in}} = i_{\text{out}} = e \frac{\gamma_{\text{empty}}\gamma_{\text{fill}}}{\gamma_{\text{empty}} + 2\gamma_{\text{fill}}}. \quad (1.78)$$

This concludes the whirlwind tour of the relevant concepts in spin physics that will be used in the coming chapters. The remainder of the chapter will focus on an overview of the progress made in the area of spintronics.

1.6 Early accomplishments in spintronics

The utility of spin dependent effects in solid state structures was first noticed three decades ago in Fe/Cr superlattices and Fe/Cr/Fe multilayers [10, 11]. It was in these structures that giant magnetoresistance (GMR) was observed by the anti-ferromagnetic coupling of the superlattices. In the following decade much effort was put into developing devices based on these spin correlations. Spin valves and magnetic tunnel junctions (MTJs) found application in magnetic field sensing, hard drive read heads, and magnetoresistive random access memory (MRAM). In both devices there is a pinned ferromagnetic (FM) layer and a free FM layer. Based on the orientation of the free layer the device can switch from a high resistance orientation to a low resistance orientation. Spin valves sandwich a thin non-magnetic conducting layer between the two ferromagnets and MTJs sandwich a thin insulating layer such as aluminum oxide. While MTJs are able to generate a larger change in resistance, they have a larger absolute resistance due to the tunneling nature of the device. The technological significance of the magnetic field dependent resistivity led Albert Fert and Peter Grünberg to the Nobel prize in physics in 2007 for the identification of magnetoresistance. Initially the MRAMs free layer was switched by the magnetic fields of two crossed current carrying lines. More modern approaches use spin polarized current through the memory cell itself via spin-transfer torque. The main benefit to magnetic storage of information is non-volatility *i.e* the memory state is preserved without a power supply. These devices are predominantly used for memory storage, but semiconducting devices are better suited for data processing.

1.7 Semiconductor spintronics

Semiconductors provide an ideal environment for data processing because their material properties can be tailored by the addition of dopants or intentional imperfections in the crystal. The conductivity can be significantly influenced by the presence of electric fields, as is seen in the field effect transistor (FET). Three decades ago a spin-based analog of the FET was proposed[12]. The model proposed by Datta and Das in 1990 relies on the spin

precession due to strong spin-orbit interactions in narrow gap semiconductors to modulate the current. The gate voltage in this case controls the degree of precession through the effective magnetic field interacting with the current [13]. By using a polarized injection contact and a polarized detection contact the degree of precession has a direct effect on the detected current [14–16].

Electrical injection and detection of spin polarized carriers via ferromagnetic (FM) contacts requires one to overcome losses due to conductance mismatch and imperfections at the interface. These obstacles make injection via metallic FM contacts inefficient [17]. Typical injection and detection efficiencies are 1%. This means that the efficiency of a FM/semiconductor/FM structure is near 0.01%. The low efficiency not only makes it very difficult to detect spin signals electrically, but also hinders one’s ability to gain any information regarding the injection or detection process. One way of combating the conductance mismatch between the FM material and the adjacent non-magnetic (NM) material is to use materials with similar conductances. The FM material in this case could be replaced by a diluted magnetic semiconductor (DMS) [14, 15, 18]. A spin analog of the bipolar junction was proposed based on the high spin polarization of DMSs [19]. Non-electrical methods for injection and detection rely on optical selection rules. In spin based light emitting diodes (LED) an intrinsic semiconducting material is sandwiched between a p-type and an n-type material to form a *pin* diode. Dilute magnetic II-VI semiconductors (n-type) or a metallic FM contact can be used to inject spin polarized electrons into the intrinsic quantum well along with unpolarized holes from a NM p-type contact [14]. The electron-hole recombination is dependent on the optical selection rules where conservation of momentum demands the change in angular momentum to be $\Delta m = \pm 1$. The degree of right-handed or left-handed circular polarization of the emitted photon reveals the degree of polarization the injected carriers possess. Injection of spin-polarized holes can be realized by using the III-V DMS GaMnAs [15]. Since detectors are emitters with opposite current, the spin LED can be used to inject polarized carriers into a device. The spin polarization of carriers in a semiconductor can also be measured by the degree of rotation an incident beam of linearly polarized

light has after reflection. This technique is called the magnetooptical Kerr effect (MOKE). Although optical generation and detection skirts the issue of inefficiency at the interface, miniaturization places a limit on the applicability.

1.8 Dots, dopants, and defects

Quantum information depends on a deep understanding of phase coherent phenomena and entangled spin states to store and manipulate quantum bits (qubits) of information. The physical realization of quantum computing challenges our understanding of quantum material science both theoretically and experimentally. In 2000 DiVincenzo published five criteria that must be met for the implementation of quantum computing to be possible [20]:

1. A scalable physical system with well characterized qubits
2. The ability to initialize the qubits
3. Long relevant decoherence times
4. A “universal” set of quantum gates
5. A qubit specific measurement capability

The starting point for implementation is a two-level system that can be used as a qubit. Single electrons act as a direct realization and can be localized for instance in quantum dots, dopants, and defects.

1.8.1 Gate controlled structures

A semiconducting heterostructure consisting of one n-doped material with a large gap and a material with a smaller gap can trap a layer of conduction electrons at the interface due to the potential well generated by the band bending. Gate controlled dots are created by electrostatically defining regions of these two-dimensional electron gases. A typical quantum dot set-up localizes electrons in a AlGaAs/GaAs two-dimensional electron gas (2DEG). Owing to the three dimensional confinement the energy levels of the dot are discrete. Manipulating

the voltage applied by individual contacts allows one to control the occupation of the dot by adjusting the potential barrier between adjacent dots. The influence of the charge state on a nearby quantum point contact can then be measured owing to the conductance quantization in units of $2e^2/h$ [21]. Conceptually, as an electron enters the dot it decreases the conductance by a discrete amount due to the electrostatic interactions at the point contact [2]. Since the small size of the dot results in a large capacitance, the energy required to add charge to the dot ranges on the order of 0.1-1 meV. For this reason, probing the occupancy of the dot requires temperatures below 4K.

Initialization of the QD can be done by a three step process: 1) empty the dot by applying a sufficient negative bias such that the QD levels are above the chemical potential of the electron reservoir μ_{res} , 2) increase the plunger bias bringing the QD levels below μ_{res} and monitor occupation via the quantum point contact, 3) wait for the spin state to relax into the ground state.

Read-out of the spin state involves an extra step required to convert the spin state into a charge state. Once the QD is singly occupied the spin state can be determined by monitoring the current through the point contact. This requires the Zeeman split levels to be tuned so that the ground state is below μ_{res} and the excited state is above. If the current experiences a jump during this part of the scheme then the spin was in the excited state. The increase in current indicating a temporary emptying of the QD resulting from a spin swap with the reservoir. The above outlines a means of achieving criterion 1-3 and 5.

Quantum dots are only one means of localizing spin states. In the next section the possibility of using defects as localized spin centers for quantum information is discussed.

1.8.2 Dopants and defects

In single spin sensors such as spin-1/2 dopants in semiconductors the physical quantity measured is the magnetic field [22]. Vacancies like the nitrogen vacancy in diamond and the divacancy in silicon carbide have a higher spin manifold which allow them to couple to strain and electric fields in addition to magnetic fields. In diamond the nitrogen vacancy

has spin sublevels that are sensitive to Zeeman, Stark, and temperature shifts allowing for a wide array of applications.

The absolute limit on device size leads one to examine the possibility of single-impurity devices. As device scale decreases the distribution of dopants cannot be assumed random and uniform. Techniques for precise placement of impurities then become critical for device performance. The influence of single impurities on device performance was first observed in the form of telegraph noise. Since then, the discrete properties of impurities have been considered in proposed device designs, including defects for quantum information. A major hurdle towards optimal single-impurity applications is the reliable construction of defects and placement of dopants. The past decades have seen remarkable progress in techniques like ion implantation [23] and nanoscale construction via scanning tunneling probe. Two groups have used ion implantation of N in diamond to generate NV^- centers with precision on the order of 20 nm [24, 25]. Atom manipulation via STM has been shown with manganese dopants on a GaAs surface with near atomic precision [26]. In silicon, using STM H-lithography, phosphorus donors have been incorporated with 1 nm precision [27]. In addition to deterministic placement of single atoms, silicon dangling bonds have been linked together through single atom depassivation techniques to form artificial molecules [28]. While much encouraging progress has been made there are still open challenges [29].

Chapter 2

Charge and Spin Phenomena in Organic Semiconductors

2.1 A brief word on organic spintronics

Organic semiconductors provide attractive opportunities for spintronic devices because of their low cost to manufacture and their long spin coherence times [30]. Their relatively weak spin-orbit coupling allows other features of the materials to influence spin dynamics, such as hyperfine interactions of carriers with nuclei, and low carrier mobility. Correlations between slow carrier transport and spin relaxation mechanisms can yield complex behavior, including room-temperature magnetic-field effects on spin and charge dynamics that greatly exceed those found within nonmagnetic inorganic semiconductors [31–33]. A detailed understanding of spin relaxation is often essential for applications [34, 35], as the introduction of spin-relaxing heavy atoms [36–38] or radical spins [39, 40] to relax excitonic or exciplex spins increases the light emission in organic light emitting diodes. Despite this, key parameters of organic material design have not yet been exploited, such as the correlated orientation of π orbitals due to steric interaction even in disordered films. Spin relaxation through the spin-orbit interaction depends on an interplay between the carrier’s orbital motion, which is influenced by the orientation of the π orbitals, and the carrier’s spin. Thus a local steric environment that limits the orientation change of neighboring π orbitals, such as affects the electronic and optical properties of conjugated polymers[41–43], should substantially change the spin relaxation in organic semiconductors. It has recently been observed[43] that reversible mechanical changes in a planar molecule’s steric environment modify the charge conductivity of an organic thin film.

2.2 Hybridization in π -conjugated polymers

Semiconducting organic polymers are based on chains of carbon atoms. Each carbon has four valence electrons (six total electrons) and in free space or a spherically symmetric potential the $1s$ and $2s$ orbitals are filled leading to a $1s^2 2s^2 2p^2$ configuration. Bonding in organics is a bit more peculiar than one might expect. For instance, the molecule CH_4 consists of four hydrogen atoms covalently bonded to a single carbon and it would be reasonable to assume that the bond between the hydrogen and the s -orbital would differ in some way from the bond with the p -orbital. In fact, it has been shown experimentally that the covalent bonds of each hydrogen are identical. This observation leads to the mixing or hybridization of s and p orbitals which can achieve a lower overall energy and allow the bonds to be spatially directed. In the case of CH_4 and traditional plastics like polyethylene the hybridization involves an s electron and and three p electrons leaving the carbon valence saturated. This theory was laid out by chemist Linus Pauling in 1931 [44].

The first step in hybridization is to promote an electron from the $2s$ orbital to the $2p$ orbital yielding a $1s^2 2s^1 2p^3$ configuration. The energy involved in this transition is $\Delta E \approx 7.8$ eV and will be made up for by bonding. The most common type of hybrid orbital is the sp^3 which forms tetrahedral bonds in diamond and saturated polymers, but since this work focuses on spin and charge transport in polymers the hybridization of interest is the sp^2 hybridization found in π -conjugated polymers. In either case the the s and p wave functions can be defined in terms of an orthogonal coordinate system:

$$\begin{aligned}
 |s\rangle &= f_s(r) \\
 |p_x\rangle &= \sqrt{3} \sin \theta \cos \phi f_p(r) \\
 |p_y\rangle &= \sqrt{3} \sin \theta \sin \phi f_p(r) \\
 |p_z\rangle &= \sqrt{3} \cos \theta f_p(r)
 \end{aligned}
 \tag{2.1}$$

where the above expressions have been normalized to $1/\sqrt{4\pi}$. Forming directional bonds is now a matter of combining the wave functions in Eq(2.1) with the desired symmetry. The resulting wave function in the case of sp^2 hybridization will have $1/3$ s character and $2/3$ p

character leaving a single unhybridized p -orbital. For example,

$$\begin{aligned}
\psi_z &= |p_z\rangle, \\
\psi_0 &= \sqrt{\frac{2}{3}} \left(\frac{\sqrt{2}}{2} |s\rangle + \frac{\sqrt{3}}{2} |p_x\rangle - \frac{1}{2} |p_y\rangle \right), \\
\psi_{120} &= \sqrt{\frac{2}{3}} \left(\frac{\sqrt{2}}{2} |s\rangle + |p_y\rangle \right), \\
\psi_{240} &= \sqrt{\frac{2}{3}} \left(\frac{\sqrt{2}}{2} |s\rangle - \frac{\sqrt{3}}{2} |p_x\rangle - \frac{1}{2} |p_y\rangle \right),
\end{aligned} \tag{2.2}$$

where the unhybridized p -orbital points along the z -axis. In the next section it will be shown that the unhybridized p -orbital is critical to conduction in π -conjugated polymers.

2.3 Conduction in π -conjugated polymers

Transport along a π -conjugated polymer requires one to consider how the orbitals discussed in the previous section influence the electronic structure of molecules. The π bond formed by the unhybridized orbital in sp^2 hybridized carbon is higher in energy than the σ bonds formed by the hybridized orbitals and because of this splitting the hybridized orbitals form a backbone along which the relatively delocalized π electrons can conduct. The relevant transport molecular orbitals are formed by the bonding of the p_z atomic orbitals. The Hamiltonian for the p_z atomic orbitals of two π -bonded carbon is

$$\mathcal{H}_\pi = \begin{pmatrix} E_{p_z} & t_\pi \\ t_\pi & E_{p_z} \end{pmatrix} \tag{2.3}$$

and by diagonalizing $\mathcal{H}_\pi$ one finds the stationary states to be the symmetric $|\pi\rangle$ and anti-symmetric $|\pi^*\rangle$ which are split in energy by $2t_\pi$:

$$\begin{aligned}
|\pi\rangle &= \frac{1}{\sqrt{2}} \left(|p_z^{(1)}\rangle + |p_z^{(2)}\rangle \right) \\
|\pi^*\rangle &= \frac{1}{\sqrt{2}} \left(|p_z^{(1)}\rangle - |p_z^{(2)}\rangle \right),
\end{aligned} \tag{2.4}$$

where $|\pi^{(i)}\rangle$ is the unhybridized orbital of the i^{th} carbon.

2.3.1 Uniform trans-(CH)_x

A simple example of a polymer is polyacetylene - a polymer composed of only carbon and hydrogen atoms. As a point of comparison consider the case where the bonds between N neighboring carbon are uniform. This will generate a single π -band which can accommodate $2N$ electrons (accounting for spin degeneracy). Each of the N carbon atoms will contribute a single electron to the π -band from the singly occupied p-orbital at each site making the π -band half filled, making it a metal. The electronic structure of this polymer can be discussed theoretically using the Su-Schrieffer-Heeger (SSH) Hamiltonian, which assumes the π -electronic structure can be treated in the tight-binding approximation and that the chain of carbon atoms couples to the local electron density through the bond length. The Hamiltonian can be written as

$$\mathcal{H}_{SSH} = \sum_{n,\sigma} [-t_0 + \alpha(u_{n+1} - u_n)] (c_{n+1,\sigma}^\dagger c_{n,\sigma} + c_{n,\sigma}^\dagger c_{n+1,\sigma}) + \sum_n \frac{p_n^2}{2m} + \frac{1}{2} K \sum (u_{n+1} - u_n)^2, \quad (2.5)$$

where $c_{n,\sigma}^\dagger$ and $c_{n,\sigma}$ are the fermion creation and annihilation operators for site n and spin σ , p_n are the nuclear momenta, u_n are the displacements from equilibrium, m is the carbon mass, and K is the effective spring constant of the σ -bond. The electronic structure is defined by the first term and the remaining terms takes into account vibrations in the lattice. The electronic structure can be analyzed by looking at the tight-binding terms in Eq(2.5) and diagonalizing gives

$$\mathcal{H} = \sum_{k,\sigma} [-I_p + (1 - 2 \cos ka)] c_{k,\sigma}^\dagger c_{k,\sigma}, \quad (2.6)$$

where I_p is the ionization potential at each site. The π -electron energies are given by

$$E_\pi = \langle H \rangle = -I_p + t_0(1 - 2 \cos ka) \quad (2.7)$$

relative to vacuum and form a single π -band of width $4t_0$.

2.3.2 Dimerized trans-(CH)_x

In reality polyacetylene is not a metal. Instead of a uniform spacing between carbon sites (CH)_x forms dimers with alternating bond length. This transition is an example of the

Peierls instability in 1D metals. Dimerization results in an overall decrease in energy and a doubling of the unit cell generates a gap making the polyacetylene a semiconductor. The reason why this distortion lowers the overall energy is because the uniform $(\text{CH})_x$ has a half filled π -band meaning that the average energy of the electrons in this band is less than the Fermi energy. This distortion opens a gap at the Fermi energy that is proportional to the size of the distortion and as a result pushes the average energy of the electrons down by an amount proportional to the distortion. This energy could of course be compensated for by the increase in lattice energy, but since the electron phonon coupling happens through the bonds it is dependent on the wave function overlap $S = \langle n \pm 1 | n \rangle$. The wave functions at a given site have a radial decay that is exponential, so

$$S \propto \exp[-(2a \pm \delta)/r_\pi], \quad (2.8)$$

where a is the lattice spacing, r_π is the localization length, and δ is the deviation from the average site position. Expanding Eq(2.8) leads to the realization that leading terms in potential energy difference between a uniform lattice and a dimerized lattice are of order δ^2 . Since the decrease in average π -electron energy is of order δ and the increase in lattice energy is of order δ^2 the distortion is energetically favorable. The dimerized form of the SSH Hamiltonian yields a dispersion expression

$$E_k = \pm \sqrt{\epsilon_k^2 + \Delta^2} \quad (2.9)$$

where $\epsilon_k = 2t_0 \cos ka$ and 2Δ is the difference in hopping energy between the two bond lengths. The solution lower in energy represents the valence π -band and is completely full. The higher energy solution is the conduction π^* -band and it is completely empty.

2.4 Application to PTMA and P3HT

2.4.1 PTMA

To investigate controversial claims that poly(2,2,6,6-tetramethylpiperidinyloxy methacrylate) (PTMA) can be conductive at room temperature, Zhang *et al.* measured the conductivity of PTMA samples fabricated using three different polymerization techniques [45].

Regardless of method PTMA was found to be highly insulating - adding clarity to the debate over PTMA as a viable material for energy storage applications.

Using Mott variable range hopping [46] as a model for charge transport in PTMA, an estimate is made on the number of sites contributing to conduction. Assuming that one monomer contains a single radical side-chain, a sample with a mass density of μ_s implies a volumetric density of states, N_{tot} , equal to

$$N_{\text{tot}} = \frac{\mu_s}{m_m}, \quad (2.10)$$

where m_m is the mass of a monomer. Spacing between radical side chains has been approximated as the average spacing between two monomers. The volume for a monomer is calculated using Eq (2.10) and the spacing between sites is extracted:

$$r = \left[\frac{3}{4\pi N_{\text{tot}}} \right]^{1/3}. \quad (2.11)$$

To arrive at the Mott expression for variable range hopping, consider phonon assisted transport as described by Miller and Abrahams in which hopping transport is modeled as current through a network of random resistors with inter-site resistance of the form

$$R_{ij} = R_{ij}^0 \text{Exp} \left(\frac{2r_{ij}}{a} + \frac{\epsilon_{ij}}{k_B T} \right), \quad (2.12)$$

where k_B is the Boltzmann constant, T is the temperature, a is the electronic wave function localization length, r_{ij} is the spacing between site i and site j , and ϵ_{ij} is the energy difference between the sites defined as

$$\epsilon_{ij} = \frac{1}{2} \left(|\epsilon_i - \epsilon_j| + |\epsilon_i - \mu| + |\epsilon_j - \mu| \right), \quad (2.13)$$

and μ is the Fermi level. At low temperatures Eq (2.12) insists that only sites near in energy contribute to the conduction and from Eq (2.13) the relevant sites are located in a narrow band symmetric about the Fermi level whose width decreases as $T \rightarrow 0$. At sufficiently low temperatures the density of states in this band is considered to be constant and the number of states within the band $|\epsilon_i - \mu| \leq \epsilon_0$ is

$$N(\epsilon_0) = 2g(\mu)\epsilon_0, \quad (2.14)$$

where $g(\mu)$ is the density of states at the Fermi level. The resistivity is now written in terms of ϵ_0 ,

$$\rho = \rho_0 \exp \left\{ \frac{1}{[g(\mu)\epsilon_0]^{1/3}a} + \frac{\epsilon_0}{k_B T} \right\}, \quad (2.15)$$

where the numerical constant $(3/4\pi)^{1/3}$ from the expression for r_{ij} has been ignored. The exponential in Eq (2.12) is controlled by the overlap term depending on r_{ij} and the activation term containing ϵ_{ij} . It is worth noting that as the energy band decreases in width the resistance decreases until the sites become so sparse that the term r_{ij} begins to dominate. This interplay suggests an optimal value for the energy band. To estimate a typical value for inter-site resistance it is assumed that $r_{ij} \propto N(\epsilon_0)^{1/3}$ and $\epsilon_{ij} = \epsilon_0$ and by substituting these values in Eq (2.12) a minimum is found at

$$\epsilon_0^{\text{opt}} = \left(\frac{k_B T}{a} \right)^{3/4} \left(\frac{1}{g(\mu)} \right)^{1/4}. \quad (2.16)$$

The band defined by $|\epsilon_i - \mu| \leq \epsilon_0^{\text{opt}}$ is known as the Mott optimal energy band and is considered to be the band responsible for conduction within the system. Eq (2.16) is dependent on the density of states and before making any statements on the system at hand an estimate of $g(\mu)$ is required. Substituting $\epsilon_0 = \epsilon_0^{\text{opt}}$ into Eq (2.15) returns resistivity characteristic of Mott variable range hopping,

$$\rho = \rho_0 \exp \left[\left(\frac{T_0}{T} \right)^{1/4} \right], \quad (2.17)$$

where $T_0 = \beta/a^3 k_B g(\mu)$ with β a percolative constant [46], and ρ_0 a material dependent constant describing the minimum resistivity based on carrier-phonon interactions [46],

$$\rho_0 = \frac{9\pi}{4} \frac{a^3 d s^5 \hbar^4 \kappa^2}{E_1^2 e^6 k_c^2} \left(\frac{\beta}{a^3 g(\mu) k_B T} \right)^{\frac{\nu+2}{4}}. \quad (2.18)$$

In the above expression d is the mass density of the dielectric, s is the speed of sound in the dielectric [47], κ is the dielectric constant [48], E_1 is the deformation potential [49], e is the electronic charge, k_c is the Coulomb constant, and ν is a numerical constant determined through percolation simulations [46].

For the anionic PTMA sample we now calculate the ratio of conducting sites to insulating sites. An estimate of the density of states is made by equating Eq (2.17) with measured values of the resistivity. For the anionic PTMA sample the measured resistivity is $\rho = 9.19 \times 10^{10} \Omega \cdot \text{cm}$ and from Eq (2.16) $\epsilon_0^{\text{opt}} = 0.5 \text{ eV}$. It is assumed that each monomer of PTMA contains one side chain and given a mass density of $\mu = 1.018 \text{ g} \cdot \text{cm}^{-3}$ and the molecular weight of a monomer $w = 240.32 \text{ Da}$, the average spacing between sites is $r = 4.5 \text{ \AA}$. The concentration of conducting sites is $N_c = 2\epsilon_0^{\text{opt}} \cdot g(\mu) = 2.7 \times 10^{20} \text{ cm}^{-3}$ and so the ratio of conducting sites to insulating sites, f , is

$$f = 2\epsilon_0^{\text{opt}} \cdot g(\mu) \cdot \frac{4\pi}{3} r^3. \quad (2.19)$$

For the parameters outlined above $f = 0.1$ suggesting that about 10% of sites within the sample contribute to charge transport. This corresponds to about 5% of sites being unoccupied, which is consistent with the attenuated total internal reflectance-Fourier transform infrared (ATR-FTIR) spectroscopy measurements [45].

β	20	percolation constant	[46] p.206
κ	3.4	dielectric constant	[48]
ν	0.9	critical exponent	[46] p.105
s	$2000 \text{ m} \cdot \text{s}^{-1}$	speed of sound	[47]
E_1	1 eV	deformation potential	[49]
T	300 K	temperature	
a	1 $\text{\AA}$	wave function localization length	

Table 2.1: Values for terms in Mott conduction model.

$$\mu_{\text{PTMA}} \mid 1018 \text{ kg} \cdot \text{m}^{-3} \mid \text{mass density of PTMA}$$

Table 2.2: Measured values for terms in Mott conduction model.

r	4.5 $\text{\AA}$	average monomer spacing
ϵ_0^{opt}	0.52 eV	Mott optimal energy band
$g(\mu)$	$2.5 \times 10^{20} \text{ eV}^{-1} \cdot \text{cm}^{-3}$	density of states
N_c	$2.5 \times 10^{20} \text{ cm}^{-3}$	vol. density of states for conducting sites
f	0.1	ratio of conducting to insulating sites

Table 2.3: Derived values in Mott conduction model.

2.4.2 P3HT

Charge storage applications take advantage of the redox-active nature of stable radical polymers. Enhanced conductivity leads to improved performance and utilizing radical polymers with a conjugated backbone should be a clear way of introducing conductivity. However, the radical concentration plays a role in the conductivity of the organic as was shown for poly(3-hexylthiophene) (P3HT) [50]. Regioregular P3HT was grown with varying concentrations of TEMPO radical content. As the concentration increased linearly, the conductivity was seen to decrease exponentially. The radical sites are seen to introduce steric effects to the conjugated backbone that limit both inter- and intrachain hopping conductivity. As charge moves throughout P3HT there are relatively low potential barriers between sites along the same polymer (intra-polymer hopping) compared to the potential barriers between polymers (inter-polymer hopping), due to the nature of the conducting backbone. This results in backbone conduction along the length of the polymer until it terminates. After traveling the length of the polymer the charge proceeds to hop to another nearby chain. Since the inter-polymer hops are unavoidable and correspond to larger barriers they are the limiting factor in the conductance. In the case of pure regioregular P3HT the polymers are able to neatly stack facilitating inter-polymer hopping. The addition of TEMPO side-chains distorts the polymer and adds to the disorder in chain-chain orientation increasing the inter-chain hopping barrier.

We now model charge conduction in P3HT as tunneling through inter-chain barriers. The average number of sites per volume, N , can be written as the product of the density of states $g(\mu)$ and the bandwidth ε_0 ,

$$N = 2g(\mu)\varepsilon_0, \quad (2.20)$$

where the factor of 2 accounts for sites in the range $\mu \pm \varepsilon_0$. The addition of TEMPO side-chains acts to reduce the number of available hopping sites by pushing them further away in position or energy. This suggests the volumetric density of states is proportional to the TEMPO percentage, f . In the simplest model, each TEMPO has a probability p for pushing inter-polymer sites out of reach. This allows one to write the density of states as a function

of f ,

$$g(\mu) = \frac{3(1 - pf)}{8\pi\ell^3\varepsilon_0}, \quad (2.21)$$

where ℓ is the average inter-site distance. Applying Eq.(2.21) to the expression for the resistance of a unit segment of a Miller-Abrahams type network allows one to write down an expression for the conductivity

$$\sigma = \sigma_0 \exp\left[\frac{-2\ell}{a(1 - pf)^{1/3}}\right], \quad (2.22)$$

where a represents the carrier wave function localization, and σ_0 is a material and temperature dependent term that describes the conductivity due to carrier-phonon interactions. To recover an exponential dependence of conductivity on TEMPO concentration, we consider the perturbative limit where $fp \ll 1$. This permits the expansion of Eq.(2.22) about $fp = 0$,

$$\sigma \approx \sigma_0 \exp\left[\frac{-2\ell(1 + \frac{pf}{3})}{a}\right]. \quad (2.23)$$

If we take $p = 1$, the slope of the fit line in Figure 2.1 near $f = 0$ gives an estimate for $\ell/a \sim 30$. Although this simple model cannot simultaneously account for the low- f and high- f behavior of σ , it qualitatively captures the trend in the data. For larger values of f , we expect that one would need to explicitly account for the dependence of ℓ on f , and there could be contributions from disorder that further decrease the level of intrachain hopping transport, which would also decrease the conductivity exponentially. We do not make these extensions because five data points cannot constrain a more complex model. Taking $a \sim 0.7 \text{ \AA}$, corresponding roughly to the covalent radius of carbon [51], we estimate that the average interchain hopping distance is around 2 nm, which is comparable to but smaller than the average chain length. This result suggests that the crystalline domains have a sizable average separation and that increasing the TEMPO percentage reduces the average number of accessible crystalline regions available for conductivity, thus exponentially suppressing the conductivity.

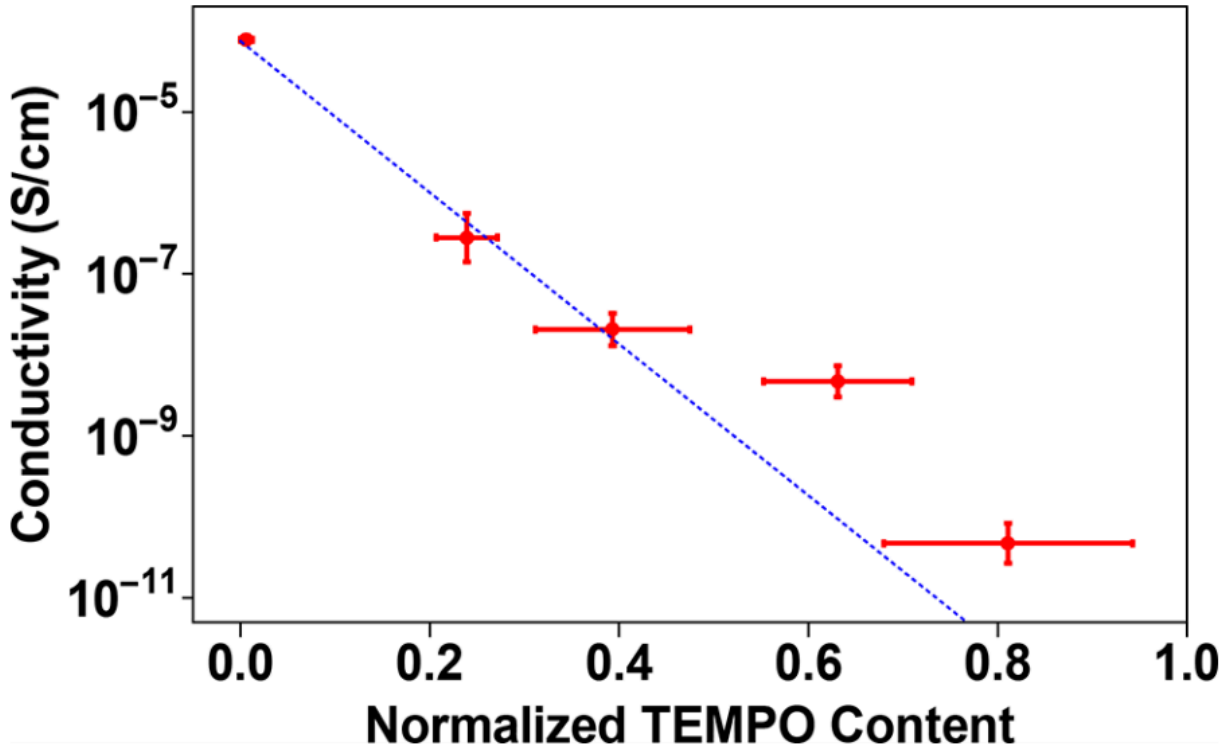


Figure 2.1: Semilog plot of the conductivity of P3HT-TEMPO-X as a function of calibrated TEMPO content. The dashed line is a fit to the simple model described by Eq(2.23)

2.5 A simple model for flexible organics

For conjugated organics the sp^2 hybridized orbitals are split in energy from the un-hybridized p- orbital. This energy splitting results in a HOMO and LUMO with p-type symmetry. In planar molecules like PTCDI-C8, the cores are composed of conjugated ring-like segments with well- defined spatial orientation along the axis of the un-hybridized orbitals. As the primary transport orbitals in these materials lack polar symmetry, the angular dependence of the wave function overlap between neighbouring sites becomes a key component in transport calculations [43]. To illustrate the role of site orientation on the inter-site hopping rate the polar degree of freedom must be included in the expression for inter-site resistance.

2.5.1 The model

For a Miller-Abrahams type network of hydrogenic sites, the inter-site resistance between sites i and j is expressed as

$$R_{ij} = R_{ij}^0 \exp(2r_{ij}/a), \quad (2.24)$$

where R_{ij}^0 describes a minimum resistance limited by material properties of the system, r_{ij} is the inter-site spacing, and a is the carrier wave function localization radius. Consider now two sites, i and j , with arbitrary orientations. Adopting the local basis of site i the states of a carrier at i and j can be expressed as

$$\begin{aligned} |\psi_i\rangle &= |p_z\rangle \\ |\psi_j\rangle &= \sin\theta \cos\phi |p_x\rangle + \sin\theta \sin\phi |p_y\rangle + \cos\theta |p_z\rangle, \end{aligned} \quad (2.25)$$

where $|p_x\rangle$ and $|p_y\rangle$ represent hybridized orbitals and $|p_z\rangle$ represents the un-hybridized transport orbital of site i . For short orbital coherence times, as is typical in organic semiconductors, the dependence of the hopping rate γ_{ij} on site orientation can be calculated through application of Fermi's golden rule,

$$\gamma_{ij} = \frac{2\pi}{\hbar} |\langle\psi_j|\hat{V}|\psi_i\rangle|^2 \rho, \quad (2.26)$$

where ρ is the density of states and $\hat{V}$ is a hopping operator representing the incoherent electron-phonon interaction which facilitates carrier transport. From Eq(2.25) and Eq(2.26) it is clear that

$$\gamma_{ij} \propto \cos^2 \theta_{ij}, \quad (2.27)$$

where θ_{ij} is the relative angle between the transport orbitals at sites i and j . Since $R_{ij} \propto 1/\gamma_{ij}$, in π -conjugated materials with well-defined site orientations it is appropriate to consider an inter-site resistance of the form

$$R_{ij}(r, \theta) = \frac{R_{ij}^0}{\cos^2 \theta_{ij}} \exp(2r_{ij}/a). \quad (2.28)$$

The next step is to consider how the parameters r_{ij} and θ_{ij} change as the organic is flexed.

For a deformed material the molecular unit cell is strained and as a result the volume is, in general, not conserved. This results in a modification to the volumetric density of

states which enters the transport equations exponentially through the inter-site spacing r_{ij} . For a change in length dx along the applied strain axis the material response along the perpendicular axes (dy) are governed by the Poisson's ratio, $dy = -\nu dx$. The fractional change in volume as a result of small deformations is expressed as

$$\frac{\Delta V}{V} = (1 - 2\nu) \frac{dx}{x_0}, \quad (2.29)$$

where ΔV is the change in volume, V is the initial volume, and x_0 is the initial length along the strain axis. The new site spacing is proportional then to the cube-root of the volume of the deformed molecular unit cell,

$$r'_{ij} \propto (V + \Delta V)^{1/3} = V^{1/3} \left[1 + (1 - 2\nu) \frac{dx}{x_0} \right]^{1/3}. \quad (2.30)$$

Rotations of single sites can be analysed by focusing on the cross-sectional plane spanned by the axis of applied strain and a single perpendicular axis ignoring the spherical angle. The local orientation of a single site is defined by a polar angle θ between the global basis vector $\hat{y}$ and the local basis vector $\hat{y}'$ [Fig(2.2)]. Normal strain on a molecular unit cell leaves the orientation of the site unchanged while shear strain causes rotation. The strain in the global coordinates (ϵ_x and ϵ_y) can be written in the local basis of a given site through the strain transformation relations:

$$\begin{aligned} \epsilon_{x'} &= \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2} \cos 2\theta + \frac{\gamma_{xy}}{2} \sin 2\theta \\ \epsilon_{y'} &= \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2} \cos 2\theta - \frac{\gamma_{xy}}{2} \sin 2\theta \\ \frac{\gamma_{x'y'}}{2} &= -\frac{\epsilon_x - \epsilon_y}{2} \sin 2\theta + \frac{\gamma_{xy}}{2} \cos 2\theta. \end{aligned} \quad (2.31)$$

The normal strain in the global coordinate system generates shear strain $\gamma_{x'y'}$ that rotates the site by an angle $\delta\theta = \gamma_{x'y'}/2$. The degree of rotation is linked to the material properties through Poisson's ratio and local shear strain can be related to the radius of curvature through

$$\epsilon_x - \epsilon_y = \frac{w}{R}(1 + \nu), \quad (2.32)$$

where R is the radius of curvature and w is the distance from the neutral axis. For two

neighboring sites (i and j) with normal vectors in the flat configuration

$$\begin{aligned}\hat{n}_\infty^i &= \sin \theta_i \hat{x} + \cos \theta_i \hat{y} \\ \hat{n}_\infty^j &= \sin \theta_j \hat{x} + \cos \theta_j \hat{y},\end{aligned}\tag{2.33}$$

and deformed configuration with radius of curvature R

$$\begin{aligned}\hat{n}_R^i &= \sin(\theta_i + \delta\theta_i) \hat{x} + \cos(\theta_i + \delta\theta_i) \hat{y} \\ \hat{n}_R^j &= \sin(\theta_j + \delta\theta_j) \hat{x} + \cos(\theta_j + \delta\theta_j) \hat{y},\end{aligned}\tag{2.34}$$

the relative angle between the two sites is

$$\theta_{ij} = \cos^{-1}[\sin(\theta_i + \delta\theta_i) \sin(\theta_j + \delta\theta_j) + \cos(\theta_i + \delta\theta_i) \cos(\theta_j + \delta\theta_j)].\tag{2.35}$$

Current through the organic layer exhibits hopping conduction along a path of sites with each hop occurring at a rate that depends on the relative position and orientation of the initial and final site. These rates are analogous to resistors in series with a value corresponding to Eq(2.28). Since the bulk resistance is determined by summing the individual resistances along the path it is clear from Eq(2.28) that hops between sites with relative orientations near perpendicular will dominate the total value. These bottlenecks that exist in the relaxed organic will be broken as the device is bent and the sites shift their relative position and orientation. Since the strain is dependent on the distance from the neutral axis the average value is considered, $S_{\text{avg}} = \frac{w_{\text{max}}}{2R}$, where we have assumed the neutral axis to be midway between the top and bottom contacts and therefore $w_{\text{max}} = 100$ nm.

For two neighboring sites i and j in a bottleneck configuration the orientation of site i and site j with respect to the global basis are related through $\theta_j = \theta_i + \pi/2 - \xi$, where ξ is a small deviation from completely perpendicular orientation. The inter-site resistance for a bottleneck in a relaxed organic can then be written as

$$R_\perp^\infty = R_{ij}^0 \frac{\exp\left(\frac{2r_{ij}}{a}\right)}{\left[\sin \theta_i \sin\left(\theta_i + \frac{\pi}{2} - \xi\right) + \cos \theta_i \cos\left(\theta_i + \frac{\pi}{2} - \xi\right)\right]^2}\tag{2.36}$$

and therefore, the resistivity of the flat organic is

$$\rho_{\perp}^{\infty} \propto \frac{\exp\left(\frac{2r_{ij}}{a}\right)}{\left[\sin\theta_i \sin\left(\theta_i + \frac{\pi}{2} - \xi\right) + \cos\theta_i \cos\left(\theta_i + \frac{\pi}{2} - \xi\right)\right]^2} \quad (2.37)$$

and the resistivity of the deformed organic is

$$\rho_{\perp}^R \propto \frac{\exp\left[\frac{2r_{ij}}{a}(1 + (1 - 2\nu)S_{\text{avg}})^{1/3}\right]}{\sin^2[\xi - S_{\text{avg}}(\nu + 1)\cos(\xi)\sin(2\theta_i - \xi)]}. \quad (2.38)$$

In comparing with experimental values, the fractional change in resistivity is considered

$$\frac{\Delta\rho}{\rho_{\perp}^{\infty}} = \frac{|\rho_{\perp}^{\infty} - \rho_{\perp}^R|}{\rho_{\perp}^{\infty}} \quad (2.39)$$

as a function of the radius of curvature. The value of the inter-site spacing is approximated from the cubed root of the volume of the measured triclinic unit cell for PTCDI-C₈ as 9.19 Å and the carrier localization radius is approximated as 1 Å. The value for the Poisson's ratio is estimated from measured values of similar materials to be 0.4 and the angular deviation ξ is left as a fitting parameter. For bottlenecks in the relaxed organic with relative angle $\theta_{\perp} = \pi/2 - \xi$ the general experimental trend can be reproduced for $\xi = 2.5 \times 10^{-6}$ rad as seen in Fig(2.2).

In this treatment the fractional change in resistivity of a single bottleneck is examined with no regard to a particular packing motif and the strain is distributed equally among the individual sites. This model is intended to motivate the impact of site orientation on carrier transport in planar π -conjugated organics.

2.5.2 Anisotropic transport

It is common for materials with π -stacking to exhibit anisotropic transport. In this case it is reasonable to consider a molecular unit cell with cylindrical geometry where the diameter defines inter-site spacing with the non-stacking plane and the length of the cylinder defines spacing along the stacking axis. Assuming these unit cells can be arbitrarily oriented in space consider a rotation by θ , once again ignoring the spherical angle. The normal strain

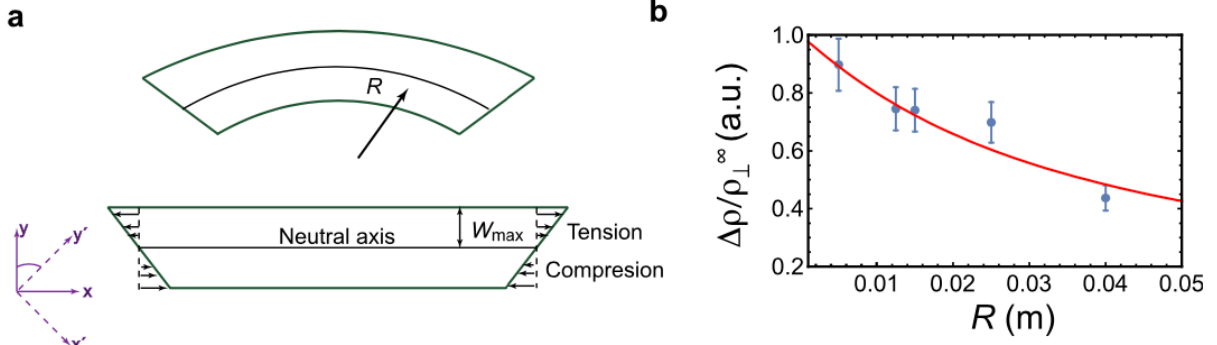


Figure 2.2: (a) Diagram of a flexed organic with radius of curvature, R (top) and an unbent representation with linearly varying force (bottom). The global basis (unprimed) and the local basis (primed) are shown with the angle, θ , defining the degree of rotation in the $\hat{x} - \hat{y}$ plane. $\hat{x}$ is the axis of applied strain, and $w_{\max}$ is the distance between the neutral axis and the contact surface (100 nm). (b) Fractional change in resistivity as a function of the radius of curvature for experimental measurements (circle) and theoretical model (solid line).

that is applied along the global x -axis will now be responsible for a shear strain in addition to normal strain along the local x' -axis and y' -axis as defined by Eq(2.31). Each of these normal strains will induce a reactionary strain in the transverse direction related by the Poisson's ratio. Taking the y -axis to be the π -stacking axis, the resistivity along this direction is expressed as

$$\rho_{\perp\pi}^R \propto \frac{\exp \left\{ 2L/a \left[1 + \frac{S_{\text{avg}}}{2} (1 - \cos 2\theta_i) - \nu \frac{S_{\text{avg}}}{2} (1 + \cos 2\theta_i) \right] \right\}}{\sin(\xi - S_{\text{avg}}(\nu + 1) \cos \xi \sin(2\theta_i - \xi))^2} \quad (2.40)$$

Where L is the length of the unit cell. The resistivity between sites in the non-stacking plane is then

$$\rho_{\perp d}^R \propto \frac{\exp \left\{ 2d/a \left[1 + \frac{S_{\text{avg}}}{2} (1 + \cos 2\theta_i) - \nu \frac{S_{\text{avg}}}{2} (1 - \cos 2\theta_i) \right] \right\}}{\sin(\xi - S_{\text{avg}}(\nu + 1) \cos \xi \sin(2\theta_i - \xi))^2} \quad (2.41)$$

where d is the diameter of the unit cell.

Plotting the fractional change in the resistivity for average strain corresponding to a radius of curvature $R = 0.05\text{m}$ and a dimensionless parameter $d/a = 4L/a = 20$ produces results that are virtually identical for both transport directions, with a percentage difference between them on the order of 0.001% as can be seen from Fig(2.3). Since the site rotation

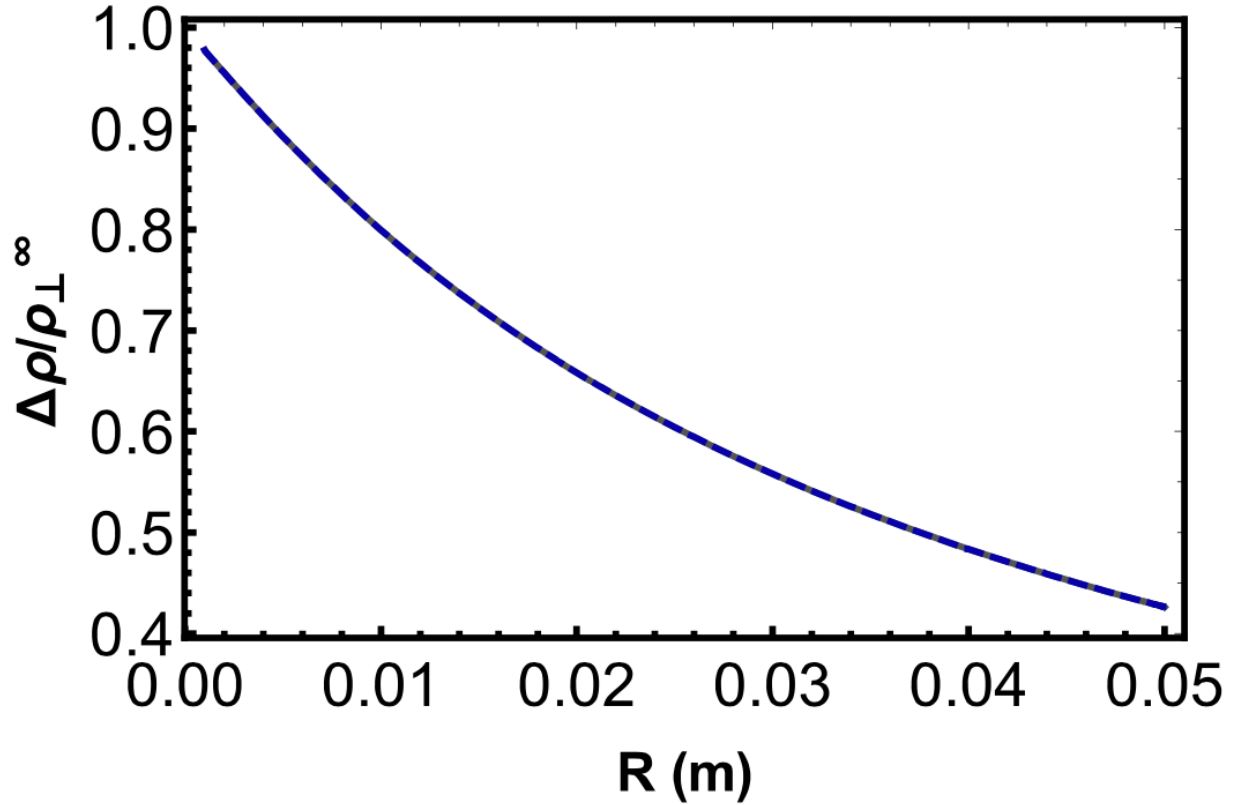


Figure 2.3: Fractional change in resistivity as a function of the radius of curvature for anisotropic systems. Since the site rotation dominates the resistivity in the bottleneck configuration, the essential behavior of the flexed materials is captured in a model that assumes isotropic transport.

dominates the resistivity in the bottleneck configuration, the essential behavior of the flexed materials is captured in a model that assumes isotropic transport.

2.6 Spin-orbit induced spin relaxation in organics

In π -conjugated materials sp^2 hybridized molecules are connected together by σ bonded hybridized orbitals and a π -bond backbone resulting from un-hybridized p_z orbitals. The

Hamiltonian for σ and π p -orbitals is

$$\begin{aligned}
\hat{H} &= H_0 + H_{\text{so}} + V \\
&= \sum_{i,n,s} \epsilon_{i,n} a_{i,n,s}^\dagger a_{i,n,s} + H_{\text{so}} \\
&\quad + \sum_{i,j,n,m,s} \langle p_{j,m} | \hat{V} | p_{i,n} \rangle a_{j,m,s}^\dagger a_{i,n,s},
\end{aligned} \tag{2.42}$$

where i and j represent molecular sites, n and m molecular orbitals, and s the spin state, $\uparrow$ or $\downarrow$. $\epsilon_{i,n}$ is the energy of a carrier at site i and orbital n , H_{so} the spin-orbit Hamiltonian, $\hat{V}$ the hopping operator, and $a_{i,n,s}^\dagger$ and $a_{i,n,s}$ are the creation and annihilation operators for site i , orbital n , and spin s . In organic materials the coupling between spin and orbital angular momentum is weak due to the low atomic masses of the constituent atoms. This allows the spin-orbit interaction, $H_{\text{so}} = (2/3)\Delta\vec{L} \cdot \vec{S}$, to be treated as a perturbation to H_0 , where Δ is the spin-orbit coupling constant and $\vec{L}$ and $\vec{S}$ are the orbital and spin angular momentum vectors. In a basis generated by the molecule's orbital states and eigenfunctions of the spin operator along the unhybridized axis, the eigenstates of this Hamiltonian are

$$\begin{aligned}
|+\rangle &= |p_z \uparrow\rangle + \frac{\Delta}{3\delta\epsilon} \left(|p_x \downarrow\rangle + i|p_y \downarrow\rangle \right) \\
|-\rangle &= |p_z \downarrow\rangle - \frac{\Delta}{3\delta\epsilon} \left(|p_x \uparrow\rangle - i|p_y \uparrow\rangle \right),
\end{aligned} \tag{2.43}$$

where $\delta\epsilon$ is the on-site energy splitting between $\epsilon_{i,\pi}$ and $\epsilon_{i,\sigma}$, the energy of the π and σ orbitals at site i respectively. The two states in Eq. (2.43) maximize and minimize $\langle \hat{S}_z \rangle$ and are thus pseudo up and down spin states [52].

For a realistic organic material, orbitals on neighboring sites will generally stray from parallel alignment. To represent more than one site at a time it is necessary to establish a global coordinate system, which will also provide a consistent spin quantization axis, which we will consider to be in the $\hat{z}$ direction. In the following we will represent the global basis without primes and local bases with primes (Fig. 2.4). Representations of the eigenstates of a given site now depend on the polar and azimuthal angle that define the orientation of the local p'_z with respect to the global basis.

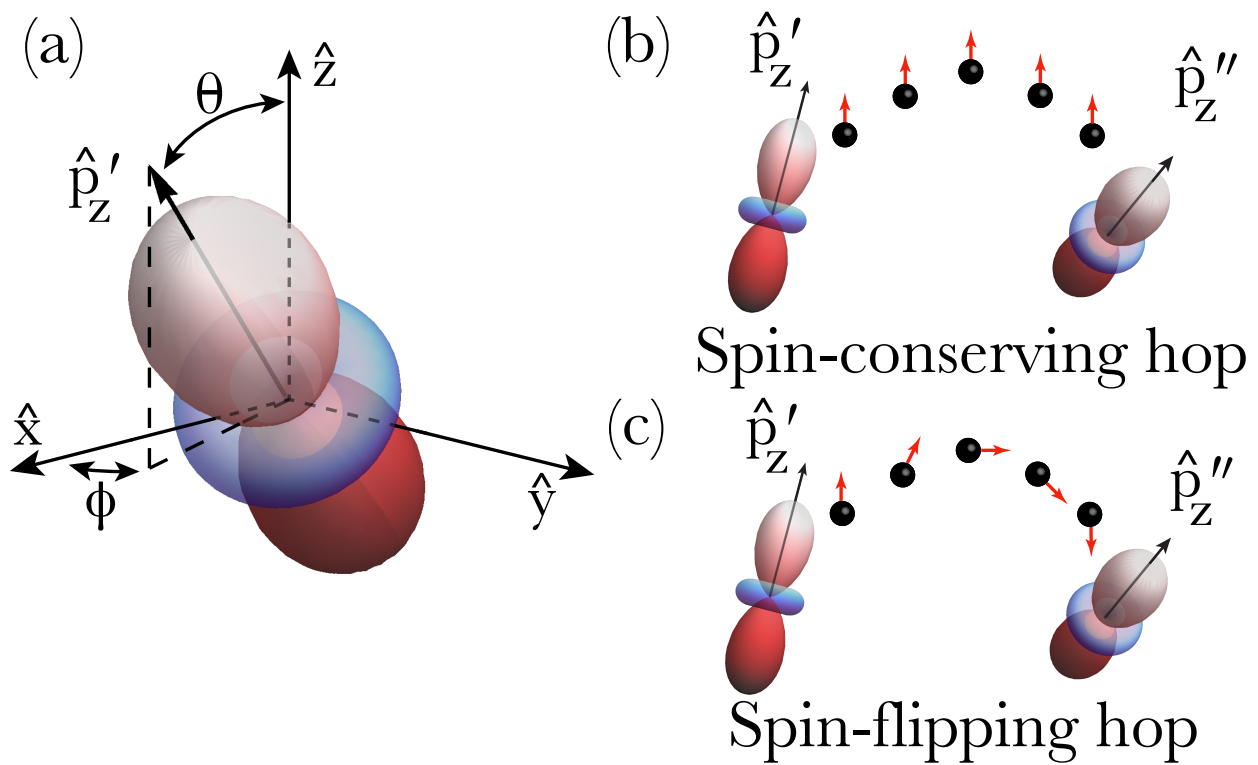


Figure 2.4: (a) Representation of an sp^2 hybridized molecule with global and local coordinates and a conceptual comparison between (b) spin-conserving and (c) spin-flipping hops.

Expanding the local spin eigenstates, Eq. (2.43), in the global basis yields

$$\begin{aligned}
|+\rangle &= |p'_z \uparrow\rangle + \lambda \left(e^{i\phi} |p'_x \downarrow\rangle \right. \\
&\quad \left. - i \sin \theta |p'_y \uparrow\rangle + i \cos \theta e^{i\phi} |p'_y \downarrow\rangle \right), \\
|-\rangle &= |p'_z \downarrow\rangle + \lambda \left(-e^{-i\phi} |p'_x \uparrow\rangle \right. \\
&\quad \left. + i \cos \theta e^{-i\phi} |p'_y \uparrow\rangle + i \sin \theta |p'_y \downarrow\rangle \right),
\end{aligned} \tag{2.44}$$

where $\lambda = \Delta/3\delta\epsilon$ and θ and ϕ are the polar and azimuthal angles respectively. In general a carrier in the $|+\rangle$ state can transition to either $|+\rangle$ or $|-\rangle$, the former representing a spin-conserving hop and the latter a spin-flipping hop (Fig. 2.4).

For spin relaxation to occur the carriers need to change their pseudospin states as they hop from site to site. The hopping operator itself is spin independent and is not able to flip spins on its own. Thus the only contributing factors to spin relaxation are the perturbative corrections to carrier eigenstates, *i.e.* due to spin-orbit coupling the up spin state contains a finite down spin component, which in general allows for overlap between the opposing spin states of two sites. Without loss of generality [52] we consider hops only between parallel orbitals, $\langle p_{j,m} | \hat{V} | p_{i,n} \rangle = v_0 \delta_{nm}$, where i and j are nearest neighbors. To calculate spin transition rates we first calculate the hopping matrix elements. The hopping matrix corresponding to a transition between two sites (site 1 and site 2) has the elements

$$\begin{aligned}
V_{++} &= v_0 \left[\cos \theta_i \cos \theta_j + \sin \theta_i \sin \theta_j \cos(\phi_i - \phi_j) \right. \\
&\quad \left. + i2\lambda \sin \theta_i \sin \theta_j \sin(\phi_i - \phi_j) \right] \\
V_{--} &= v_0 \left[\cos \theta_i \cos \theta_j + \sin \theta_i \sin \theta_j \cos(\phi_i - \phi_j) \right. \\
&\quad \left. - i2\lambda \sin \theta_i \sin \theta_j \sin(\phi_i - \phi_j) \right] \\
V_{-+} &= -2\lambda v_0 \left(e^{i\phi_i} \sin \theta_i \cos \theta_j - e^{i\phi_j} \cos \theta_i \sin \theta_j \right) \\
V_{+-} &= 2\lambda v_0 \left(e^{-i\phi_i} \sin \theta_i \cos \theta_j - e^{-i\phi_j} \cos \theta_i \sin \theta_j \right).
\end{aligned} \tag{2.45}$$

The rates for spin-conserving hops, $\nu_c = \nu_{\uparrow\uparrow} = \nu_{\downarrow\downarrow}$, and spin-flipping hops, $\nu_{sf} = \nu_{\downarrow\uparrow} = \nu_{\uparrow\downarrow}$,

are calculated via Fermi's golden rule,

$$\begin{aligned}\nu_c &= \frac{\nu_{tot} |\langle +'' | \hat{V} | +' \rangle|^2}{(1/4\pi) \int_{\Omega} \left[|\langle +'' | \hat{V} | +' \rangle|^2 + |\langle +'' | \hat{V} | -' \rangle|^2 \right]} \\ \nu_{sf} &= \frac{\nu_{tot} |\langle -'' | \hat{V} | +' \rangle|^2}{(1/4\pi) \int_{\Omega} \left[|\langle +'' | \hat{V} | +' \rangle|^2 + |\langle +'' | \hat{V} | -' \rangle|^2 \right]},\end{aligned}\tag{2.46}$$

where ν_{tot} is the sum of ν_c and ν_{sf} , averaged over all orientations (Ω). This approach was used in [52–54] with ν_c and ν_{sf} each averaged independently over all orientations. For each possible orientation of the relevant sites, ν_c and ν_{sf} take on correlated values. These expressions are cumbersome, but can be simplified for a fixed initial orientation, to clarify the origin of steric tunability of spin-orbit relaxation.

2.7 Steric influences on spin relaxation in π -conjugated polymers

Spin relaxation through the spin-orbit interaction depends on an interplay between the carrier's orbital motion, which is influenced by the orientation of the π orbitals, and the carrier's spin. Thus a local steric environment that limits the orientation change of neighboring π orbitals, such as affects the electronic and optical properties of conjugated polymers[41–43], should substantially change the spin relaxation in organic semiconductors. It has recently been observed[43] that reversible mechanical changes in a planar molecule's steric environment modify the charge conductivity of an organic thin film.

Here we calculate the effect of neighboring π orbital orientation on spin and charge motion via incoherent hopping through an organic material using a theory that includes steric effects. The spin relaxation is calculated using a continuous time random walk approach that is well suited to ensembles of sites [55, 56]. Highly aligned π orbitals dramatically suppress the spin relaxation rate, which suggests layers of aligned organic molecules should have longer spin relaxation times than layers of disordered organic molecules. Steric effects are strongest in quasi-1D systems, which should influence spin transport along the backbone of a polymer with radical side-chains, such as 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) [57, 58]. We also find that spin relaxation calculations for completely disordered organic molecules depend

on subtle features that can produce surprisingly large changes in the relaxation rate compared to previous calculations [52, 53]. These subtleties result from the correlation between the rate for a spin-flip hopping event and a spin-preserving hopping event for any specific instance of an initial and final site. The dependence of the relaxation on the steric environment can lead to organic materials with longer spin relaxation times, and the spin relaxation rate (and its temperature dependence) could provide insight into the orbital orientation landscape of a given organic semiconductor. Opportunities are also likely via nanometer-scale patterning using phase-separated diblock copolymers[59–61] to provide tailored orbital orientation landscapes.

To illustrate the dependence of the hopping rates on molecular orientation and the correlation between the two rates themselves, consider the case for an initial site fixed parallel to the spin quantization ($\hat{z}$) axis and a final site’s p''_z axis in the xz -plane. Fig. 2.5 shows that the hopping rates are complementary in nature,

$$\begin{aligned}\nu_c &\propto \cos^2 \theta_{\Omega'\Omega''} \\ \nu_{sf} &\propto \sin^2 \theta_{\Omega'\Omega''},\end{aligned}\tag{2.47}$$

where the polar angle is the directional cosine between p'_z (direction Ω') and p''_z (direction Ω''). When p'_z and p''_z are parallel or anti-parallel to the spin quantization axis the spin-flipping rate vanishes, eliminating transitions from the $|+'(-'')\rangle$ state to the $|-''(+'')\rangle$ state. Conversely when $p'_z(p''_z)$ is parallel or anti-parallel to the quantization axis, and $p''_z(p'_z)$ is perpendicular to the quantization axis the spin-conserving rate vanishes, eliminates transitions from the $|+'(-'')\rangle$ state to the $|+''(-'')\rangle$ state. This results in a process we call “carrier funneling”, wherein carriers spend little time in regions of the material with nearly parallel alignments, in favor of regions within which the sites’ spin quantization axes are near perpendicular, promoting faster spin flipping behavior.

Assuming the orientations of molecular sites are chosen uniformly over the unit sphere, carrier funneling generates faster relaxation times compared to the relaxation times seen when simply averaging the hopping rates over all possible site orientations. The increase in relaxation rate is due to correlations between the spin-conserving and spin-flipping rates, so

we refer to this as the *correlated approximation*. If each rate is averaged over all orientations independent of the specific orientations of the two sites, then we recover the *uniform approximation* [52, 53]. A comparison of these two approximations and the relevance of the selection rules is depicted in Fig. 2.5. Orientations represented by polar angles 0 and π (parallel and anti-parallel sites respectively) yield only spin conserving transitions, whereas at polar angle $\pi/2$ only spin-flipping transitions are allowed. In the case of uni-directional transport (a system with one nearest neighbor), and π orbitals completely free to move, the spin relaxation time from the correlated approximation is $\sim 2/3$ that predicted by the uniform rate approximation.

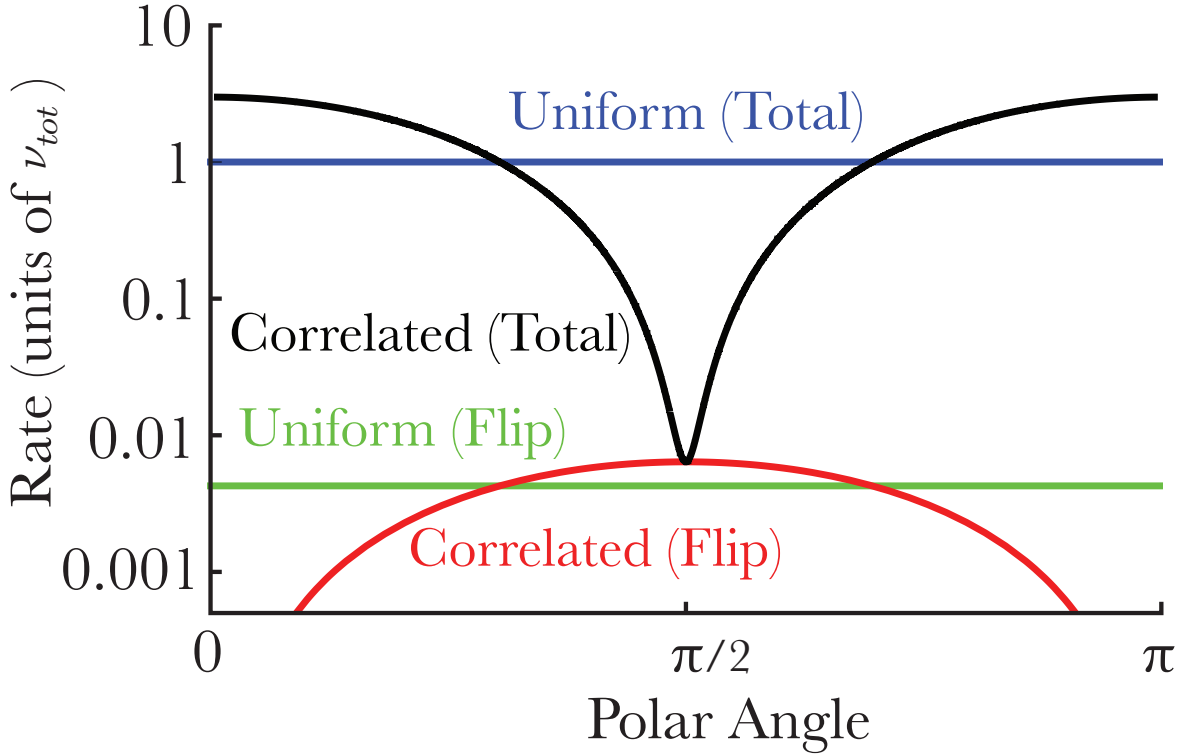


Figure 2.5: Uniform total rate (blue) and flipping rate (green) compared to correlated total rate (black) and flipping rate (red) as a function of polar angle with initial site parallel to the global spin quantization axis. The increased total rate between sites that are parallel or anti-parallel acts to funnel carriers towards sites with orientations that promote spin-flip events.

The tendency for the orbitals at neighboring sites to stray from parallel alignment can either be due to a frozen disorder or from a thermal distribution in a potential well that

favors parallel orientation. We assume the potential energy varies as the cosine of the angle $\theta_{\Omega'\Omega''}$ between p'_z and p''_z . The probability of two sites having a relative polar angle of θ is described by

$$P(\theta) \propto e^{s_H \cos \theta}, \quad (2.48)$$

where s_H is a steric hindrance factor[41, 42]. This distribution is analogous to the normal distribution on the two dimensional sphere, a special case of the Kent distribution [62]. Here we focus on a thermal distribution in a potential well, as it provides an effective temperature parameter to tune steric effects, but the results obtained are equally valid for glassy systems with frozen disorder. In the latter case the static disorder often can be characterized by the glass transition temperature $T = T_g$, a quantity defining the threshold temperature below which the positional relaxation time of sites in the material is, for practical purposes, infinite. For this model

$$s_H = U_0/k_B T \quad (2.49)$$

with U_0 the difference in potential energy between the parallel and perpendicular configurations, k_B the Boltzmann constant, and T the relevant temperature of the system ($= T_g$ for frozen glassy disorder). For example, $U_0 \approx 65$ meV for deviations from the optimal bond angle for polyacetylene[41]. At room temperature this corresponds to a steric constant of $s_H = 2.5$ and an average relative angle between nearest neighbors of 48° .

To calculate the spin-orbit induced relaxation under these conditions we implement a continuous time random walk simulation aimed at expressing the spin polarization as a function of time. We assume a negligible energy disorder and three simple geometries: linear, planar, and cubic. The linear geometry corresponds to unidirectional transport, implying a single nearest neighbor. This can be viewed as a primitive model for transport along a chain of molecules with an external electric field parallel to its length. Whether the hop is spin-conserving or spin-flipping depends on the spatial orientation of the initial and final sites [Eq. (2.44)]. The first site orientation is selected randomly and the nearest neighbor orientations are subject to probability distributions of the form in Eq. (2.48) centered about the p'_z -axis. No memory of the previous site's spatial orientation is retained from one hop to another,

meaning the sequence of hops from sites $A \rightarrow B \rightarrow C$ is statistically equivalent to hops from $A \rightarrow B \rightarrow A$. Approximations of this type are referred to as “multiple trapping” theories [63–67] which often apply to systems with a preferential transport direction, such as occurs with large electric fields. Since the carriers themselves do not interact with each other, we allow a solitary spin to hop from site to site, with the number of nearest neighbors reflecting the dimensionality, while recording the spin projection along the global z -axis at specified time intervals. This process is then averaged over many temporal instances, the number of which (usually more than 10^5) depends on statistical convergence of the spin relaxation rate. The length of time spent at a given site is determined by the wait-time distribution, which is assumed to be exponential with a characteristic rate equal to the sum of the rates calculated for each nearest neighbor. Hyperfine effects are neglected, so relaxation only occurs during the hopping process. The number of hops each carrier makes is determined by statistical convergence towards exponential spin polarization decay and the length of time sampled is on the order of the ratio of hops to average hopping rate.

The rate at which the spin polarization decays depends on the steric constant s_H . As s_H increases, the orbitals experience larger correlation and, as suggested by Fig. 2.5, reduced spin-flip probabilities in this highly constrained system. We thus predict that by controlling the variation in site orientation the spin-orbit induced relaxation can be tuned. This can be seen from Fig. 2.6, a plot of the spin polarization as a function of time in a unidirectional linear geometry for different values of s_H . For a free system ($s_H = 0$) the average relative angle between nearest neighbors is 90° , and (using spin-orbit parameters consistent with Alq3 [52]) we calculate the spin lifetime to be 38 ns, which is in rough agreement with experimental measurements[68]. *We predict that by restricting the relative angle to an average difference of 27.5° ($s_H = 7$) the spin lifetime will double to 77 ns and an average difference of 14.4° ($s_H = 25$) will increase the spin-lifetime by an order of magnitude to 400 ns.* If the orbitals of the molecular sites are all parallel, then there is no spin relaxation due to the spin-orbit mechanism. The temperature dependence shown in Fig. 2.6 and Fig. 2.7 implies that a high temperature (including a higher effective temperature T_g associated with

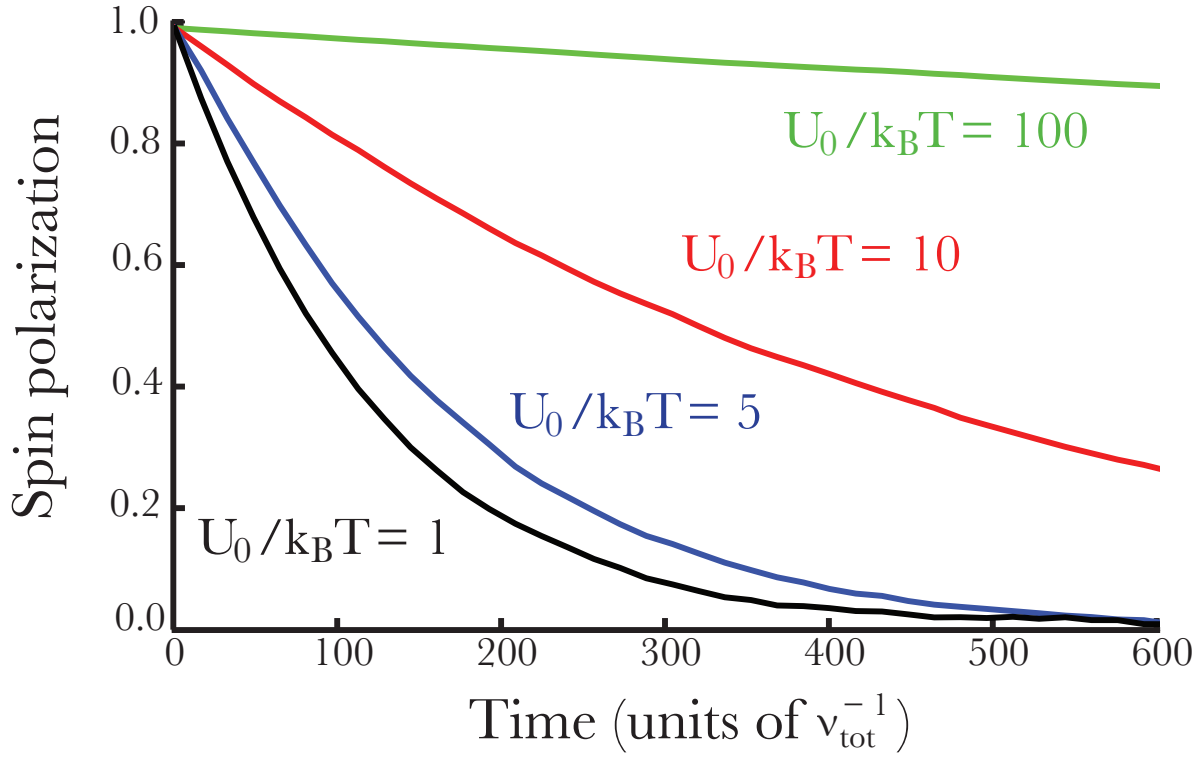


Figure 2.6: Spin polarization vs time for different steric hindrance constants in a unidirectional linear geometry. Large steric hindrance $s_H = U_0/k_B T$ implies little variation in molecular orientation from site to site, drastically reducing the spin-orbit induced relaxation.

the “glassy phase” with frozen disorder) corresponds to a small steric constant, resulting in shorter spin-relaxation times. For these effects to influence the actual temperature dependence of spin relaxation the organic material must not be in the glassy phase, in which the disorder is frozen below a glass transition temperature T_g . Thus these effects likely do not influence the temperature dependence in devices with such organic layers[69, 70]. The overall spin relaxation could be affected by the disorder characterized by the effective temperature T_g , however a comparison with experiment would require structural characterization of the disorder in the specific organic layers.

A similar effect is present with additional nearest neighbors, suggesting that the spin-orbit relaxation is tunable for systems of varying dimensions. Adding nearest neighbors (increasing the coordination number) in general decreases the spin relaxation rate, and when $s_H \sim 0$ as well, then the rate approaches the value for the uniform rate approximation. This is to be expected since the uniform rate approximation uses average values of the rates from a uniform distribution about the unit sphere, which is recovered for $s_H = 0$.

We have shown that the orientation of sites within π -conjugated organics can have appreciable effects on the spin relaxation time of carriers. We predict that by aligning the π orbitals parallel that the spin-orbit relaxation would be minimized. Conversely, orbital orientation landscapes with 90° rotations of neighboring π orbitals, so that the orientations are staggered with each nearest neighbor orbital perpendicular, would maximize the spin-orbit relaxation. Correlations between the spin-conserving and spin-flipping rate establish hop selection rules whereby certain hop types are forbidden for certain orbital orientations. This correlation between hopping rate and site orientation allows tuning of the spin-orbit induced relaxation within a system, allowing for spin lifetime optimization for spintronic applications.

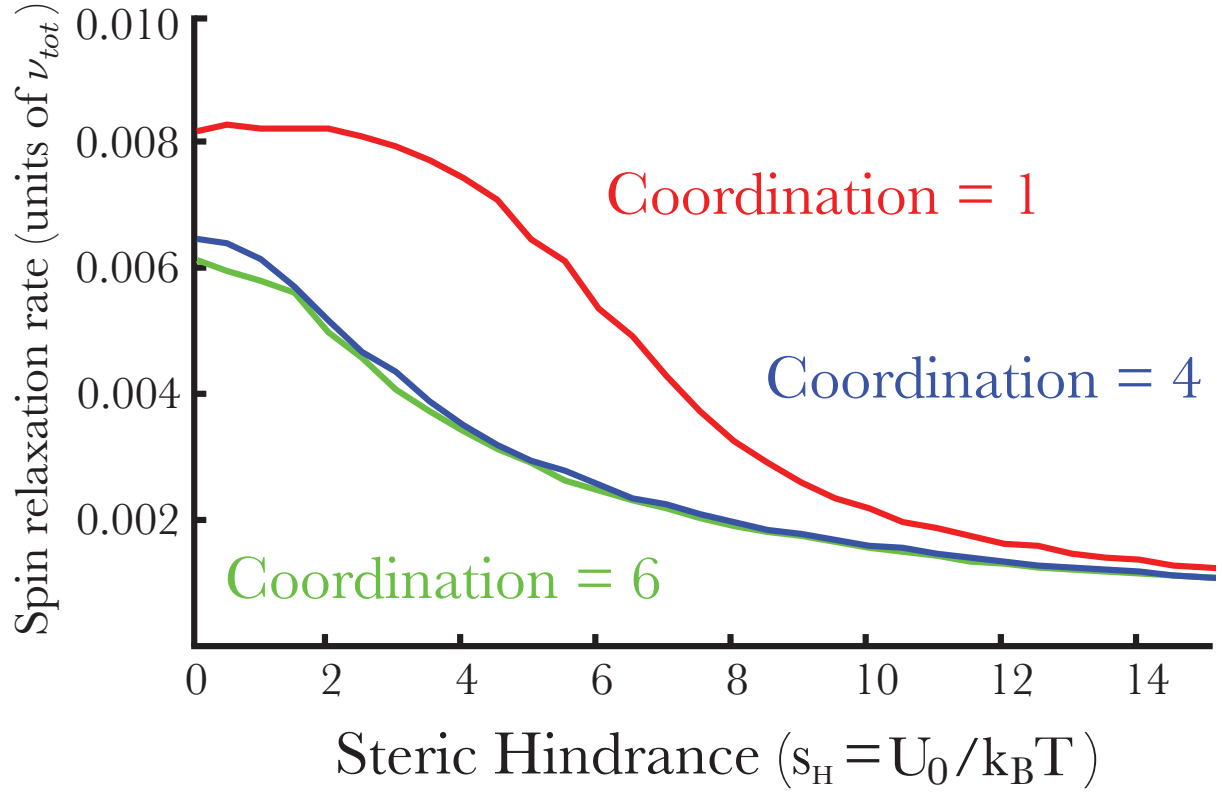


Figure 2.7: Spin relaxation rate vs steric constant for one (red), four (blue) and six (green) nearest neighbors. The relaxation rates differ as the dimensionality of the system changes. The reduction in spin-orbit relaxation rate as the variation in orientation is decreased appears less sensitive to the number of neighbors.

Chapter 3

Probing Spin Dynamics of Isolated spin centers using coherence and correlations

Localized spin states with long spin coherence times are a fundamental element of the quantum coherent spin systems[71, 72] that underlie quantum sensing[4] and quantum information processing[73]. Dopants, and some other controlled defects, in semiconductor or insulator hosts provide a robust realization of these localized coherent spins[74, 75]. Optical probes have revealed the spin dynamics of such spin centers, along with their coherent interactions with neighboring spins[76–81]. Similar evidence of all-electrical probing of coherent interactions has been achieved at low temperatures in dopants in large magnetic fields[82, 83], and with electron spin resonance combined with spin-polarized scanning tunneling microscopy (STM-ESR)[84–87]. Low-field magnetoresistance without rf fields in much higher temperature dc electrical measurements, however, has been found in transport from a magnetic contact[88, 89] through traps to a nonmagnetic region, as well as in transport through traps in semiconductor devices[90]. These effects can be understood as originating from coherent spin dynamics in resonant[91] or non-resonant (incoherent hopping) transport[89], and are largely limited to zero-field peaks or dips in the electrical resistance unless the spin polarization of a contact is time dependent[92].

In this chapter we begin with a general analysis of resonant signatures between two exchange coupled spin-1/2 centers embedded near the surface of a non-conducting host. We then derive the bottleneck state ϕ_b when T_+ and S are degenerate in Sec. 3.2 and present an approximate analytic model (Sec. 3.3) based on this emerging state that allows greater understanding of the evolution of the system from the two-spin subspace to the single-spin

subspace and how the polarizations emerge on resonance. In the appendix we provide the (very lengthy) general analytic expression for the current and show its limiting expressions near the zero-field and finite-field resonances, in Appendix E.1 in addition to the lengthy general analytic expressions for the spectator spin polarizations and two-spin product state populations in Appendix E.2.

3.1 Exchange coupled spin-1/2 dopants

Here we predict that the *dc* magnetoresistance of electrical current through the spin state of a single dopant, which can be measured e.g. using spin-polarized STM (SP-STM) [93–95], should directly image the coherent effects of the environment on the dopant’s spin dynamics, *e.g.* exchange or hyperfine interactions, even at temperatures far larger than the energy scales of those interactions. These sharp environmental features in the *dc* magnetoresistance will occur in the absence of spin-orbit interaction in the system, and will persist even when the transport process is charge incoherent (spin orientation is preserved when hopping to and from the dopant’s spin state). We consider the single dopant to reside in the surface region of a weakly conducting nonmagnetic host, and under the influence of a second nuclear or electron spin, which affects the first spin through isotropic exchange or isotropic hyperfine interactions. The charge and spin magnitude of the second spin is fixed, as would be appropriate for the electron spin of a deep trap or for a nuclear spin. Strong *dc* magnetoresistance features are seen when the dopant’s spin precession due to the applied magnetic field cancels the precession due to the exchange or hyperfine interaction with the second spin. These features can be resolved at very small magnetic fields and thus should permit the measurement of milliTesla hyperfine interactions and μeV exchange interaction strengths, such as are predicted for spin centers in wide-gap semiconductors such as diamond[96]. Double quantum dot measurements of spin bottlenecks[97] would not see these features, as the spin polarization *transverse* to the applied magnetic field must be measured. These coherent features in the magnetoresistance will remain detectable, due to the persistent coherence of the spin, even when the thermal energy exceeds the energy scales of spin interactions with

the environment by orders of magnitude, so long as the dopant’s spin coherence time exceeds the nanosecond transport times of the junction. This differs from SP-STM measurements of spin excitations during inelastic tunneling spectroscopy[98, 99], which cannot be resolved when the temperature exceeds the energy of the spin excitation. Similar features appear to occur in the photoluminescence from organic molecules when the applied field brings a triplet exciton state into resonance with the singlet exciton state[100], although these occur at far larger magnetic fields ($\gg$ Tesla). More generally, these coherent features may emerge in any flow of current through localized states in nanoscale devices containing contacts with spin polarization oblique to an external field.

The detailed configuration is shown in Fig. 3.1. Here we focus on the concrete situation in which the dopant through which transport occurs (transport site, in green) possesses two states, one of which is spin-0 and “empty”, and the other is spin-1/2 and “full”, corresponding to an additional electron (spectator site, in red); similar features are expected for a wide variety of transport and spectator spin states. The charge state of the spectator spin is unaffected during transport. The details of the magnetoresistive curves, and the visibility of coherent features within them, depend on the relative strength of the exchange interaction J and the hopping rates, here γ_{on} from the nonmagnetic host to the transport site, and γ_{off} from the transport site to the SP-STM tip. The use of the STM provides atomic-scale resolution and permits direct selection of the dopant spin to be probed, and it also provides direct means to independently tune γ_{on} and γ_{off} . By moving the tip towards or away from the surface the γ_{off} can be adjusted[101–103], and through tip-induced band bending both γ_{off} and γ_{on} can be modified[104, 105]. We find the most sensitive configurations correspond to $\gamma_{off} \ll \gamma_{on}$, which is the expected experimental situation under ordinary conditions. We consider the SP-STM to be polarized along the positive z -axis, which can point in any direction relative to the surface normal in Fig. 3.1.

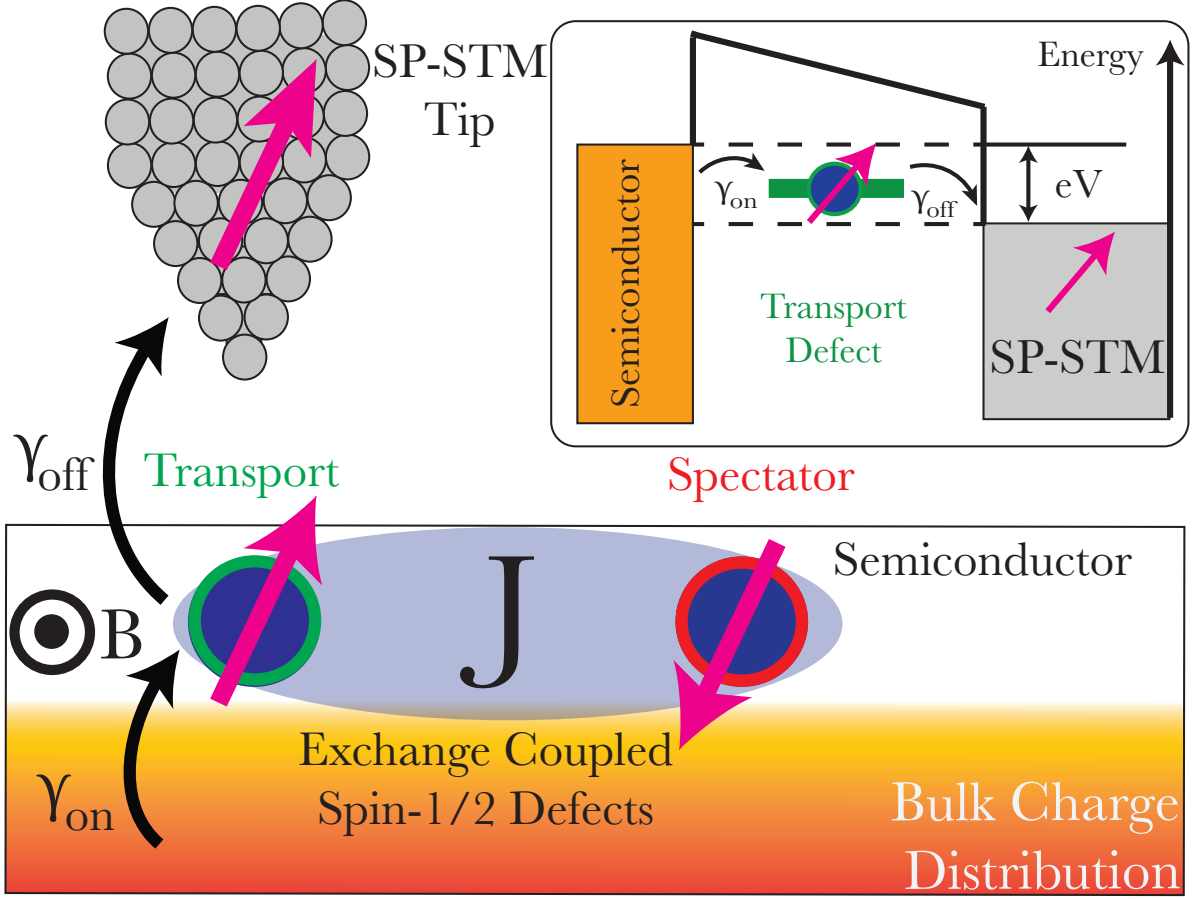


Figure 3.1: Schematic current path for an electron through a single dopant (transport site, in green) that is exchange-coupled to another spin-1/2 (spectator site, in red). The transport site has two charge states, empty (spin-0) and full (spin-1/2), and the spectator site's charge state is stable. These spins reside near the surface of the nonmagnetic host and the Coulomb repulsion at the transport site is assumed to be sufficiently large to prevent double occupation (see inset). Hopping from the transport site to a spin polarized scanning tunneling microscope (SP-STM) tip occurs with rate γ_{off} , which is controllable by moving the STM tip relative to the surface. Replenishment of the occupation of the transport site from the nonmagnetic host occurs with a rate γ_{on} . If the host is a semiconductor the charge distribution can be adjusted by the STM voltage V through tip induced band bending, which adjusts γ_{on} and γ_{off} .

Our main results are the predicted finite-field dips (resonances) in the dc magnetoresistance in Fig. 3.2 that occur due to degeneracies in the transport states and the formation of bottlenecks. Coupling dopants to spin-polarized contacts has been shown to lead to zero-field current bottlenecks resulting from dopant polarization anti-parallel to the magnetization of the contact[89]. The application of an external magnetic field, $\mathbf{B}$, transverse to the contact

spin polarization, releases the bottleneck by precessing the spin and produces a characteristic dip feature in the zero-field dc magnetoresistance[88, 89]. Figure 3.2, however, shows two additional features at finite field when the Zeeman energy of the transport spin equals its exchange energy with the spectator spin. New bottlenecks emerge under these conditions, because the triplet state polarized parallel or antiparallel to the external field (depending on the sign of J) becomes degenerate with the singlet state (S) of these two spins at this external field. We find the dip in current when the Zeeman energy equals J has a maximal value of $1/3$ of the $B = 0$ dip. Although our description here is for the case of a SP-STM tip that is 100% spin polarized, these same features will occur for any finite spin polarization, with a reduced amplitude dependent on the tip's spin polarization (as seen for zero-field dips in previous work[89]).

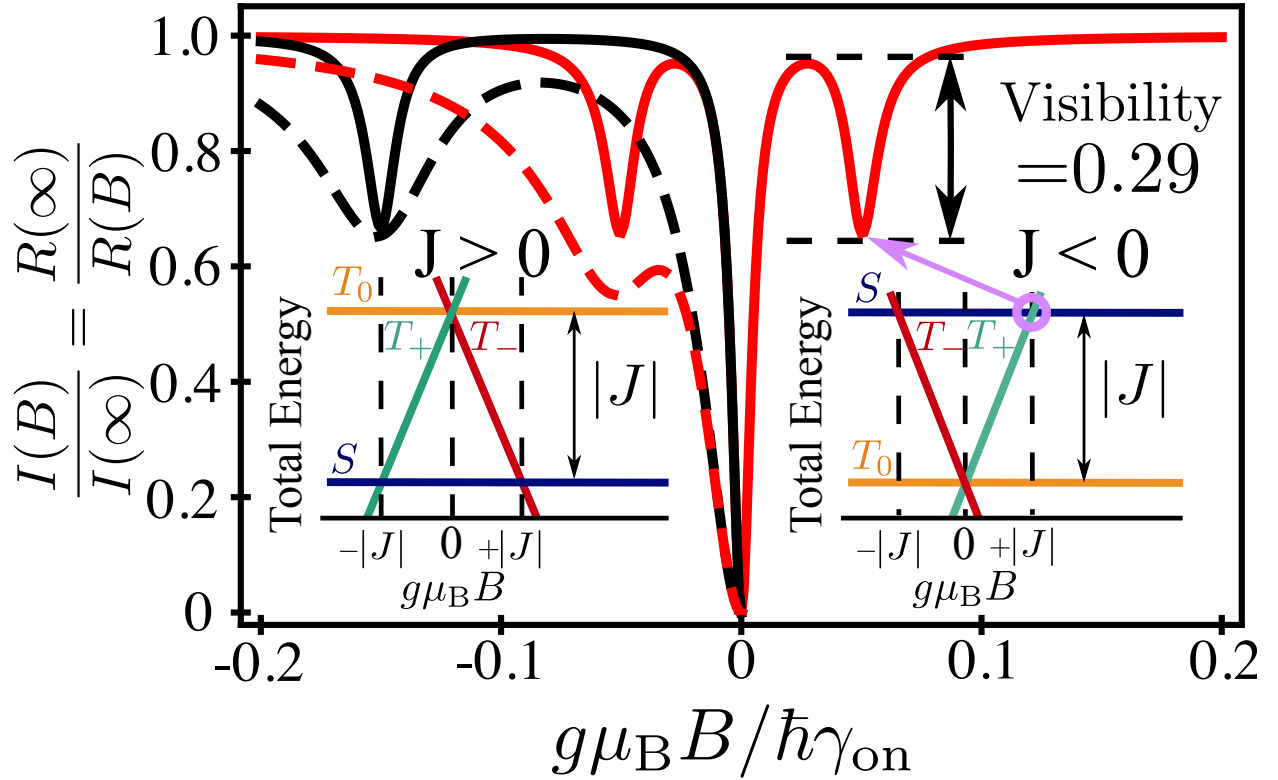


Figure 3.2: Magnetoresistance of current through spin-1/2 transport dopant for exchange splitting of $0.05 \hbar\gamma_{\text{on}}$ (red) and $0.15 \hbar\gamma_{\text{on}}$ (black). Features broaden for moderate extraction rates (dashed) compared to slow extraction (solid).

The density matrix requires time evolution terms describing coherent evolution of the

spins, originating from the magnetic field and exchange, and incoherent processes that inject or extract carriers from the transport site (thus producing current). We use the stochastic Liouville formalism[106, 107], which has successfully predicted the manipulation of electroluminescence from organic LEDs[108], exciplex light emission[109], spin-pumping[110], and nuclear spin influences on electron transport[111]. The Hamiltonian describing coherent evolution of the spins at both the spectator and transport sites in an external magnetic field $\mathbf{B}$ and an exchange interaction J between them is

$$\hat{H}_{st} = g\mu_B \mathbf{B} \cdot (\mathbf{S}_s + \mathbf{S}_t) + J\mathbf{S}_s \cdot \mathbf{S}_t, \quad (3.1)$$

where $\mathbf{S}_{s(t)}$ represents the spin operator at the spectator (transport) site, g is the Landé g -factor (assumed 2), and μ_B is the Bohr magneton. When the transport site is unoccupied the Hamiltonian becomes

$$\hat{H}_s = g\mu_B \mathbf{B} \cdot \mathbf{S}_s. \quad (3.2)$$

The coherent evolution of the density operator is

$$\left(\frac{\partial \rho_i(t)}{\partial t} \right)_{\text{coh}} = -\frac{i}{\hbar} [\hat{H}_i, \rho_i(t)], \quad (3.3)$$

where the subscript i is used to denote either the operator for the isolated spectator ($i = s$) or the combined spectator and transport spin ($i = st$) and $[\cdot, \cdot]$ represents the commutator.

Transport occurs when the dopant at the transport site couples to the unpolarized bulk states and the SP-STM resulting in the generation and extraction of carriers. Processes of this type will not preserve the trace of the individual sub-spaces of the system, but will preserve the total trace, $\text{Tr}\rho_s(t) + \text{Tr}\rho_{st}(t) = 1$, and they will decohere off-diagonal elements of ρ_{st} . The polarization of the SP-STM tip can be written as an operator $\hat{M}_{\text{off}}(\mathbf{n})$, where $\mathbf{n}$ is the orientation of the SP-STM polarization. In general, for a single spin-1/2 manifold [112] the interaction operator can be written as a linear superposition of oppositely polarized projection operators $\hat{\pi}(\mathbf{n})$ and $\hat{\pi}(-\mathbf{n})$,

$$\begin{aligned} \hat{m}(\mathbf{n}) &= a\hat{\pi}(\mathbf{n}) + b\hat{\pi}(-\mathbf{n}) \\ &= \frac{1}{2}\{(a+b)\hat{\sigma}_0 + (a-b)\mathbf{n} \cdot \hat{\sigma}\} \end{aligned} \quad (3.4)$$

where a and b are interaction amplitudes, and $\mathbf{n}$ is the orientation of the polarization. In the case of a manifold describing two spin-1/2 dopants the operator results from the direct product of two single spin operators,

$$\hat{M}(\mathbf{n}) = \hat{m}_t(\mathbf{n}) \otimes \hat{m}_s(\mathbf{n}) \quad (3.5)$$

where $\hat{m}_t(\mathbf{n}) = a_t \hat{\pi}(\mathbf{n}) + b_t \hat{\pi}(-\mathbf{n})$ and $\hat{m}_s(\mathbf{n}) = a_s \hat{\pi}(\mathbf{n}) + b_s \hat{\pi}(-\mathbf{n})$. The coefficients for $\hat{M}_{\text{off}}$ are $a_t^{\text{off}} = 1$, $b_t^{\text{off}} = 0$, $a_s^{\text{off}} = b_s^{\text{off}} = 1/2$. $\hat{M}_{\text{on}}$ couples the bulk to the dopant, and for unpolarized bulk states has the coefficients $a_t^{\text{on}} = a_s^{\text{on}} = b_t^{\text{on}} = b_s^{\text{on}} = 1/2$. We assume a large on-site Coulomb repulsion that excludes double occupation of the dopant sites. The spin state of the carrier entering the transport site from the bulk is determined by the spin polarization of the bulk ($\hat{M}_{\text{on}}$) and the spin state of the spectator is determined by the density matrix for the 2×2 manifold, $\rho_s(t)$. These considerations allow one to write down a current operator for the reoccupation of the transport site,

$$\hat{i}_{\text{on}} = \frac{e\gamma_{\text{on}}}{2} \hat{m}_t^{\text{on}}(\mathbf{n}) \otimes \rho_s(t), \quad (3.6)$$

where e is the electronic charge.

The current from the dopant to the SP-STM depends on the availability of spin states in the tip ($\hat{M}_{\text{off}}$) and the two-dopant density matrix ρ_{st} . The current operator for dopant-to-tip transport is

$$\hat{i}_{\text{off}} = \frac{e\gamma_{\text{off}}}{2} \{\hat{M}_{\text{off}}, \rho_{st}(t)\}, \quad (3.7)$$

where the anticommutator $\{, \}$ guarantees the operator is Hermitian and properly decoheres off-diagonal elements of ρ_{st} [113]. The current onto and off of the transport site is found from the trace of the current operators. The total current onto and off of the transport site must be equal: $\text{Tr} \hat{i}_{\text{on}} = \text{Tr} \hat{i}_{\text{off}} = i$, however the spin current $i_{\text{on(off)}}^{\text{spin}} = \text{Tr}[\hat{i}_{\text{on(off)}} \sigma]$ is not conserved due to decoherence.

The appropriate term can now be added to the stochastic Liouville equation for carrier generation

$$\left(\frac{\partial \rho_{st}(t)}{\partial t} \right)_{\text{gen}} = \gamma_{\text{on}} \hat{m}_t^{\text{on}}(\mathbf{n}) \otimes \rho_s(t) \quad (3.8)$$

and extraction

$$\left(\frac{\partial \rho_{st}(t)}{\partial t}\right)_{\text{ext}} = -\gamma_{\text{off}}\{\hat{M}_{\text{off}}, \rho_{st}(t)\}. \quad (3.9)$$

The total equation for the 4×4 density operator ρ_{st} is then

$$\frac{\partial \rho_{st}(t)}{\partial t} = -\frac{i}{\hbar}[\hat{H}_{st}, \rho_{st}(t)] + \gamma_{\text{on}}\mathbf{I}_t \otimes \rho_s(t) - 2\gamma_{\text{off}}\{\hat{M}_{\text{off}}, \rho_{st}(t)\}, \quad (3.10)$$

where $\mathbf{I}_t$ is the identity matrix in the transport subspace and the relation $2\hat{n}_t^{\text{on}}(\mathbf{n}) = \mathbf{I}_t$ has been used. Eq. (3.10) is coupled to the single spin density operator ρ_s through the generation term. The trace of the total system is preserved and thus $\text{Tr}\rho_s + \text{Tr}\rho_{st} = 1$; generation(decay) of the 4×4 density matrix must be balanced by the decay(generation) of the 2×2 density matrix allowing one to derive for the isolated spectator,

$$\frac{\partial \rho_s(t)}{\partial t} = -\frac{i}{\hbar}[\hat{H}_s, \rho_s(t)] + 2\gamma_{\text{off}}\text{Tr}_t\{\hat{M}_{\text{off}}, \rho_{st}(t)\} - 2\gamma_{\text{on}}\rho_s(t), \quad (3.11)$$

where Tr_t is a partial trace over the transport subspace.

These expressions can be solved analytically, but are very cumbersome. A full analytic solution is presented in the Supplemental Material, along with a more intuitive derivation under some approximations. The expressions for the near zero field feature agree with calculations of current through mediating dopants in insulating spacers, during either resonant[91] or incoherent[89] transport. However we are interested in the expressions for conditions where the Zeeman energy ($\tilde{B} = g\mu_B|\mathbf{B}|$) approaches the exchange energy J . An approximate expression valid for $\gamma_{\text{on}} \gg \tilde{B}, J \gg \gamma_{\text{off}}$ near the $\tilde{B} = J$ resonance is

$$i = \left(\frac{\gamma_{\text{off}}}{2}\right) \frac{6J^2\gamma_{\text{off}}^2 + 4(\tilde{B}^2 - J^2)^2}{9J^2\gamma_{\text{off}}^2 + 4(\tilde{B}^2 - J^2)^2} \quad (3.12)$$

where the prefactor is the limiting value away from the resonant features. Thus the *relative* current has a limiting value for $\tilde{B} = J$ of $2/3$, corresponding to a visibility of $1/3$ as seen below in numerical calculations. Figure 3.2 shows numerical values for the magnetoresistance, for a 100% spin-polarized STM tip.

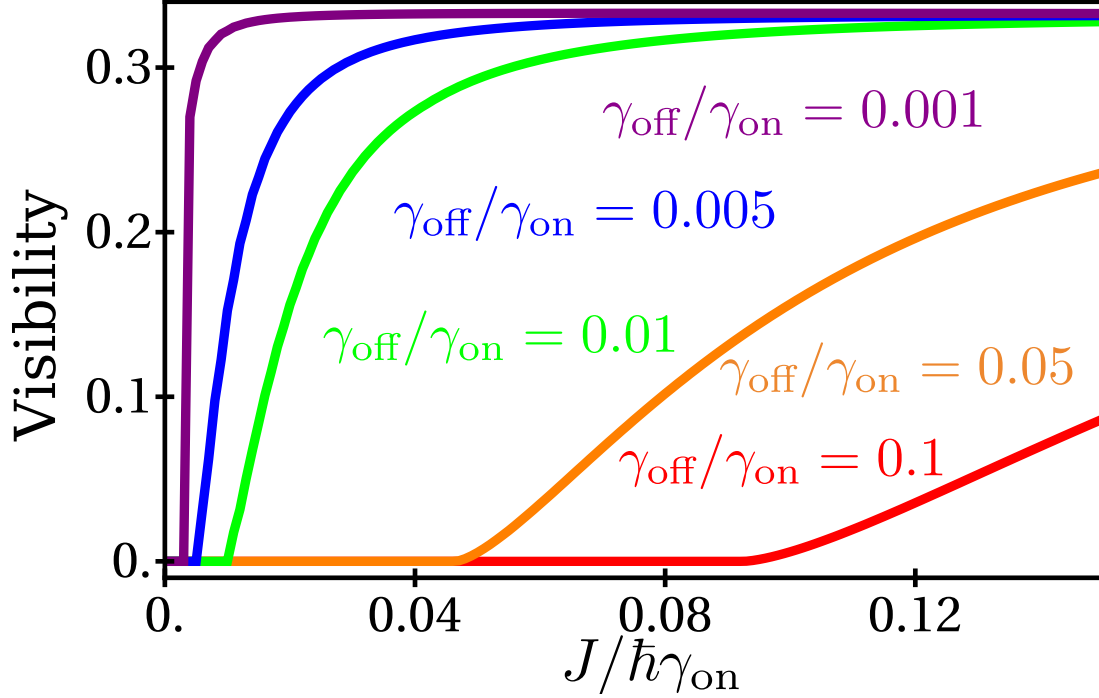


Figure 3.3: Visibility of the finite-field feature (resonance) as a function of the exchange parameter plotted for different values of the hopping ratio. Smaller hopping ratios yield higher visibility due to extended spin evolution in the exchange field.

The visibility for different hopping ratios is plotted as a function of the exchange coupling parameter in Fig. 3.3. At a finite-field dip in current one of the triplet states (T_+ or T_- along the direction of $\mathbf{B}$, depending on the sign of J) becomes degenerate with the singlet state S . The other two triplet states are singly degenerate in the combined Zeeman-exchange Hamiltonian and thus do not participate in a bottleneck configuration. The bottleneck in current manifests as a reduced rate, for one of the two degenerate states, for the transport site's electron to hop to the SP-STM tip. The linear combination of those two states that has the lowest transport-site tunneling probability to the SP-STM tip is $|\phi_b\rangle$, and the orthogonal state is $|\overline{\phi_b}\rangle$, where

$$\begin{aligned} |\phi_b\rangle &= 6^{-1/2}|\uparrow\uparrow\rangle + 6^{-1/2}|\downarrow\downarrow\rangle + (2/3)^{1/2}|\downarrow\uparrow\rangle, \\ |\overline{\phi_b}\rangle &= (12)^{-1/2}(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle - |\downarrow\uparrow\rangle) + (3/4)^{1/2}|\uparrow\downarrow\rangle. \end{aligned} \quad (3.13)$$

The first spin (single arrow) represents the transport site and the second (double arrow) represents the spectator site, and the axes are along the SP-STM spin polarization. ϕ_b has

a transport spin polarization of $2/3$, which produces a maximum dip in the current of $1/3$ as derived in the Supplemental Material, as plotted in Figs. 3.2 and 3.3, and as apparent in Eq. (3.12) for $\tilde{B} = J$.

Figure 3.3 also illustrates the increase in resolution and precision gained by minimizing the hopping ratio. For a dopant-to-tip hopping rate of $\gamma_{\text{off}} = 1 \text{ ns}^{-1}$ we predict exchange energy resolution $\sim \mu\text{eV}$. If the energy splittings to double occupancy of the transport site greatly exceed the thermal energy, and the spin coherence times of the spins exceed the timescales set by the smallest energies in the problem (Zeeman energy, exchange energy, or hopping) then the magnetoresistance features survive; observations of zero-field dips are seen at room temperature[90]. The energy sensitivity surpasses the practical limits of STM spectroscopic probes of exchange interactions in semiconductor dopant pairs[26, 114], and could be measured at temperatures far above where inelastic spin excitations can be imaged for adatoms[98, 99].

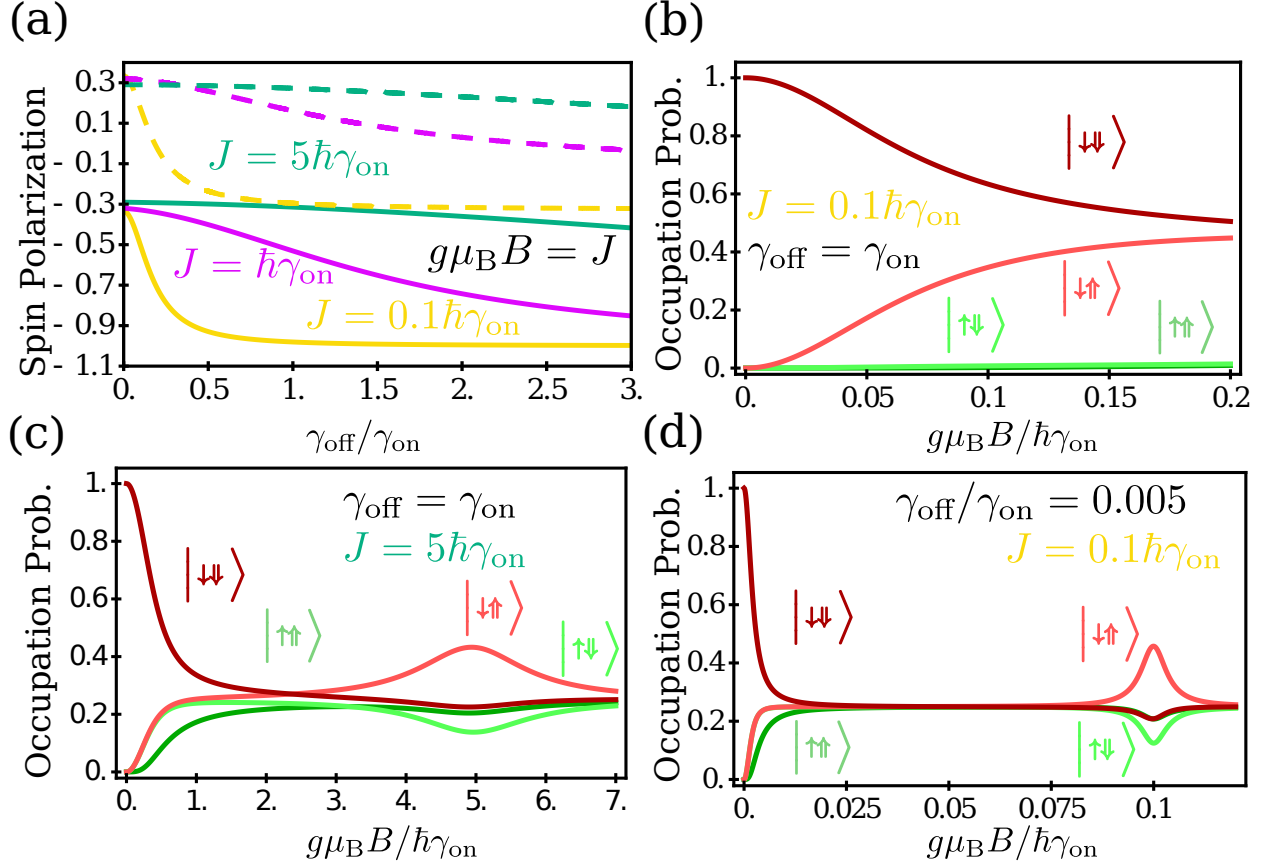


Figure 3.4: (a) Spin polarization as a function of hopping ratio plotted for different values of the exchange coupling. The dashed lines represent the spectator spin and solid lines represent the transport spin. (b-d) Occupation probability for the different states in the product basis as a function of the transverse magnetic field. Shown are (b) $\gamma_{\text{off}} = \gamma_{\text{on}}$, $\gamma_{\text{off}} \gg J$, (c) $\gamma_{\text{off}} = \gamma_{\text{on}}$, $\gamma_{\text{off}} \ll J$, and (d) $\gamma_{\text{on}} \gg J \gg \gamma_{\text{off}}$. Good visibility is realized when $\gamma_{\text{off}} \ll J$, and especially for (d).

Figure 3.4(a) shows calculations for the degree of spin polarization as a function of the hopping ratio for different values of the junction parameters. The dashed lines represent the spectator spin and the solid lines represent the transport spin. In Fig 3.4(b-d) shows the occupation probability for each state in the product basis as a function of B . An analytic treatment for $\tilde{B} = J$ and the limit $\gamma_{\text{off}} \ll J \ll \gamma_{\text{on}}$ [similar to panel (d)], described in the Supplemental Material, predicts a polarization for $|\downarrow\uparrow\rangle$ of $11/24$, for $|\uparrow\downarrow\rangle$ of $1/8$, and for $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ of $5/24$, which are all different from the random average of $1/4$. These results are very similar to those shown in Fig. 3.4(d). The spectator spin is also dynamically

polarized during this process, even when the transport spin state is empty; we find for the limiting conditions above, similar to Fig. 3.4,(d) a polarization of 1/2 antiparallel to the magnetic field direction and 1/4 parallel to the SP-STM polarization direction for a total polarization of $\sqrt{5}/4$.

3.2 Determination of the Bottleneck State of the Two-Spin System

To find the bottleneck state we consider the condition when one of the three triplet states is degenerate with the singlet state, and find the state that has the smallest overlap of the transport spin with the STM tip spin polarization. Because the magnetic field is oriented transverse to the STM tip polarization we must work both in the basis where the spin quantization axis is parallel to the STM tip (*i.e.* $\uparrow_{\text{STM}}$ and $\downarrow_{\text{STM}}$) and the basis where the axis is parallel to the magnetic field (*i.e.* $\uparrow_B$ and $\downarrow_B$).

The four eigenstates of $H_{\text{st}} = g\mu_B \mathbf{B} \cdot (\mathbf{S}_t + \mathbf{S}_s) + J\mathbf{S}_t \cdot \mathbf{S}_s$ with quantization axis parallel to the magnetic field are

$$\begin{aligned} |T_+\rangle &= |\uparrow\uparrow\rangle_B \\ |T_-\rangle &= |\downarrow\downarrow\rangle_B \\ |T_0\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle_B + |\downarrow\uparrow\rangle_B) \\ |S\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle_B - |\downarrow\uparrow\rangle_B) \end{aligned} \tag{3.14}$$

with energies E_+ , E_- , E_0 , and E_S . For a positive exchange coupling, J , at $g\mu_B B = J$ the $|T_+\rangle$ and $|S\rangle$ states are degenerate, *i.e.* $E_+ = E_S$. To evaluate the bottleneck in the two-state degenerate manifold we consider the states in a product basis parallel to the SP-STM polarization:

$$\begin{aligned} |T_+\rangle &= \frac{1}{2}(|\uparrow\uparrow\rangle_{\text{STM}} + |\downarrow\downarrow\rangle_{\text{STM}} + |\uparrow\downarrow\rangle_{\text{STM}} + |\downarrow\uparrow\rangle_{\text{STM}}) \\ |T_-\rangle &= \frac{1}{2}(|\uparrow\uparrow\rangle_{\text{STM}} + |\downarrow\downarrow\rangle_{\text{STM}} - |\uparrow\downarrow\rangle_{\text{STM}} - |\downarrow\uparrow\rangle_{\text{STM}}) \\ |T_0\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle_{\text{STM}} - |\downarrow\downarrow\rangle_{\text{STM}}) \\ |S\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle_{\text{STM}} - |\downarrow\uparrow\rangle_{\text{STM}}). \end{aligned} \tag{3.15}$$

We cannot find a linear combination of $|T_+\rangle$ and $|S\rangle$ that has no component with transport spin polarized down; the best we can do is find that has the smallest projection of the transport spin along $|\uparrow\rangle_{\text{STM}}$ from this linear combination will have no $|\uparrow\downarrow\rangle_{\text{STM}}$ component. The bottleneck state is then

$$|\phi_b\rangle = \frac{1}{\sqrt{6}}(|\uparrow\uparrow\rangle_{\text{STM}} + |\downarrow\downarrow\rangle_{\text{STM}}) + \sqrt{\frac{2}{3}}|\downarrow\uparrow\rangle_{\text{STM}} \quad (3.16)$$

and the orthogonal state in the degenerate subspace is

$$|\bar{\phi}_b\rangle = \frac{1}{\sqrt{12}}(|\uparrow\uparrow\rangle_{\text{STM}} + |\downarrow\downarrow\rangle_{\text{STM}} - |\downarrow\uparrow\rangle) + \sqrt{\frac{3}{4}}|\uparrow\downarrow\rangle_{\text{STM}}. \quad (3.17)$$

3.3 Simplified Model for evolution of the two-spin and single-spin subspaces

Although the analytic solution provides the full results, it does not provide as much insight as one might wish to the evolution of the coupled two-spin and single-spin manifolds. A simplified model can be solved analytically at the finite-field resonance, although it does not include off-diagonal density matrix elements between the two degenerate states in the bottleneck. Thus the results do not precisely reproduce the analytic full solution, but they are close and provide further insight.

3.3.1 Rate of Current Flow for each of the Two-Spin States

For the general case of an STM tip that is not 100% spin polarized there will be a different rate for extracting a spin-up carrier from the transport site ($\Gamma_\uparrow$) and a spin-down carrier from the transport site ($\Gamma_\downarrow$). For the plots in the paper we have chosen $\Gamma_\downarrow = 0$. We now consider the transport rates for the four possible states, T_- , T_0 , ϕ_b and $\bar{\phi}_b$. As there is no state with that forms a complete bottleneck (total rate = 0) under ideal polarization conditions ($\Gamma_\downarrow = 0$) the visibility can not be 100%. Instead, the visibility can be determined by comparing the time-averaged rate for a steady-state solution to the occupations and coherences of these four states to the rate without bottleneck.

Both $|T_-\rangle_{\text{STM}}$ and $|T_0\rangle_{\text{STM}}$ are stationary, non-degenerate solutions for H_{st} and have equal probability density of spin-up and spin-down along the STM spin quantization axis.

Thus any coherences between one of these states and any other state would dephase very rapidly and can be ignored. For both of these states the total hopping rate to the SP-STM tip is $(\Gamma_{\uparrow} + \Gamma_{\downarrow})/2$. The two degenerate states require more careful treatment. For $|\phi_b\rangle$ the rate for hopping to the SP-STM tip is $\Gamma_{\uparrow}/6 + 5\Gamma_{\downarrow}/6$ and for $|\bar{\phi}_b\rangle$ the rate is $5\Gamma_{\uparrow}/6 + \Gamma_{\downarrow}/6$. We find calculating the steady-state including the coherence between ϕ_b and $\bar{\phi}_b$ is intractable, so we neglect this coherence and still find results very close to the numerical calculations that include this coherence.

The residence time in each of the states, *once* that state has been populated, is

$$\begin{aligned} T_{T_0} &= \frac{2}{\Gamma_{\uparrow} + \Gamma_{\downarrow}} \\ T_{T_-} &= \frac{2}{\Gamma_{\uparrow} + \Gamma_{\downarrow}} \\ T_b &= \frac{6}{\Gamma_{\uparrow} + 5\Gamma_{\downarrow}} \\ T_{\bar{b}} &= \frac{6}{5\Gamma_{\uparrow} + \Gamma_{\downarrow}}. \end{aligned} \tag{3.18}$$

For the rest of our derivation we will assume $\Gamma_{\downarrow} = 0$, and $\Gamma_{\uparrow} = \gamma_{\text{off}}$.

3.3.2 Reduction of current due to bottleneck at $g\mu_B B = J$ (maximum visibility)

We evaluate the steady-state occupation by considering transition rates from the two-spin manifold to the single-spin states and back.

Transitions from $|T_- \rangle$ State

Transitions from the $|T_- \rangle$ state in Eq (3.15) to the single-spin manifold leave the spectator in the spin-down eigenstate of the spin operator parallel to the magnetic field:

$$\begin{aligned} \langle \uparrow_{\text{STM}} | T_- \rangle_{\text{STM}} &= \frac{1}{2} (| \uparrow_{\text{STM}} \rangle - | \downarrow_{\text{STM}} \rangle) \\ &= | \downarrow_B \rangle / \sqrt{2} \end{aligned} \tag{3.19}$$

Thus 100% of the time the transition occurs to the single-spin state $| \downarrow_B \rangle$, and the rate of hopping is $\gamma_{\text{off}}/2$.

To determine the branching ratio for the transition back from the single-spin state to the two-spin manifold, we write the four two-spin states with the spectator spin in the basis parallel to $\mathbf{B}$:

$$\begin{aligned} |T_-\rangle &= \{ |\uparrow_{\text{STM}}\rangle - |\downarrow_{\text{STM}}\rangle \} |\downarrow_B\rangle / \sqrt{2} \\ &= \{ |\uparrow_{\text{STM}}\downarrow_B\rangle - |\downarrow_{\text{STM}}\downarrow_B\rangle \} / \sqrt{2} \end{aligned} \quad (3.20)$$

$$\begin{aligned} |T_0\rangle &= \frac{1}{\sqrt{2}} |\uparrow_{\text{STM}}\rangle \left(\frac{|\uparrow_B + \downarrow_B\rangle}{\sqrt{2}} \right) - \frac{1}{\sqrt{2}} |\downarrow_{\text{STM}}\rangle \left(\frac{|\uparrow_B - \downarrow_B\rangle}{\sqrt{2}} \right) \\ &= \frac{1}{2} \{ |\uparrow_{\text{STM}}\uparrow_B\rangle + |\uparrow_{\text{STM}}\downarrow_B\rangle - |\downarrow_{\text{STM}}\uparrow_B\rangle + |\downarrow_{\text{STM}}\downarrow_B\rangle \} \end{aligned} \quad (3.21)$$

$$\begin{aligned} |\phi_b\rangle &= \frac{1}{\sqrt{6}} |\uparrow_{\text{STM}}\rangle \left(\frac{|\uparrow_B + \downarrow_B\rangle}{\sqrt{2}} \right) + \frac{1}{\sqrt{6}} |\downarrow_{\text{STM}}\rangle \left(\frac{|\uparrow_B - \downarrow_B\rangle}{\sqrt{2}} \right) + \sqrt{\frac{2}{3}} |\downarrow_{\text{STM}}\rangle \left(\frac{|\uparrow_B + \downarrow_B\rangle}{\sqrt{2}} \right) \\ &= \frac{1}{\sqrt{12}} |\uparrow_{\text{STM}}\uparrow_B\rangle + \frac{1}{\sqrt{12}} |\uparrow_{\text{STM}}\downarrow_B\rangle + \frac{3}{\sqrt{12}} |\downarrow_{\text{STM}}\uparrow_B\rangle + \frac{1}{\sqrt{12}} |\downarrow_{\text{STM}}\downarrow_B\rangle \end{aligned} \quad (3.22)$$

and

$$|\bar{\phi}_b\rangle = \frac{2}{\sqrt{6}} |\uparrow_{\text{STM}}\uparrow_B\rangle - \frac{1}{\sqrt{6}} |\uparrow_{\text{STM}}\downarrow_B\rangle - \frac{1}{\sqrt{6}} |\downarrow_{\text{STM}}\downarrow_B\rangle \quad (3.23)$$

For the case where the $|T_-\rangle$ transitions to the $|\downarrow_B\rangle$ there are four possibilities for the next evolution:

$$\begin{aligned} |\downarrow_B\rangle &\xrightarrow{1} |T_-\rangle \text{ at rate } \overbrace{\frac{1}{2} \left(\frac{1}{2} \right)}^{\text{spin-up}} + \overbrace{\frac{1}{2} \left(\frac{1}{2} \right)}^{\text{spin-down}} = \frac{1}{2} \\ &\xrightarrow{2} |T_0\rangle \text{ at rate } \frac{1}{2} \left(\frac{1}{4} \right) + \frac{1}{2} \left(\frac{1}{4} \right) = \frac{1}{4} \\ &\xrightarrow{3} |\phi_b\rangle \text{ at rate } \frac{1}{2} \left(\frac{1}{12} \right) + \frac{1}{2} \left(\frac{1}{12} \right) = \frac{1}{12} \\ &\xrightarrow{4} |\bar{\phi}_b\rangle \text{ at rate } \frac{1}{2} \left(\frac{1}{6} \right) + \frac{1}{2} \left(\frac{1}{6} \right) = \frac{1}{6} \end{aligned} \quad (3.24)$$

where the coefficients of $1/2$ represent the unpolarized spin injection from the bulk and the terms in the parentheses corresponds to $|\langle \uparrow_{\text{STM}}\downarrow_B | \psi \rangle|^2$ and $|\langle \downarrow_{\text{STM}}\downarrow_B | \psi \rangle|^2$, the probability of the particular transport spin orientation paired with a spin-down spectator along the field axis, respectively where $\psi = \{T_-, T_0, \phi_b, \bar{\phi}_b\}$ and the rates add to unity.

Transitions from $|T_0\rangle$ State

The state $|T_0\rangle$ will transition to $|\uparrow_{\text{STM}}\rangle$ every time at a rate $\gamma_{\text{off}}/2$ and the transitions back to the two-spin manifold are:

$$\begin{aligned}
|\uparrow_{\text{STM}}\rangle &\xrightarrow{1} |T_-\rangle \text{ at rate } \overbrace{\frac{1}{2}\left(\frac{1}{4}\right)}^{\text{spin-up}} + \overbrace{\frac{1}{2}\left(\frac{1}{4}\right)}^{\text{spin-down}} = \frac{1}{4} \\
&\xrightarrow{2} |T_0\rangle \text{ at rate } \frac{1}{2}\left(\frac{1}{2}\right) = \frac{1}{4} \\
&\xrightarrow{3} |\phi_b\rangle \text{ at rate } \frac{1}{2}\left(\frac{1}{6}\right) + \frac{1}{2}\left(\frac{2}{3}\right) = \frac{5}{12} \\
&\xrightarrow{4} |\bar{\phi}_b\rangle \text{ at rate } \frac{1}{2}\left(\frac{1}{12}\right) + \frac{1}{2}\left(\frac{1}{12}\right) = \frac{1}{12}
\end{aligned} \tag{3.25}$$

Transitions from $|\phi_b\rangle$ State

The state $|\phi_b\rangle$ transitions to the single-spin state $|\uparrow_{\text{STM}}\rangle$ every time with a rate $\gamma_{\text{off}}/6$. The transitions back to the two-spin manifold are the same as shown above for the $|T_0\rangle$ state.

Transitions from $|\bar{\phi}_b\rangle$ State

Finally we consider the state $|\bar{\phi}_b\rangle$. This state will transition to the single-spin state $(1/\sqrt{10})|\uparrow_{\text{STM}}\rangle + (3/\sqrt{10})|\downarrow_{\text{STM}}\rangle = |\uparrow_{\hat{n}}\rangle$, where the superposition in the SP-STM basis is written as a pure state along a basis parallel to the unit vector $\hat{n}$. The rate is $5\gamma_{\text{off}}/6$. The two spin states in the $\hat{n}$ basis are:

$$\begin{aligned}
|\uparrow_{\hat{n}}\rangle &= \frac{|\uparrow_{\text{STM}}\rangle}{\sqrt{10}} + \frac{3|\downarrow_{\text{STM}}\rangle}{\sqrt{10}} \\
|\downarrow_{\hat{n}}\rangle &= \frac{3|\uparrow_{\text{STM}}\rangle}{\sqrt{10}} - \frac{|\downarrow_{\text{STM}}\rangle}{\sqrt{10}}
\end{aligned} \tag{3.26}$$

and for convenience

$$\begin{aligned}
|\uparrow_{\text{STM}}\rangle &= \frac{3|\downarrow_{\hat{n}}\rangle + |\uparrow_{\hat{n}}\rangle}{\sqrt{10}} \\
|\downarrow_{\text{STM}}\rangle &= \frac{3|\uparrow_{\hat{n}}\rangle - |\downarrow_{\hat{n}}\rangle}{\sqrt{10}}.
\end{aligned} \tag{3.27}$$

The basis states are then written with the spectator spin in the $\hat{S}_n$ basis:

$$\begin{aligned}
|T_-\rangle &= -\frac{|\uparrow_{\text{STM}}\uparrow_{\hat{n}}\rangle}{\sqrt{10}} + \frac{2|\uparrow_{\text{STM}}\downarrow_{\hat{n}}\rangle}{\sqrt{10}} + \frac{|\downarrow_{\text{STM}}\uparrow_{\hat{n}}\rangle}{\sqrt{10}} - 2\frac{|\downarrow_{\text{STM}}\downarrow_{\hat{n}}\rangle}{\sqrt{10}} \\
|T_0\rangle &= \frac{|\uparrow_{\text{STM}}\uparrow_{\hat{n}}\rangle}{\sqrt{20}} + \frac{3|\uparrow_{\text{STM}}\downarrow_{\hat{n}}\rangle}{\sqrt{20}} - \frac{3|\downarrow_{\text{STM}}\uparrow_{\hat{n}}\rangle}{\sqrt{20}} + \frac{|\downarrow_{\text{STM}}\downarrow_{\hat{n}}\rangle}{\sqrt{20}} \\
|\phi_b\rangle &= \frac{|\uparrow_{\text{STM}}\uparrow_{\hat{n}}\rangle}{\sqrt{60}} + \frac{3|\uparrow_{\text{STM}}\downarrow_{\hat{n}}\rangle}{\sqrt{60}} + \frac{5|\downarrow_{\text{STM}}\uparrow_{\hat{n}}\rangle}{\sqrt{60}} + \frac{5|\downarrow_{\text{STM}}\downarrow_{\hat{n}}\rangle}{\sqrt{60}} \\
|\bar{\phi}_b\rangle &= \frac{5|\uparrow_{\text{STM}}\uparrow_{\hat{n}}\rangle}{\sqrt{30}} + \frac{|\downarrow_{\text{STM}}\uparrow_{\hat{n}}\rangle}{\sqrt{30}} - \frac{2|\downarrow_{\text{STM}}\downarrow_{\hat{n}}\rangle}{\sqrt{30}}.
\end{aligned} \tag{3.28}$$

Now the antibottleneck state, $|\bar{\phi}_b\rangle$, will transition to the single-spin state $|\uparrow_{\hat{n}}\rangle$ 100% of the time and then back to the two-spin manifold with the following rates:

$$\begin{aligned}
|\uparrow_{\hat{n}}\rangle &\xrightarrow{1} |T_-\rangle \text{ at rate } \overbrace{\frac{1}{2}\left(\frac{1}{10}\right)}^{\text{spin-up}} + \overbrace{\frac{1}{2}\left(\frac{1}{10}\right)}^{\text{spin-down}} = \frac{1}{10} \\
&\xrightarrow{2} |T_0\rangle \text{ at rate } \frac{1}{2}\left(\frac{1}{20}\right) + \frac{1}{2}\left(\frac{9}{20}\right) = \frac{1}{4} \\
&\xrightarrow{3} |\phi_b\rangle \text{ at rate } \frac{1}{2}\left(\frac{1}{60}\right) + \frac{1}{2}\left(\frac{25}{60}\right) = \frac{13}{60} \\
&\xrightarrow{4} |\bar{\phi}_b\rangle \text{ at rate } \frac{1}{2}\left(\frac{25}{30}\right) + \frac{1}{2}\left(\frac{1}{30}\right) = \frac{13}{30}
\end{aligned} \tag{3.29}$$

Steady-state occupation of the Two-Spin States

Next we solve for the steady-state occupation of the four states. The hopping rates from the two-spin state to the SP-STM tip under ideal polarization ($\Gamma_{\uparrow} = 1$) are

$$\begin{aligned}
\Gamma_{T_-} &= \frac{1}{2} \\
\Gamma_{T_0} &= \frac{1}{2} \\
\Gamma_{T_b} &= \frac{1}{6} \\
\Gamma_{T_{\bar{b}}} &= \frac{5}{6}
\end{aligned} \tag{3.30}$$

and the rate equations are

$$\begin{aligned}
\frac{dP_{T_-}}{dt} &= 0 = \left(\frac{1}{2} - 1\right)P_{T_-}\Gamma_{T_-} + \frac{1}{4}P_{T_0}\Gamma_{T_0} + \frac{1}{4}P_b\Gamma_b + \frac{1}{10}P_{\bar{b}}\Gamma_{\bar{b}} \\
&= -\frac{P_{T_-}}{4} + \frac{P_{T_0}}{8} + \frac{P_{T_b}}{24} + \frac{P_{T_{\bar{b}}}}{12} \\
\frac{dP_{T_0}}{dt} &= 0 = \frac{P_{T_-}}{8} - \frac{3P_{T_0}}{8} + \frac{P_{T_b}}{24} + \frac{5P_{T_{\bar{b}}}}{24} \\
\frac{dP_b}{dt} &= 0 = \frac{P_{T_-}}{24} + \frac{5P_{T_0}}{24} - \frac{7P_{T_b}}{72} + \frac{65P_{T_{\bar{b}}}}{360} \\
\frac{dP_{\bar{b}}}{dt} &= 0 = \frac{P_{T_-}}{12} + \frac{5P_{T_0}}{24} + \frac{P_{T_b}}{72} - \frac{17P_{T_{\bar{b}}}}{36}
\end{aligned} \tag{3.31}$$

which yields the following occupation probabilities: $P_{T_-} = 1/5$, $P_{T_0} = 1/6$, $P_b = 17/30$, and $P_{\bar{b}} = 1/15$. In the product basis we calculate

$$\begin{aligned}
P_{\uparrow\uparrow} &= \frac{1}{4}P_{T_-} + \frac{1}{2}P_{T_0} + \frac{1}{6}P_b + \frac{1}{12}P_{\bar{b}} = \frac{7}{30} \\
P_{\downarrow\downarrow} &= \frac{1}{4}P_{T_-} + \frac{1}{2}P_{T_0} + \frac{1}{6}P_b + \frac{1}{12}P_{\bar{b}} = \frac{7}{30} \\
P_{\uparrow\downarrow} &= \frac{1}{4}P_{T_-} + \frac{3}{4}P_{\bar{b}} = \frac{1}{10} \\
P_{\downarrow\uparrow} &= \frac{1}{4}P_{T_-} + \frac{2}{3}P_b + \frac{1}{12}P_{\bar{b}} = \frac{13}{30}.
\end{aligned} \tag{3.32}$$

These results are quite close to the results obtained in the full description in Sec. E.2.

The visibility is calculated by first considering the current, I ,

$$\begin{aligned}
I_{\text{res}} &= P_{T_-}\Gamma_{T_-} + P_{T_0}\Gamma_{T_0} + P_b\Gamma_b + P_{\bar{b}}\Gamma_{\bar{b}} \\
&= \frac{1}{5} \cdot \frac{1}{2} + \frac{1}{6} \cdot \frac{1}{2} + \frac{17}{30} \cdot \frac{1}{6} + \frac{1}{15} \cdot \frac{5}{6} = \frac{1}{3}
\end{aligned} \tag{3.33}$$

$$I_{\text{off-res}} = \left(\sum P\right) \frac{\Gamma_{\uparrow}}{2} = \frac{1}{2}. \tag{3.34}$$

The ratio of the two currents is $I_{\text{res}}/I_{\text{off-res}} = 2/3$, so the visibility is $1/3$.

Lastly we consider the polarization of the spectator site. There are three states that describe the spectator: $|\uparrow_{\text{STM}}\rangle$, $|\downarrow_B\rangle$, and $|\uparrow_{\hat{n}}\rangle$ with occupation probabilities P_z , P_{-x} ,

and $P_{\hat{n}}$ respectively. The transitions are as follows:

$$\begin{aligned}
| \uparrow_{\text{STM}} \rangle &\xrightarrow{1} | T_- \rangle \rightarrow | \downarrow_B \rangle \text{ at rate } \frac{1}{4} \\
&\xrightarrow{2} | T_0 \rangle \rightarrow | \uparrow_{\text{STM}} \rangle \text{ at rate } \frac{1}{4} \\
&\xrightarrow{3} | \phi_b \rangle \rightarrow | \uparrow_{\text{STM}} \rangle \text{ at rate } \frac{5}{12} \\
&\xrightarrow{4} | \bar{\phi}_b \rangle \rightarrow | \uparrow_{\hat{n}} \rangle \text{ at rate } \frac{1}{12}
\end{aligned} \tag{3.35}$$

$$\begin{aligned}
| \uparrow_{\hat{n}} \rangle &\xrightarrow{1} | T_- \rangle \rightarrow | \downarrow_B \rangle \text{ at rate } \frac{1}{10} \\
&\xrightarrow{2} | T_0 \rangle \rightarrow | \uparrow_{\text{STM}} \rangle \text{ at rate } \frac{1}{4} \\
&\xrightarrow{3} | \phi_b \rangle \rightarrow | \uparrow_{\text{STM}} \rangle \text{ at rate } \frac{13}{60} \\
&\xrightarrow{4} | \bar{\phi}_b \rangle \rightarrow | \uparrow_{\hat{n}} \rangle \text{ at rate } \frac{13}{30}
\end{aligned} \tag{3.36}$$

$$\begin{aligned}
| \downarrow_B \rangle &\xrightarrow{1} | T_- \rangle \rightarrow | \downarrow_B \rangle \text{ at rate } \frac{1}{2} \\
&\xrightarrow{2} | T_0 \rangle \rightarrow | \uparrow_{\text{STM}} \rangle \text{ at rate } \frac{1}{4} \\
&\xrightarrow{3} | \phi_b \rangle \rightarrow | \uparrow_{\text{STM}} \rangle \text{ at rate } \frac{1}{12} \\
&\xrightarrow{4} | \bar{\phi}_b \rangle \rightarrow | \uparrow_{\hat{n}} \rangle \text{ at rate } \frac{1}{6}
\end{aligned} \tag{3.37}$$

and the rate equations are

$$\begin{aligned}
\frac{dP_z}{dt} &= \left(\frac{2}{3} - 1 \right) P_z + \frac{1}{3} P_{-x} + \frac{7}{15} P_{\hat{n}} = 0 \\
\frac{dP_{-x}}{dt} &= \left(\frac{1}{2} - 1 \right) P_{-x} + \frac{1}{4} P_z + \frac{1}{10} P_{\hat{n}} = 0 \\
\frac{dP_{\hat{n}}}{dt} &= \left(\frac{13}{30} - 1 \right) P_{\hat{n}} + \frac{1}{12} P_z + \frac{1}{6} P_{-x} = 0,
\end{aligned} \tag{3.38}$$

which yield occupation probabilities of $P_z = 8/15$, $P_{-x} = 3/10$, $P_{\hat{n}} = 1/6$. The resulting spin polarizations are $2/5$ along the SP-STM and $-1/5$ along the magnetic field.

These polarizations for the spectator spin differ from those obtained in Sec. E.2, suggesting that the coherence between ϕ_b and $\bar{\phi}_b$ plays a more substantial role in determining this spin polarization than the two-spin product populations.

Chapter 4

Extension to spin-1 defects

4.1 The spin-1 manifold: a look at divacancies in silicon carbide

4.1.1 The case for NV centers in diamond

When considering a prime defect for spin applications there are a number of favorable characteristics to search for. For instance, attributes that limit decoherence would be weak spin-orbit interaction, a wide band gap, and the option for isotopic purification. The NV center in diamond has all of these qualities and as a result has been highly studied for the past two decades as a viable spin center for quantum sensing and quantum information applications. In addition to room temperature coherence, the nuclear states of the NV provide potential for intrinsic quantum registers.

4.1.2 Optical approaches to NV dynamics

Accurate measure of the NV center spin state is arguably the most important step towards sensing and information application. The question of how one measures a quantum entity is in itself a subtle matter. The NV center in diamond is optically active and as a result the traditional readout scheme involves an analysis of the spin-dependent photoluminescence (PL). The defect is aligned along the $\langle 111 \rangle$ axis of the diamond and consists of a substitutional nitrogen adjacent to a vacancy. The ground state is a spin triplet and due to the lack of inversion symmetry the zero-field splitting lifts the degeneracy between the $m_s = 0$ state and the $m_s = \pm 1$ states. Additional splitting is possible through Zeeman coupling with an external field applied along the $\langle 111 \rangle$ axis. The finite spin-orbit interaction leads to an intersystem crossing (ISC) between the triplet and singlet states that is highly spin dependent in the

excited state manifold. This spin dependence allows one to probe the spin state of the NV optically by paying attention to the relative PL. Decay through the singlet states is non-radiative and the $m_s = \pm$ excited states have a much higher rate of decaying through this channel than the $m_s = 0$ state. This means that the pumping from $|T_0^{(gs)}\rangle \rightarrow |T_0^{(es)}\rangle$ will continue to fluoresce through the radiative channel while the excitation from $|T_{\pm}^{(gs)}\rangle \rightarrow |T_{\pm}^{(es)}\rangle$ will transition to the non-radiative channel via the ISC. After illumination the ground state has an occupation probability of 80% in the $m_s = 0$ state so this scheme is often used for state initialization. The relevant timescale for identifying contrast in PL is based on the singlet decay time (300 ns).

4.1.3 Looking ahead to the neutral divacancy in SiC

While the NV center in diamond has a strong history of experimental and theoretical interest, there are other systems with the similar properties that are attracting interest [115]. The one that we will focus on is the divacancy in SiC. Unlike diamond, SiC has over 250 polytypes allowing for a higher degree of customization. The divacancy has demonstrated key attributes for quantum application: long spin-coherence [116], and single defect isolation [116]. However, most of the work done on divacancies has been optical. By applying the spin-coherence and correlated-dynamics concepts discussed in Chapter 3 we plan to develop a theoretical argument for room temperature electrical measurements of the neutral divacancy in SiC. In the following sections we provide a brief discussion on the various components necessary for theoretical analysis of the coherence correlation properties of a spin-1 ground state like the divacancy.

4.2 Electronic dipolar coupling: zero-field interaction

The neutral divacancy has a electronic spin-1 ground state and as a result the two electronic magnetic moments interact through dipolar fields. In the absence of an applied field the crystal field will split the triply degenerate states into a doublet, comprised of the T_+ and T_- states, and the T_0 state. A magnetic moment in an inhomogeneous magnetic field will

feel a force equal to the derivative of the field parallel to the moment,

$$\mathbf{F} = \nabla(\mu \cdot \mathbf{B}). \quad (4.1)$$

This relation was used by Stern and Gerlach to show the quantization of spin angular momenta in the valence electron of Ag atoms. The force exerted on the magnetic moment has no curl since $\nabla \times \nabla\psi = 0$. This implies that the force is conservative and therefore derivable from a potential,

$$\mathbf{F} = -\nabla\hat{V}_{12}, \quad (4.2)$$

where in this case $\hat{V}_{12}$ is the interaction potential between two spin magnetic moments $\mathbf{S}_1$ and $\mathbf{S}_2$.

The dipolar field from electron 1 at r_1 felt by electron 2 at r_2 is given by the usual expression,

$$\hat{V}_{12} = \frac{\mu_0\gamma_1\gamma_2}{4\pi} \left[\frac{\mathbf{S}_1 \cdot \mathbf{S}_2}{r_{12}^3} - \frac{3(\mathbf{S}_1 \cdot \mathbf{r}_{12})(\mathbf{S}_2 \cdot \mathbf{r}_{12})}{r_{12}^5} \right], \quad (4.3)$$

where $\gamma_{1(2)}$ is the gyromagnetic ratio for electron 1 (2) and μ_0 is the permeability of free space. Expanding Eq(4.3) in Cartesian coordinates using

$$\begin{aligned} \mathbf{S}_1 &= S_{x1}\hat{i} + S_{y1}\hat{j} + S_{z1}\hat{k} \\ \mathbf{S}_2 &= S_{x2}\hat{i} + S_{y2}\hat{j} + S_{z2}\hat{k} \\ \mathbf{r} &= x\hat{i} + y\hat{j} + z\hat{k} \\ \hat{\mathbf{r}} &= \hat{i} + \hat{j} + \hat{k}. \end{aligned} \quad (4.4)$$

and grouping like terms results in

$$\begin{aligned} \mathbf{D} = \omega_D \bigg[& \left(1 - \frac{3x^2}{r^2}\right) S_{x1}S_{x2} + \left(1 - \frac{3y^2}{r^2}\right) S_{y1}S_{y2} + \left(1 - \frac{3z^2}{r^2}\right) S_{z1}S_{z2} \\ & - 3\frac{xy}{r^2} S_{x1}S_{y2} - 3\frac{xz}{r^2} S_{x1}S_{z2} - 3\frac{yz}{r^2} S_{y1}S_{z2} \\ & - 3\frac{yx}{r^2} S_{y1}S_{x2} - 3\frac{zx}{r^2} S_{z1}S_{x2} - 3\frac{zy}{r^2} S_{z1}S_{y2}, \end{aligned} \quad (4.5)$$

where $\omega_D = \mu_0\gamma_1\gamma_2/4\pi r^3$. Eq(4.5) can be written in matrix form coupling the spin operators

of both electrons

$$\mathcal{H}_D = \omega_D \mathbf{S}_1 \cdot \begin{pmatrix} 1 - 3x^2/r^2 & -3xy/r^2 & -3xz/r^2 \\ -3yx/r^2 & 1 - 3y^2/r^2 & -3yz/r^2 \\ -3zx/r^2 & -3zy/r^2 & 1 - 3z^2/r^2 \end{pmatrix} \cdot \mathbf{S}_2 = \mathbf{S}_1 \cdot \mathbf{D} \cdot \mathbf{S}_2. \quad (4.6)$$

$\mathbf{D}$ is often referred to as the dipole operator. The dipole operator is symmetric and it can be easily verified that this operator is traceless meaning it is diagonalizable. Due to these properties one can recast Eq(4.6) in terms of the total spin operator $\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2$,

$$\mathcal{H}_D = \mathbf{S} \cdot \mathbf{D} \cdot \mathbf{S} \quad (4.7)$$

4.3 The hyperfine interaction

The coupling between a nuclear moment and electronic moment is called the hyperfine interaction. While the energy scales of this term are relatively small, the hyperfine interaction leads to spin decoherence and in some cases limits the operational time of a spin qubit [97]. Three interactions contribute to the hyperfine Hamiltonian in general, (i) dipolar interaction between the nuclear moment and electronic spin angular momenta valid at distances larger than the scale of the nucleus, (ii) the contact interaction valid at the position of the nucleus itself, and (iii) the interaction between the nucleus and the orbital momenta of the electron.

Natural SiC Hyperfine Isotopes		
Isotope	Spin	Natural Abundance
^{13}C	1/2	1.07%
^{29}Si	1/2	4.68%

Nuclear moments interact with the electrons through the angular momentum. The energy of this interaction has the form $-\mu \cdot \mathbf{B}_{electron}$ where the magnetic field due to the electrons is proportional to the total electronic angular momentum. This leads one to a simple form for the hyperfine Hamiltonian,

$$\mathcal{H}_{\text{hf}} = \mathbf{I} \mathbf{A} \mathbf{J}, \quad (4.8)$$

where the nuclear moment is $\mathbf{I}$, the electronic angular momentum is $\mathbf{J}$, and the coupling tensor is $\mathbf{A}$ which can be measured experimentally.

The divacancy in SiC is a localized spin-1 center with three nearest neighbor carbon atoms, three nearest neighbor silicon atoms, and nine second nearest neighbor silicon atoms. The spin density as calculated using density functional theory has been found to exist primarily on the nearest neighbor carbon [117]. The number of important nuclei sites leading to resolvable hyperfine interactions are limited to the second nearest neighbor of the divacancy. Given the natural abundance of ^{13}C and ^{29}Si for a single divacancy there will most likely (55% of the time) be no influential neighbors with a magnetic moment. The most probable hyperfine situation is to have a single ^{29}Si and the chance of that is 32%. First principles calculations for the hyperfine tensor of nearby Si atoms show relatively isotropic coupling in the range of 1-10 MHz [117]. The influence of the hyperfine interaction will depend on the emptying rate of the divacancy, γ_e , and in situations where $\gamma_e \gg \gamma_{\text{hfi}}$, which for hyperfine rates on the order of 10 MHz would correspond to tunneling rates of 100 MHz, the influence of the hyperfine coupling on the current is irrelevant. Due to the low natural abundance of influential atoms and the ability to tune the system to a transport range where this interaction becomes irrelevant, we ignore this term at this time.

4.4 The stochastic Liouville equation

4.4.1 The Hamiltonian

The spin-1 ground state of the neutral divacancy couples to electric and strain fields in addition to the magnetic fields. In this chapter the hh and kk divacancies will be considered. Since these defects are aligned with the c-axis of the crystal the zero-field splitting breaks the degeneracy between the T_0 state and the $T_{+/-}$ states defined along the c-axis and allowing for a convenient choice of the z-axis,

$$\mathcal{H}_{\text{zfs}} = D\hat{\Sigma}_z^2, \quad (4.9)$$

where $\hat{\Sigma}_z$ is the z-component of the spin-1 analog to the Pauli matrices defined as

$$\hat{\Sigma}_i \equiv \hat{\sigma}_i \otimes \mathbf{I}_{2 \times 2} + \mathbf{I}_{2 \times 2} \otimes \hat{\sigma}_i, \quad (4.10)$$

with $\hat{\sigma}_i$ representing the i^{th} Pauli matrix.

The applied magnetic field couples to the neutral divacancy through the Zeeman term

$$\mathcal{H}_B^{(1)} = g\mu_B \hbar \mathbf{B} \cdot \mathbf{\Sigma}, \quad (4.11)$$

where $\mathbf{\Sigma} = \{\hat{\Sigma}_x, \hat{\Sigma}_y, \hat{\Sigma}_z\}$. The corresponding term for the positively charged divacancy is

$$\mathcal{H}_B^{(1/2)} = \frac{g\mu_B \hbar}{2} \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (4.12)$$

The total effective spin Hamiltonians are

$$\mathcal{H}_1 = \mathcal{H}_{\text{zfs}} + \mathcal{H}_B^{(1)} \quad (4.13)$$

and

$$\mathcal{H}_{1/2} = \mathcal{H}_B^{(1/2)}. \quad (4.14)$$

This analysis considers incoherent transport and therefore the Hamiltonians do not contain terms relating to the movement of charge into or out of the defect. These Hamiltonians describe the coherent evolution of each manifold and is accounted for in the stochastic Liouville equations through the terms

$$\left[\frac{d}{dt} \rho_i(t) \right]_{\text{coh}} = -\frac{i}{\hbar} [\mathcal{H}_i, \rho_i(t)], \quad (4.15)$$

where $i = \{1/2, 1\}$ denotes the electronic spin of the corresponding manifold.

4.4.2 Generation and dissipation due to incoherent charge transport

Transport through the divacancy does not conserve spin locally and requires terms in the SLE that correspond to incoherent evolution. Since the trace of the entire Fock space is conserved the term corresponding to generation in the spin-1 manifold will necessarily be balanced by the charge-based dissipation term in the spin-1/2 manifold. Transitions of this

type are labeled (1/2:1). The same is true in reverse - generation of the spin-1/2 manifold is balanced by charge-based dissipation of the spin-1 manifold. Analogously, these transitions will be labeled (1:1/2). The terms for (1/2:1) can be written as

$$\left[\frac{d}{dt} \rho_1(t) \right]_{\text{gen}} = +2\gamma_f \hat{P}_T (\mathbf{I}/2 \otimes \rho_{1/2}(t)) \hat{P}_T \quad (4.16)$$

and

$$\left[\frac{d}{dt} \rho_{1/2}(t) \right]_{\text{dis}} = -\frac{3}{2} \gamma_f \rho_{1/2}(t), \quad (4.17)$$

where $\hat{P}_T$ is the projection operator onto the triplet basis expressed in the product basis. The factor of $\mathbf{I}/2$ represents the mixed state of the spins hopping onto the divacancy from the SiC bulk and γ_f defines the rate for this transport process per spin channel. This means that a bulk electron with $m_s = +1/2$ will hop to the divacancy occupied by a single electron with $m_s = +1/2$ at a rate γ_f . The only way for a bulk electron with $m_s = -1/2$ to hop to the same divacancy is for the bulk spin to be entangled with the resident spin forming a T_0 state. The generation of the entangled state involves generating equal occupancy of four product states which results in a generation rate of $\gamma_f/4$ for each product state.

Terms for the (1:1/2) transition depend on the polarization of the cathode. The description of the polarization follows closely the approach taken for the exchange coupled spin-1/2 system. The operators are defined as

$$\hat{M}_e^{(i)} = \frac{1}{4} (\mathbf{I} + \mathbf{P} \cdot \boldsymbol{\Sigma}_i), \quad (4.18)$$

where $\mathbf{P}$ is the polarization vector of the contact, $\boldsymbol{\Sigma}$ is the spin-1 analog of the Pauli matrices, and $i = \{a, b\}$ corresponds to the two subspaces of the two-spin Hilbert space. The SLE terms are then

$$\left[\frac{d}{dt} \rho_1(t) \right]_{\text{dis}} = -2\gamma_e (\{\hat{M}_e^{(a)}, \rho_1(t)\} + \{\hat{M}_e^{(b)}, \rho_1(t)\}) \quad (4.19)$$

and

$$\left[\frac{d}{dt} \rho_{1/2}(t) \right]_{\text{gen}} = +2\gamma_e (\text{Tr}_a[\{\hat{M}_e^{(a)}, \rho_1(t)\}] + \text{Tr}_b[\{\hat{M}_e^{(b)}, \rho_1(t)\}]). \quad (4.20)$$

In the next section we investigate the steady-state current with two different polarization/magnetic field configurations.

4.5 Magnetoresistance calculations

Calculations of the steady-state ($d\rho/dt = 0$) current through the spin-1 center shows both a zero-field feature and a finite-field feature depending on the relative configuration of the crystal, tip polarization, and applied magnetic field. The first configuration considered here is one with the tip polarization along the c-axis of the crystal and an applied transverse magnetic field as shown in figure 4.1(a). The hopping rate from the bulk of the semiconductor to the spin-1 is labeled γ_f and the rate from the spin-1 to the tip is labeled γ_e . Here the tip is assumed to be perfectly polarized. Figure 4.1(b) shows the magnetoresistance for carriers through the spin-1. At zero applied field the current is quenched, corresponding to the occupation of the triplet bottleneck state T_- in the basis of the STM. The applied magnetic field evolves the bottleneck state through the intermediate T_0 state that is split from the bottleneck state by an energy D_{zfs} . As shown in figure 4.1(b) the width of the feature depends on the emptying rate γ_e . This is to be expected since for rapid transport off of the site the spin has less time to precess in the magnetic field. This means the same field magnitude will result in less current for faster hopping rates. Recall that the key to generating current is efficient evolution out of the bottleneck configuration.

In the second configuration discussed, the magnetic field is along the c-axis of the semiconductor and the tip polarization is transverse. The dynamics in this case are different from the previous case, as might be expected by comparing figure 4.1(b) with figure 4.2(b). The energy structure here is degenerate at zero applied field in the STM basis and due to the zero-field splitting along the axis transverse to the STM, the zero-field bottleneck is absent. In a similar way to the exchange coupled system in chapter 3, in the basis of the applied magnetic field the $T_{+/-}$ states are split from the T_0 state and Zeeman splitting brings the T_+ or T_- state into resonance once the Zeeman energy is near the zero-field splitting energy. Once again the width of the finite-field features is dependent on the emptying rate. Furthermore, a bias field can be applied to the system, centering the finite-field feature about zero applied field. This may provide a path toward sensing minuscule magnetic fields.

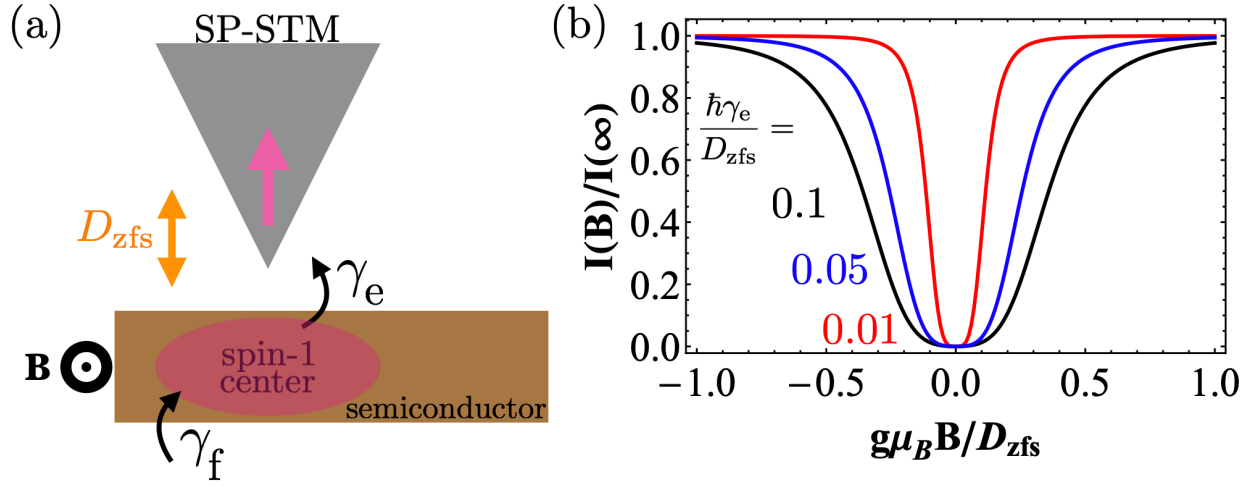


Figure 4.1: (a) Schematic of the set-up for a zero-field measurement of the zero-field splitting. The zero-field splitting axis is and (b) the steady-state current through the spin-1 center as a function of applied magnetic field. The ratio of the hopping rates $\gamma_e/\gamma_f \ll 1$ to ensure the spin-1 state is primarily occupied.

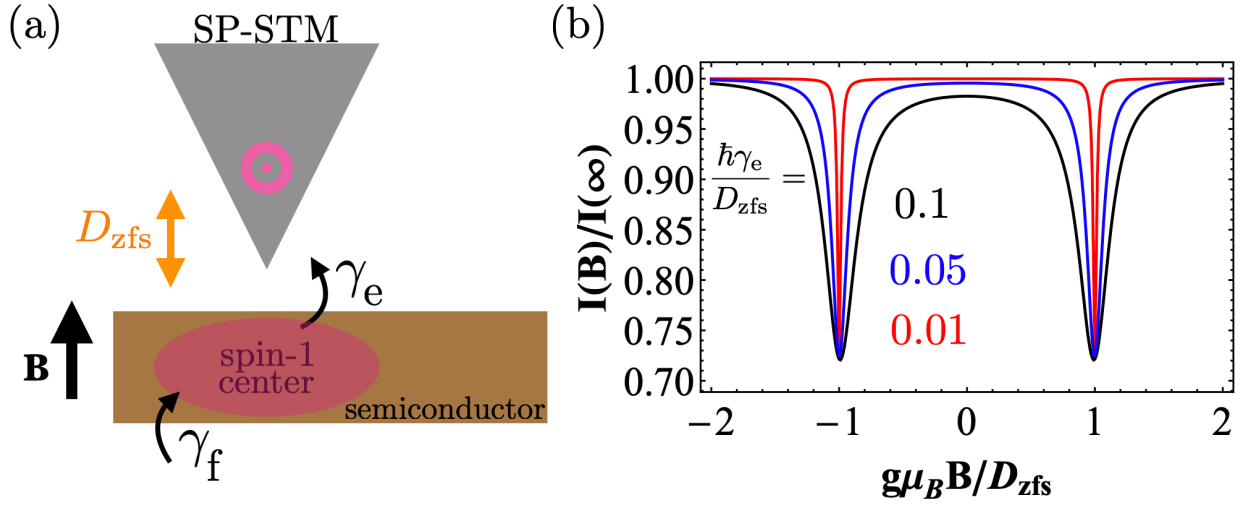


Figure 4.2: (a) Schematic of the set-up for a finite-field measurement of the zero-field splitting and (b) the steady-state current through the spin-1 center as a function of applied magnetic field. The ratio of the hopping rates $\gamma_e/\gamma_f \ll 1$ to ensure the spin-1 state is primarily occupied.

Chapter 5

Outlook

Incoherent spin-correlated transport is emerging as an exciting area of research to consider for those interested in harnessing the spin degree of freedom. In the case of organic π -conjugated polymers we predict that steric influences on the spin-flip probability provide a means of tailoring the spin-orbit induced relaxation. For instance, by relative alignment of localized sites with the spin quantization axis, the relaxation associated with spin-orbit coupling can be quenched. Organics have long touted excellent spin-coherence times, but at the expense of low carrier mobilities. Quenching the relaxation mechanisms through steric considerations should lead to longer spin diffusion lengths, increasing the utility of these materials. Additionally, the charge transport trends seen in flexed planar organics analyzed in Section 2.5 suggest a means of potentially increasing both the spin mobility and spin lifetimes through local strain.

In addition to planar organics, there are a number of defect and dopant based localized spins centers that remain coherent at room temperature. The work presented here was focused on addressing these centers with a spin-polarized STM, but one could just as easily (or more easily) imagine a spin-polarized planar contact. This realization would package the physics in a way that would be robust enough to use in real-world application. Aside from addressing these spin centers directly, the spin-dependent correlations between shallow and deep levels can provide information about the local environment of the defects. A Pauli blockade between the shallow level and the deep level would prevent double occupancy of the deep level. Due to differences in the local environment the two states would independently evolve opening channels for double occupancy. Once the deep level is doubly occupied, the

process can be measured electrically due to electron-hole recombination. At low applied fields the weak interactions should be detectable. These types of processes are not limited to dopants and defects. It is completely reasonable to consider how these interactions would allow one to probe molecules at the surface of a substrate. Opening the door to molecular systems provides an unparalleled level of tailorability. For example, one can make use of choosing the relevant spin manifold based on the elements of a porphyrin derivative and by controlling the inter-molecular interactions through clever chemistry.

The work presented here only touches on the magnetoresistive signatures of spin-1/2 and spin-1 centers. Within the spin-1 manifold there are other interactions to be explored like strain and electric field coupling. The same type of analysis can be done for higher spin manifolds where more complex Hamiltonians should lead to a wide range of possibilities for coupling spin coherent evolution with spin-spin correlation effects in dynamics, transport, or even reactions.

Appendix A

Random walks

A.1 Introduction to random walks through spin relaxation processes

Spin transport is inherently non-deterministic and as a result the position of a carrier at a given time can be described by a random walk. This approach will be utilized later to calculate spin relaxation in π -conjugated organics, but for now we outline the basic principles.

In general, relaxation occurs both while a carrier is stationary and in motion. The former represents the interaction between the magnetic moment of the carrier with the nuclear field of the host molecule. An interaction of this kind is termed hyper-fine and is not considered in this treatment. Relaxation in the latter case is a result of the carrier’s motion through an environment with non-zero net charge. In the carrier’s rest-frame this motion will generate a magnetic field that couples to its magnetic moment. This is, of course, the spin-orbit interaction and is the only relaxation mechanism considered here. A model considering both interactions has been developed semiclassically by Harmon *et al.* in which rotation matrices act on a 3-dimensional spin vector [118].

Application to spin relaxation

As an illustrative example we will consider now the quantum mechanical treatment of spin-relaxation. To describe the hopping process of a carrier it is necessary to define the amount of time spent at a site. The mathematical description of this quantity will be assumed to be exponential and will be referred to as the wait-time distribution (WTD), $\psi(t)$. Conversely the probability of a carrier staying at a site must also be defined. This will be referred to as

the survival probability, $\Phi(t)$. There is a differential relationship between the two,

$$\frac{d\Phi(t)}{dt} = -\psi(t). \quad (\text{A.1})$$

During a hop the spin of the carrier can either be conserved or flipped. The probability of each possible outcome can be represented by the normalized rates. The relaxation operator for an ensemble undergoing a single hop can be represented by the matrix $\hat{Q}$:

$$\hat{Q} = \begin{pmatrix} \frac{\omega_{\uparrow\uparrow}}{\omega_{\uparrow\uparrow} + \omega_{\downarrow\uparrow}} & \frac{\omega_{\downarrow\uparrow}}{\omega_{\uparrow\uparrow} + \omega_{\downarrow\uparrow}} \\ \frac{\omega_{\downarrow\uparrow}}{\omega_{\uparrow\uparrow} + \omega_{\downarrow\uparrow}} & \frac{\omega_{\uparrow\uparrow}}{\omega_{\uparrow\uparrow} + \omega_{\downarrow\uparrow}} \end{pmatrix}. \quad (\text{A.2})$$

From this operator one can construct the relaxation operator for n hops. For the case where no hopping takes place,

$$\hat{Q}_0(t) = \Phi(t') = 1 - \int_0^t dt' \psi(t'). \quad (\text{A.3})$$

For one hop occurring at $t = t'$ the procedure is: 1) wait for time $t = t'$, 2) execute hop, 3) survive at new site. This is represented by

$$\hat{Q}_1(t) = \int_0^t dt' \Phi(t - t') \hat{Q} \psi(t'). \quad (\text{A.4})$$

In general, for n hops the operator is

$$\begin{aligned} \hat{Q}_n(t) = \int_0^t dt_n \int_0^t dt_{n-1} \dots \int_0^t dt_1 \Phi(t - t_n) \hat{Q} \psi(t_n - t_{n-1}) \\ \times \hat{Q} \psi(t_{n-1} - t_{n-2}) \dots \hat{Q} \psi(t_1). \end{aligned} \quad (\text{A.5})$$

To construct the relaxation operator, $\hat{P}(t)$, we consider all the carriers that have made n hops, where $n \in \mathbf{Z}_{\geq 0}$:

$$\hat{P}(t) = \sum_{i=0}^{\infty} \hat{Q}_i(t). \quad (\text{A.6})$$

Next we will calculate this operator for a system lacking spatial correlation and extract an expression for the spin relaxation time.

The importance of spatial correlation between spin-flipping and spin-conserving rates can be demonstrated through the following counterpoint. Assume the transition rates to be statistically independent of one another. This simplification allows the substitution of $\omega_{\uparrow\uparrow}$

and $\omega_{\downarrow\uparrow}$ with a constant equal to the value of the respective rates after averaging over all possible initial and final orientations,

$$\begin{aligned}\left\langle \omega_{\uparrow\uparrow}(\theta_1, \phi_1, \theta_2, \phi_2) \right\rangle &= \gamma_c \\ \left\langle \omega_{\downarrow\uparrow}(\theta_1, \phi_1, \theta_2, \phi_2) \right\rangle &= \gamma_{sf},\end{aligned}\tag{A.7}$$

where $\langle \dots \rangle$ represents the spatial averaging. The relaxation operator for a single hop, Eq. (A.2), is now

$$\hat{Q} = \begin{pmatrix} \frac{\gamma_c}{\gamma_c + \gamma_{sf}} & \frac{\gamma_{sf}}{\gamma_c + \gamma_{sf}} \\ \frac{\gamma_{sf}}{\gamma_c + \gamma_{sf}} & \frac{\gamma_c}{\gamma_c + \gamma_{sf}} \end{pmatrix}.\tag{A.8}$$

For convenience the first four hopping operators are listed below:

$$\begin{aligned}\hat{Q}_0(t) &= e^{-\gamma_{\text{tot}} t} \\ \hat{Q}_1(t) &= \hat{Q} \gamma_{\text{tot}} t e^{-\gamma_{\text{tot}} t} \\ \hat{Q}_2(t) &= \hat{Q}^2 \gamma_{\text{tot}}^2 \frac{t^2}{2} e^{-\gamma_{\text{tot}} t} \\ \hat{Q}_3(t) &= \hat{Q}^3 \gamma_{\text{tot}}^3 \frac{t^3}{3!} e^{-\gamma_{\text{tot}} t}\end{aligned}\tag{A.9}$$

where $\gamma_{\text{tot}} = \gamma_c + \gamma_{sf}$.

In order to consider the average relaxation of an ensemble of spins we construct the relaxation operator as in Eq.(A.6):

$$\begin{aligned}\hat{P}(t) &= \sum_{i=0}^{\infty} \hat{Q}_i(t) \\ &= e^{-\gamma_{\text{tot}} t} + \hat{Q} \gamma_{\text{tot}} t e^{-\gamma_{\text{tot}} t} + \dots\end{aligned}\tag{A.10}$$

The above expression is a power series expansion of the exponential multiplied by an additional function of time. This sum simplifies to

$$\hat{P}(t) = e^{\hat{Q} \gamma_{\text{tot}} t} e^{-\gamma_{\text{tot}} t} = e^{(\hat{Q}-1) \gamma_{\text{tot}} t}.\tag{A.11}$$

Physically, this operator represents the time evolution of the ensemble averaged spinor of a system. To find the spin relaxation time requires first evolving an ensemble of polarized spins then taking the expectation value of the spin operator corresponding to the quantization axis,

$$\langle \hat{S}_z(t) \rangle = \langle 0(t) | \hat{\sigma}_z | 0(t) \rangle,\tag{A.12}$$

where $|0\rangle$ represents the initial ensemble averaged spinor (at time $t = 0$) and $|0(t)\rangle$ represents the same ensemble averaged spinor after a time t has passed,

$$\begin{aligned} |0(t)\rangle &= \hat{P}(t)|0\rangle \\ &= \exp\left[\begin{pmatrix} -\gamma_{\text{sf}} & \gamma_{\text{sf}} \\ \gamma_{\text{sf}} & -\gamma_{\text{sf}} \end{pmatrix} t\right] \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \end{aligned} \quad (\text{A.13})$$

Expanding $\hat{P}(t)$ and writing out the first few terms yields

$$\begin{aligned} |0(t)\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -\gamma_{\text{sf}} \\ \gamma_{\text{sf}} \end{pmatrix} t + \begin{pmatrix} 2\gamma_{\text{sf}}^2 \\ -2\gamma_{\text{sf}}^2 \end{pmatrix} \frac{t^2}{2!} + \dots \\ &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \begin{pmatrix} \gamma_{\text{sf}}^n \\ -\gamma_{\text{sf}}^n \end{pmatrix} \frac{t^n}{n!} 2^n - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} [1 + e^{-2\gamma_{\text{sf}} t}] \\ \frac{1}{2} [1 - e^{-2\gamma_{\text{sf}} t}] \end{pmatrix}. \end{aligned} \quad (\text{A.14})$$

From this result it follows that the spin expectation value along the quantization axis after time t is

$$\langle 0(t) | \hat{\sigma}_z | 0(t) \rangle = e^{-2\gamma_{\text{sf}} t} \quad (\text{A.15})$$

with a relaxation rate

$$\frac{1}{T_1} = \frac{1}{T_2} = 2\gamma_{\text{sf}} = \langle \omega_{\uparrow\downarrow} \rangle + \langle \omega_{\downarrow\uparrow} \rangle \quad (\text{A.16})$$

It is worth mentioning that the lack of an external magnetic field allows us to write the equivalence between spin-lattice and spin-spin relaxation rates. The result of Eq. (A.16) will be the measuring stick by which we characterize the importance of considering spatial correlations between transition rates in π -conjugated materials.

Appendix B

Density matrix formalism

B.1 The density matrix

The state of a system at a given time is known with certainty if at that time a series of measurements is made corresponding to a complete set of commuting observables. Consider the state describing a beam of photons emitted from an unpolarized source that have passed through a polarizer aligned with the x -axis. The transmitted beam is known with certainty to exist in the state $|e_x\rangle$. On the other hand, much less is known about the incident beam of photons. The task is then to make the most of our incomplete knowledge of the physical system. For this the density matrix is a wonderful tool. The description below follows that of Cohen-Tannoudji [119].

In general, uncertainty is dealt with in terms of probabilities. Whether one is attempting to predict the outcome of a coin flip or the polarization of an arbitrary photon there is a distribution that describes the probability of each possible outcome. It is often the case in quantum mechanics that the possible states form a finite basis set $\{|e_1\rangle, |e_2\rangle, \dots, |e_N\rangle\}$ and probabilities can be assigned to each option, $\{p_1, p_2, \dots, p_N\}$ such that

$$\sum_i p_i = 1. \tag{B.1}$$

At this point we are dealing with a *statistical mixture* of states. It is important to point out that probabilities are entering from two different points. There is an uncertainty in the initial information about the system and there is uncertainty in the outcome of a particular measurement made on the system. The latter form of uncertainty is not the topic of this section. The former is our focus and it at its core has nothing to do with quantum mechanics

itself.

A pure state at time t is expressed as

$$|\psi(t)\rangle = \sum_n c_n |u_n\rangle, \quad (\text{B.2})$$

where the set $\{|u_n\rangle\}$ form an orthonormal basis on the Hilbert space and $|\psi(t)\rangle$ is assumed to be normalized,

$$1 = \sum_n |c_n(t)|^2 \quad (\text{B.3})$$

The mean value of an operator A is given by

$$\langle A \rangle(t) = \sum_{n,m} c_n^* c_m A_{nm}. \quad (\text{B.4})$$

and the evolution of Eq(B.2) obeys the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle, \quad (\text{B.5})$$

where $H(t)$ is the Hamiltonian of the system.

The right-hand side of Eq(B.4) says that the average value of the operator A is dependent on the product of coefficients $c_n^* c_m$, which are the elements of the matrix $|\psi(t)\rangle \langle \psi(t)|$ - the projection onto the ket $|\psi(t)\rangle$. This matrix is called the density matrix:

$$\rho(t) = |\psi(t)\rangle \langle \psi(t)|. \quad (\text{B.6})$$

To show that this is a valid representation of the state vector Eq(B.2) it can be shown that Eq(B.3-B.5) can be obtained using the density matrix.

Using Eq(B.6) the normalization condition Eq(B.3) becomes

$$\sum_n |c_n(t)|^2 = \sum_n \rho_{nn}(t) = \text{Tr} \rho(t) = 1, \quad (\text{B.7})$$

the expectation value becomes

$$\begin{aligned} \langle A \rangle(t) &= \sum_{n,p} \langle u_m | \rho(t) | u_n \rangle \langle u_n | A | u_m \rangle \\ &= \text{Tr}(\rho(t) A) \end{aligned} \quad (\text{B.8})$$

and the Schrödinger equation applied to the density matrix shows that the time evolution of the matrix is expressed as

$$\frac{d}{dt}\rho(t) = \frac{-i}{\hbar}[H(t), \rho(t)]. \quad (\text{B.9})$$

It is noted that the density matrix is Hermitian and in the case of a pure state only the density matrix is idempotent ($\rho^2(t) = \rho(t)$) and of course then $\text{Tr}\rho^2(t) = 1$.

Let us now turn our attention back to the statistical mixture introduced at the beginning of the section. Let the probability of finding a_n if the state vector is $|\psi_k\rangle$ be expressed as

$$\mathcal{P}(a_n) = \langle \psi_k | P_n | \psi_k \rangle. \quad (\text{B.10})$$

The expression for the general probability $\mathcal{P}(a_n)$ of measuring a_n requires statistical weights p_k for each state vector:

$$\begin{aligned} \mathcal{P}(a_n) &= \sum_k p_k \mathcal{P}_k(a_n) \\ &= \sum_k p_k \text{Tr}(\rho_k P_n) \\ &= \text{Tr} \left(\sum_k p_k \rho_k P_n \right) \\ &= \text{Tr}(\rho P_n), \end{aligned} \quad (\text{B.11})$$

where $\rho_k = |\psi_k\rangle\langle\psi_k|$ and

$$\rho = \sum_k p_k \rho_k. \quad (\text{B.12})$$

The matrix ρ is by definition the density matrix of the system. The general density matrix has the following properties

1. $\text{Tr}\rho = \sum_k p_k \text{Tr}\rho_k = \sum_k p_k = 1$
2. $\langle A \rangle = \text{Tr}(\rho A)$
3. $i\hbar \frac{d}{dt}\rho(t) = [H(t), \rho(t)]$
4. $\text{Tr}\rho^2 \leq 1$
5. ρ is a *positive* operator meaning $\langle u | \rho | u \rangle \geq 0$

We will conclude the introduction to density matrices by discussing the physical significance of the terms. From Eq(B.12) and using the components $c_n^{(k)} = \langle u_n | \psi_k \rangle$ of $|\psi_k\rangle$ in the $\{u_n\}$ basis

$$\rho_{nn} = \sum_k p_k |c_n^{(k)}|^2. \quad (\text{B.13})$$

From this we can see that the diagonal elements of the density matrix correspond to the average probability of finding the system in the state $|u_n\rangle$. Analogously one can calculate the off-diagonal elements

$$\rho_{nm} = \sum_k p_k c_n^{(k)} c_m^{(k)*}. \quad (\text{B.14})$$

The terms $c_n^{(k)} c_m^{(k)*}$ are cross terms that give rise to interference phenomena between states $|u_n\rangle$ and $|u_m\rangle$ which arise when the state $|\psi_k\rangle$ is a coherent superposition of these states. These terms can be zero, positive, negative, and are in general complex. The element ρ_{nm} is the average of these cross terms taken over all possible states of the mixture. If these elements are zero then that means that the averaging has canceled out any interference effects between $|u_n\rangle$ and $|u_m\rangle$; if it is non-zero then some coherences remain between these states. For this reason the off-diagonal terms of the density matrix are called coherences. It is important to realize that the distinction between populations and coherences is basis dependent. Simply by rotating to a basis in which ρ is diagonal removes all coherences and changes the definition of the populations. Additionally, if the kets $|u_n\rangle$ are eigenvectors of the time-independent Hamiltonian H , then the diagonal elements are constant in time and the coherences oscillate at the Bohr frequency $(\Delta E_{nm}/\hbar)$. Lastly one can prove the inequality $\rho_{nn}\rho_{mm} \geq |\rho_{nm}|^2$.

Appendix C

Time-dependent perturbations

C.1 The set-up

Here we discuss the derivation of the transition probability between two steady states of an unperturbed system given a time dependent perturbation. The overview here follows the detailed treatment given by Cohen-Tannoudji [120].

Consider a system with Hamiltonian H_0 , which is time-independent, and the eigenfunctions $|\phi_i\rangle$ such that

$$H_0|\phi_i\rangle = E_i|\phi_i\rangle. \quad (\text{C.1})$$

If the system is initially in the state $|\phi_i\rangle$ and a time-dependent perturbation is introduced, $W(t)$, proportional to some observable $\hat{W}$ by a real number $\lambda \ll 1$, then the eigenfunctions of H_0 are no longer steady-state solutions. The evolution of the quantum state $|\psi(t)\rangle$ between the time 0 when the perturbation is turned on and time t is determined by the Schrödinger equation,

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = [H_0 + \lambda \hat{W}(t)]|\psi(t)\rangle, \quad (\text{C.2})$$

subject to the boundary condition

$$|\psi(0)\rangle = |\phi_i\rangle. \quad (\text{C.3})$$

The probability of transitioning from the initial state to some final state $|\phi_f\rangle$ is given by

$$P_{if} = |\langle \phi_f | \psi(t) \rangle|^2. \quad (\text{C.4})$$

The problem then amounts to solving Eq(C.2) for $|\psi(t)\rangle$ given the boundary condition Eq(C.3). This process is not soluble in general and will lead to using approximation methods.

C.2 Coupled linear differential equations

The aim is to derive the general expression for transition probabilities. Since the quantity of interest is dependent on the eigenfunctions of the unperturbed Hamiltonian, expanding the state $|\psi(t)\rangle$ in this basis is where we start,

$$|\psi(t)\rangle = \sum_n c_n(t) |\phi_n\rangle, \quad (\text{C.5})$$

with

$$c_n(t) = \langle \phi_n | \psi(t) \rangle. \quad (\text{C.6})$$

Plugging this into Eq(C.2) and projecting both sides onto $|\phi_n\rangle$ leads to a coupled set of linear differential equations,

$$i\hbar \frac{d}{dt} c_n(t) = E_n c_n(t) + \sum_k \lambda \hat{W}_{nk}(t) c_k(t), \quad (\text{C.7})$$

with

$$\hat{W}_{nk} = \langle \phi_n | \hat{W}(t) | \phi_k \rangle. \quad (\text{C.8})$$

Here it is worth noting that the coupling is purely a result of the perturbing term $W(t)$.

For the case where $\lambda = 0$, Eq(C.7) reduces to the easily soluble set of non-coupled equations with time dependence

$$c_n(t) = b_n e^{-iE_n t/\hbar}, \quad (\text{C.9})$$

where b_n is a constant that depends on initial conditions.

Now if we consider λ to be small such that $|W| \ll |H_0|$ then we can safely predict that the solution for $c_n(t)$ will be similar to Eq(C.9) since the contribution to evolution is small, *i.e.* in the expression

$$c_n(t) = b_n(t) e^{-iE_n t/\hbar}, \quad (\text{C.10})$$

$b_n(t)$ is a slowly varying function of time. Substituting Eq(C.10) into Eq(C.7) and multiplying both sides by $e^{iE_n t/\hbar}$ yields

$$i\hbar \frac{d}{dt} b_n(t) = \lambda \sum_k e^{i\omega_{nk} t} \hat{W}_{nk}(t) b_k(t), \quad (\text{C.11})$$

where $\omega_{nk} = (E_n - E_k)/\hbar$ is the Bohr angular frequency.

At this point no approximations have been made, but if we assume that λ is sufficiently small, then we can expand the coefficients $b_n(t)$ in a power series of λ and retain only the low order terms.

C.3 Power series expansion

By substituting

$$b_n(t) = b_n^{(0)}(t) + \lambda b_n^{(1)}(t) + \lambda^2 b_n^{(2)}(t) + \dots \quad (\text{C.12})$$

into Eq(C.11) leads to the recursion relation

$$i\hbar \frac{d}{dt} b_n^{(r)}(t) = \sum_k e^{i\omega_{nk}t} \hat{W}_{nk}(t) b_k^{(r-1)}(t) \quad (\text{C.13})$$

with

$$i\hbar \frac{d}{dt} b_n^{(0)}(t) = 0. \quad (\text{C.14})$$

Recognizing that the system is in the initial state $|\phi_i\rangle$ allows one to write $b_n^{(0)}(t) = \delta_{ni}$ and leads to a first order solution,

$$b_n^{(1)}(t) = \frac{1}{i\hbar} \int_0^t e^{i\omega_{ni}t'} \hat{W}_{ni}(t') dt'. \quad (\text{C.15})$$

C.4 Transition probability

The transition probability from Eq(C.4) and Eq(C.10) is determined by the squared modulus of $b_f(t)$ where

$$b_f(t) = b_f^{(0)}(t) + \lambda b_f^{(1)}(t) + \dots \quad (\text{C.16})$$

Assuming the final state is different from the initial state we set $b_f^{(0)}(t) = 0$ and consequently the transition probability

$$P_{if}(t) = \frac{1}{\hbar^2} \left| \int_0^t e^{i\omega_{fi}t'} W_{fi}(t') dt' \right|. \quad (\text{C.17})$$

Appendix D

Introduction to open quantum systems

D.1 Open quantum systems

In many physically important systems a complete mathematical model is intractable. This motivates the importance of discussing *open quantum systems*. This is an extension of the familiar dynamics associated with closed systems - examples of which were discussed in section B. In a closed system the evolution of the density matrix is unitary and in the Schrödinger picture the evolution obeys the Liouville-von Neumann equation [Eq(B.9)]. In an open system we consider a joint Hilbert space formed by the “environment” and the “system” with Hamiltonians $\mathcal{H}_B$ and $\mathcal{H}_S$ respectively. The total Hamiltonian $\mathcal{H}(t)$ is then expressed as

$$\mathcal{H}(t) = \mathcal{H}_S \otimes I_B + I_S \otimes \mathcal{H}_B + \mathcal{H}_I(t), \quad (\text{D.1})$$

where I_i is the identity matrix of the subspace labeled i and $\mathcal{H}_I(t)$ is the Hamiltonian describing the interaction between the system and the environment.

The system S is chosen in such a way that all relevant observables are contained within the subspace. This means that all of the relevant operators have the form $A \otimes I_B$, where A is an operator acting in the Hilbert space of the system S . If the total system is described by the density matrix ρ then the expectation value of the operator A can be expressed as

$$\langle A \rangle = \text{tr}_S \{ A \rho_S \}, \quad (\text{D.2})$$

where

$$\rho_S = \text{tr}_B \rho \quad (\text{D.3})$$

and tr_i is the partial trace over the subspace of i . The density matrix of the open system, ρ_S , is the quantity of interest and since the total system evolves unitarily the time dependence of the open system is

$$\rho_S(t) = \text{tr}_B\{U(t, t_0)\rho(t_0)U^\dagger(t, t_0)\}, \quad (\text{D.4})$$

where $U(t, t_0)$ is the unitary operator of the total system. The equation of motion for the open system is then found by taking the partial trace over the environment on both sides of the Liouville-von Neumann equation for the total system,

$$\frac{d}{dt}\rho_S(t) = -i\text{tr}_B[H(t), \rho(t)]. \quad (\text{D.5})$$

This equation is not directly solvable in most cases and is therefore approximated. The non-unitary evolution of the open system can often be represented by a coherent unitary contribution (the Liouville-von Neumann term) and the addition of dissipator and generator terms which may or may not depend on the open system density matrix. The exact forms of these additional terms depend on the details of the system as will be shown in the following chapters.

Appendix E

Exact expressions for exchange coupled spin-1/2 dopant

E.1 Analytic expressions for current

Here we report the analytic expression for the current with exchange as a function of magnetic field, exchange, and hopping rates. We define $\tilde{B}$ to be the Zeeman energy for the transport spin, so $\tilde{B} = g\mu_B B$.

The current as a function of magnetic field is

$$\begin{aligned}
 I = \tilde{B}^2 \Bigg[& 2\Gamma_{\text{on}} \left\{ 4\Gamma_{\text{on}}^2 \left(4\tilde{B}^2\Gamma_{\text{off}} + \Gamma_{\text{off}}^3 \right) \left[\left(\tilde{B}^2 + 2J^2 \right) \Gamma_{\text{off}}^2 + 2 \left(\tilde{B}^2 - J^2 \right)^2 \right] \right. \\
 & + \tilde{B}^2\Gamma_{\text{off}}^2\Gamma_{\text{on}} \left[\left(22\tilde{B}^2 + 7J^2 \right) \Gamma_{\text{off}}^2 + 12 \left(2\tilde{B}^2 - J^2 \right) \left(\tilde{B}^2 - J^2 \right) \right] \\
 & \left. + 2\tilde{B}^2\Gamma_{\text{off}} \left[\left(7\tilde{B}^4 + 3J^4 \right) \Gamma_{\text{off}}^2 + 4\tilde{B}^2 \left(\tilde{B}^2 - J^2 \right)^2 + \left(5\tilde{B}^2 + 4J^2 \right) \Gamma_{\text{off}}^4 \right] \right\} \Bigg] / \\
 & \left[\left\{ 4\Gamma_{\text{off}}\Gamma_{\text{on}}^2 \left(4\tilde{B}^2(10\tilde{B}^2 - 7J^2)(\tilde{B}^2 - J^2) + 24\tilde{B}^2\Gamma_{\text{off}}^4 + \left(58\tilde{B}^4 + 15\tilde{B}^2J^2 + 3J^4 \right) \Gamma_{\text{off}}^2 \right) \right. \right. \\
 & + \Gamma_{\text{on}} \left(4 \left(10\tilde{B}^2 + J^2 \right) \Gamma_{\text{off}}^6 + 32\tilde{B}^4 \left(\tilde{B}^2 - J^2 \right)^2 + \left(140\tilde{B}^4 + 82\tilde{B}^2J^2 + 3J^4 \right) \Gamma_{\text{off}}^4 \right. \\
 & + 8\tilde{B}^2 \left(17\tilde{B}^4 - 11\tilde{B}^2J^2 + 8J^4 \right) \Gamma_{\text{off}}^2 \\
 & + 4\tilde{B}^2\Gamma_{\text{off}} \left(\left(7\tilde{B}^4 + 3J^4 \right) \Gamma_{\text{off}}^2 + 4\tilde{B}^2 \left(\tilde{B}^2 - J^2 \right)^2 + \left(5\tilde{B}^2 + 4J^2 \right) \Gamma_{\text{off}}^4 \right) \\
 & + 32\Gamma_{\text{on}}^3 \left(4\tilde{B}^2 \left(\tilde{B}^2 - J^2 \right)^2 + \left(2\tilde{B}^2 + J^2 \right) \Gamma_{\text{off}}^4 + \left(6\tilde{B}^4 + 2\tilde{B}^2J^2 + J^4 \right) \Gamma_{\text{off}}^2 \right) \\
 & + 8\Gamma_{\text{on}}^3\Gamma_{\text{off}}^2 \left(4 \left(\tilde{B}^2 - J^2 \right)^2 + 2\Gamma_{\text{off}}^4 + \left(6\tilde{B}^2 + 2J^2 \right) \Gamma_{\text{off}}^2 \right) \Big\} \tilde{B}^2\hbar \\
 & \left. + 8\Gamma_{\text{on}}^3\Gamma_{\text{off}}^4J^2\hbar \left(\Gamma_{\text{off}}^2 + J^2 \right) \right], \tag{E.1}
 \end{aligned}$$

where, to obtain the most compact expressions, we introduce the energies $\Gamma_{\text{on}} = \hbar\gamma_{\text{on}}$ and $\Gamma_{\text{off}} = \hbar\gamma_{\text{off}}$.

This expression simplifies for $|\tilde{B}| \ll \Gamma_{\text{on}}, \Gamma_{\text{off}}, |J|$, and $\Gamma_{\text{on}} \gg |J| \gg \Gamma_{\text{off}}$ to

$$I = \frac{2\tilde{B}^2\Gamma_{\text{off}}}{\hbar(4\tilde{B}^2 + \Gamma_{\text{off}}^2)} \quad (\text{E.2})$$

The relative current to the current under these conditions for large magnetic field ($|\tilde{B}| \gg \Gamma_{\text{on}}, \Gamma_{\text{off}}, |J|$), $I_\infty \sim \Gamma_{\text{off}}/2\hbar$, is

$$\frac{I}{I_\infty} = \frac{4\tilde{B}^2}{4\tilde{B}^2 + \Gamma_{\text{off}}^2} \quad (\text{E.3})$$

which is a Lorentzian centered at $\tilde{B} = 0$ with width $\Gamma_{\text{off}}/2$. Note that the expression is independent of Γ_{on} as the rate limiting step for current is the hop from the transport spin to the spin-polarized STM (Γ_{off}).

We are also interested in regimes in which $\Gamma_{\text{on}} \gg |\tilde{B}|$, $|J| \gg \Gamma_{\text{off}}$, and under those circumstances Eq. (E.1) can be simplified to

$$I = \frac{\tilde{B}^4\Gamma_{\text{off}} \left\{ \left(\tilde{B}^2 + 2J^2 \right) \Gamma_{\text{off}}^2 + 2 \left(\tilde{B}^2 - J^2 \right)^2 \right\}}{\hbar \left[\left\{ \left(4\tilde{B}^2 \left(\tilde{B}^2 - J^2 \right)^2 + \left(6\tilde{B}^4 + 2\tilde{B}^2 J^2 + J^4 \right) \Gamma_{\text{off}}^2 \right) \right\} \tilde{B}^2 + 8\Gamma_{\text{off}}^4 J^4 \right]} \quad (\text{E.4})$$

This approximation is not valid for $|\tilde{B}| \ll \Gamma_{\text{on}}, \Gamma_{\text{off}}, |J|$, as the approximation requires $|\tilde{B}| \gg \Gamma_{\text{off}}$, but this expression is valid for $|\tilde{B}| \sim |J|$. Note that the expression is independent of Γ_{on} and the width of the resonances at $|\tilde{B}| \sim |J|$ are determined by Γ_{off} . If we consider $|\tilde{B}| \sim |J|$, which is the case near the finite-field features, we obtain

$$I = \frac{\Gamma_{\text{off}} \left[3J^2\Gamma_{\text{off}}^2 + 2 \left(\tilde{B}^2 - J^2 \right)^2 \right]}{\hbar \left[4 \left(\tilde{B}^2 - J^2 \right)^2 + 9J^2\Gamma_{\text{off}}^2 \right]} \quad (\text{E.5})$$

The relative current to the current under these conditions for large magnetic field (which is $I_\infty \sim \Gamma_{\text{off}}/2\hbar$), is

$$\frac{I}{I_\infty} = \frac{6J^2\Gamma_{\text{off}}^2 + 4 \left(\tilde{B}^2 - J^2 \right)^2}{9J^2\Gamma_{\text{off}}^2 + 4 \left(\tilde{B}^2 - J^2 \right)^2} \quad (\text{E.6})$$

which has a limiting value for $|\tilde{B}| = |J|$ of $2/3$, corresponding to a visibility of $1/3$.

E.2 Analytic expressions for spectator spin polarization and two-spin product state populations

Direct solution for the density matrix provides the following expressions for the spin polarization of the spectator spin along the STM quantization direction, the magnetic field direction, and the orthogonal direction. To specify these directions within a right-handed coordinate system we use $\hat{z}$ for the direction of the STM quantization ($\hat{n}_{\text{STM}}$ in the manuscript), $\hat{x}$ for the direction of the magnetic field ($\hat{n}_{\perp}$ in the manuscript), and $\hat{y}$ for the orthogonal direction. In order to specify the spin orientation along the $\hat{x}$ direction we generalize $\tilde{B}$ such that the sign of $\tilde{B}$ is opposite that of B , so for a positive magnetic field along the $\hat{x}$ direction, $\tilde{B} < 0$.

We find the magnetization M of the spectator spin along $\hat{x}$, $\hat{y}$, and $\hat{z}$, not normalized to the occupation of the single-spin manifold, to be

$$\begin{aligned}
M_{\text{spec},x} = & \tilde{B}^3 J \Gamma_{\text{off}}^3 \left[8\Gamma_{\text{on}}^2 (4\tilde{B}^2 + \Gamma_{\text{off}}^2) (2\tilde{B}^2 + J^2 + 4\Gamma_{\text{off}}^2) + 4\Gamma_{\text{on}} \Gamma_{\text{off}} \left[20\tilde{B}^4 - \tilde{B}^2 J^2 + (22\tilde{B}^2 - 2J^2) \Gamma_{\text{off}}^2 \right] \right. \\
& + 16\tilde{B}^4 (\tilde{B}^2 + J^2) + \tilde{B}^2 \Gamma_{\text{off}}^2 (80\tilde{B}^2 + 11J^2 + 48\Gamma_{\text{off}}^2) \left. \right] / \\
& \left[2(\tilde{B}^2 + \Gamma_{\text{off}}^2) \left\{ 8\Gamma_{\text{on}}^3 (4\tilde{B}^2 + \Gamma_{\text{off}}^2) \left[4\tilde{B}^2 (\tilde{B}^2 - J^2)^2 + (6\tilde{B}^4 + 2\tilde{B}^2 J^2 + J^4) \Gamma_{\text{off}}^2 + (2\tilde{B}^2 + J^2) \Gamma_{\text{off}}^4 \right] \right. \right. \\
& + 4\Gamma_{\text{on}}^2 \tilde{B}^2 \Gamma_{\text{off}} \left[4(10\tilde{B}^2 - 7J^2) (\tilde{B}^2 - J^2) \tilde{B}^2 + (58\tilde{B}^4 + 15\tilde{B}^2 J^2 + 3J^4) \Gamma_{\text{off}} + 24\tilde{B}^2 \Gamma_{\text{off}}^4 \right] \\
& + \Gamma_{\text{on}} \tilde{B}^2 \left[32\tilde{B}^4 (\tilde{B}^2 - J^2)^2 + 8\tilde{B}^2 \Gamma_{\text{off}}^2 (17\tilde{B}^4 - 11\tilde{B}^2 J^2 + 8J^4) \right. \\
& + \Gamma_{\text{off}}^4 (140\tilde{B}^4 + 82\tilde{B}^2 J^2 + 3J^4) \Gamma_{\text{off}}^4 + 4(10\tilde{B}^2 + J^2) \Gamma_{\text{off}}^6 \left. \right] \\
& \left. \left. + \tilde{B}^4 \Gamma_{\text{off}} \left[4\tilde{B}^2 (\tilde{B}^2 - J^2)^2 + (7\tilde{B}^4 + 3J^4) \Gamma_{\text{off}}^2 + (5\tilde{B}^2 + 4J^2) \Gamma_{\text{off}}^4 \right] \right\} \right] \quad (E.7)
\end{aligned}$$

$$\begin{aligned}
M_{\text{spec},y} = & \left\{ 4\tilde{B}^3 J^2 \Gamma_{\text{off}}^2 \Gamma_{\text{on}} \left[(3J^2 - 7\tilde{B}^2) \Gamma_{\text{off}} + 4\Gamma_{\text{off}}^3 + \Gamma_{\text{on}} (11\Gamma_{\text{off}}^2 - 8(\tilde{B}^2 - J^2)) \right] \right\} / \\
& \left\{ 8\Gamma_{\text{on}}^3 (4\tilde{B}^2 + \Gamma_{\text{off}}^2) \left[4\tilde{B}^2 (\tilde{B}^2 - J^2)^2 + (6\tilde{B}^4 + 2\tilde{B}^2 J^2 + J^4) \Gamma_{\text{off}}^2 + (2\tilde{B}^2 + J^2) \Gamma_{\text{off}}^4 \right] \right. \\
& + 4\Gamma_{\text{on}}^2 \tilde{B}^2 \Gamma_{\text{off}} \left[4(10\tilde{B}^2 - 7J^2)(\tilde{B}^2 - J^2) \tilde{B}^2 + (58\tilde{B}^4 + 15\tilde{B}^2 J^2 + 3J^4) \Gamma_{\text{off}} + 24\tilde{B}^2 \Gamma_{\text{off}}^4 \right] \\
& + \Gamma_{\text{on}} \tilde{B}^2 \left[32\tilde{B}^4 (\tilde{B}^2 - J^2)^2 + 8\tilde{B}^2 \Gamma_{\text{off}}^2 (17\tilde{B}^4 - 11\tilde{B}^2 J^2 + 8J^4) \right. \\
& \left. + \Gamma_{\text{off}}^4 (140\tilde{B}^4 + 82\tilde{B}^2 J^2 + 3J^4) \Gamma_{\text{off}}^4 + 4(10\tilde{B}^2 + J^2) \Gamma_{\text{off}}^6 \right] \\
& \left. + \tilde{B}^4 \Gamma_{\text{off}} \left[4\tilde{B}^2 (\tilde{B}^2 - J^2)^2 + (7\tilde{B}^4 + 3J^4) \Gamma_{\text{off}}^2 + (5\tilde{B}^2 + 4J^2) \Gamma_{\text{off}}^4 \right] \right\} \quad (\text{E.8})
\end{aligned}$$

$$\begin{aligned}
M_{\text{spec},z} = & 2\Gamma_{\text{on}} \tilde{B}^2 J^2 \Gamma_{\text{off}}^2 \left[\tilde{B}^2 (19\Gamma_{\text{off}}^2 - 8(\tilde{B}^2 - J^2)) - 4\Gamma_{\text{on}} \Gamma_{\text{off}} (2J^2 - 5\tilde{B}^2 + 2\Gamma_{\text{off}}^2) \right] / \\
& \left\{ 8\Gamma_{\text{on}}^3 (4\tilde{B}^2 + \Gamma_{\text{off}}^2) \left[4\tilde{B}^2 (\tilde{B}^2 - J^2)^2 + (6\tilde{B}^4 + 2\tilde{B}^2 J^2 + J^4) \Gamma_{\text{off}}^2 + (2\tilde{B}^2 + J^2) \Gamma_{\text{off}}^4 \right] \right. \\
& + 4\Gamma_{\text{on}}^2 \tilde{B}^2 \Gamma_{\text{off}} \left[4(10\tilde{B}^2 - 7J^2)(\tilde{B}^2 - J^2) \tilde{B}^2 + (58\tilde{B}^4 + 15\tilde{B}^2 J^2 + 3J^4) \Gamma_{\text{off}} + 24\tilde{B}^2 \Gamma_{\text{off}}^4 \right] \\
& + \Gamma_{\text{on}} \tilde{B}^2 \left[32\tilde{B}^4 (\tilde{B}^2 - J^2)^2 + 8\tilde{B}^2 \Gamma_{\text{off}}^2 (17\tilde{B}^4 - 11\tilde{B}^2 J^2 + 8J^4) \right. \\
& \left. + \Gamma_{\text{off}}^4 (140\tilde{B}^4 + 82\tilde{B}^2 J^2 + 3J^4) \Gamma_{\text{off}}^4 + 4(10\tilde{B}^2 + J^2) \Gamma_{\text{off}}^6 \right] \\
& \left. + \tilde{B}^4 \Gamma_{\text{off}} \left[4\tilde{B}^2 (\tilde{B}^2 - J^2)^2 + (7\tilde{B}^4 + 3J^4) \Gamma_{\text{off}}^2 + (5\tilde{B}^2 + 4J^2) \Gamma_{\text{off}}^4 \right] \right\} \quad (\text{E.9})
\end{aligned}$$

On resonance, $|\tilde{B}| = |J|$, and for $\Gamma_{\text{on}} \gg |J| \gg \Gamma_{\text{off}}$, these reduce to

$$M_{\text{spec},x} = \text{sgn} \tilde{B} \frac{\Gamma_{\text{off}}}{6\Gamma_{\text{on}}} \quad (\text{E.10})$$

$$M_{\text{spec},y} = \text{sgn} \tilde{B} \frac{11\Gamma_{\text{off}}^2}{72\Gamma_{\text{on}} J} \sim 0 \quad (\text{E.11})$$

$$M_{\text{spec},z} = \frac{\Gamma_{\text{off}}}{12\Gamma_{\text{on}}} \quad (\text{E.12})$$

To find the polarization of the spectator spin when transport through the other spin has been shut off and the transport spin is unoccupied, such as would occur in an initialization

process, we divide by the occupation of the single-spin manifold, which is

$$\begin{aligned}
P_{\text{spec}} = & \left[4\tilde{B}^4\Gamma_{\text{off}}(4\tilde{B}^2(\tilde{B}^2 - J^2)^2 + (7\tilde{B}^4 + 3J^4)\Gamma_{\text{off}}^2 + (5\tilde{B}^2 + 4J^2)\Gamma_{\text{off}}^4) \right. \\
& + 2\Gamma_{\text{on}}\tilde{B}^4\Gamma_{\text{off}}^2(12(2\tilde{B}^2 - J^2)(\tilde{B}^2 - J^2) + (22\tilde{B}^2 + 7J^2)\Gamma_{\text{off}}^2) \\
& \left. + 8\Gamma_{\text{on}}^2\tilde{B}^2(2(\tilde{B}^2 - J^2)^2 + (\tilde{B}^2 + 2J^2)\Gamma_{\text{off}}^2)(4\tilde{B}^2\Gamma_{\text{off}} + \Gamma_{\text{off}}^3) \right] \\
& / \left\{ 8\Gamma_{\text{on}}^3(4\tilde{B}^2 + \Gamma_{\text{off}}^2) \left[4\tilde{B}^2(\tilde{B}^2 - J^2)^2 + (6\tilde{B}^4 + 2\tilde{B}^2J^2 + J^4)\Gamma_{\text{off}}^2 + (2\tilde{B}^2 + J^2)\Gamma_{\text{off}}^4 \right] \right. \\
& + 4\Gamma_{\text{on}}^2\tilde{B}^2\Gamma_{\text{off}} \left[4\tilde{B}^2(10\tilde{B}^2 - 7J^2)(\tilde{B}^2 - J^2) + (58\tilde{B}^4 + 15\tilde{B}^2J^2 + 3J^4)\Gamma_{\text{off}}^2 + 24\tilde{B}^2\Gamma_{\text{off}}^4 \right] \\
& + \Gamma_{\text{on}}\tilde{B}^2 \left[32\tilde{B}^4(\tilde{B}^2 - J^2)^2 + 8\tilde{B}^2(17\tilde{B}^4 - 11\tilde{B}^2J^2 + 8J^4)\Gamma_{\text{off}}^2 \right. \\
& \left. + (140\tilde{B}^4 + 82\tilde{B}^2J^2 + 3J^4)\Gamma_{\text{off}}^4 + 4(10\tilde{B}^2 + J^2)\Gamma_{\text{off}}^6 \right] \\
& \left. + 4\tilde{B}^4\Gamma_{\text{off}} \left[4\tilde{B}^2(\tilde{B}^2 - J^2)^2 + (7\tilde{B}^4 + 3J^4)\Gamma_{\text{off}}^2 + (5\tilde{B}^2 + 4J^2)\Gamma_{\text{off}}^4 \right] \right\}
\end{aligned} \tag{E.13}$$

On resonance, $|\tilde{B}| = |J|$, and for $\Gamma_{\text{on}} \gg |J| \gg \Gamma_{\text{off}}$, this reduces to

$$P_{\text{spec}} = \frac{\Gamma_{\text{on}}}{3\Gamma_{\text{off}}} \tag{E.14}$$

and the normalized polarization of the spectator spin becomes

$$\tilde{M}_{\text{spec},x} = \text{sgn}\tilde{B}\frac{1}{2} \tag{E.15}$$

$$\tilde{M}_{\text{spec},y} = 0 \tag{E.16}$$

$$\tilde{M}_{\text{spec},z} = \frac{1}{4} \tag{E.17}$$

for a total polarization of $\tilde{M}_{\text{spec}} = \sqrt{5}/4$. Note that for an electron the magnetic moment is negative, so the sign of $\tilde{B}$ is opposite to that of B

We now turn to the probabilities of the product states in the two-spin manifold. These are:

$$\begin{aligned}
P_{\uparrow\uparrow} = & \tilde{B}^4 \left\{ \Gamma_{\text{on}}^2 \left[4\Gamma_{\text{off}}^4 + 32(\tilde{B}^2 - J^2)^2 + 12\Gamma_{\text{off}}^2(2\tilde{B}^2 + 3J^2) \right] \right. \\
& + \Gamma_{\text{on}}\Gamma_{\text{off}} \left[12(2\tilde{B}^2 - J^2)(\tilde{B}^2 - J^2) + (22\tilde{B}^2 + 15J^2)\Gamma_{\text{off}}^2 \right] \\
& + 2\Gamma_{\text{off}}^4(5\tilde{B}^2 + 2J^2) + 8\tilde{B}^2(\tilde{B}^2 - J^2)^2 + \Gamma_{\text{off}}^2(14\tilde{B}^4 + 7\tilde{B}^2J^2 + 3J^4) \left. \right\} / \\
& \left\{ 8\Gamma_{\text{on}}^2(4\tilde{B}^2 + \Gamma_{\text{off}}^2) \left[4\tilde{B}^2(\tilde{B}^2 - J^2)^2 + (6\tilde{B}^4 + 2\tilde{B}^2J^2 + J^4)\Gamma_{\text{off}}^2 + (2\tilde{B}^2 + J^2)\Gamma_{\text{off}}^4 \right] \right. \\
& + 4\Gamma_{\text{on}}\tilde{B}^2\Gamma_{\text{off}} \left[12\tilde{B}^2(2\tilde{B}^2 - J^2)(\tilde{B}^2 - J^2) + (46\tilde{B}^2 + 7\tilde{B}^2J^2 - J^4)\Gamma_{\text{off}}^2 + (22\tilde{B}^2 - 4J^2)\Gamma_{\text{off}}^4 \right] \\
& + \tilde{B}^2 \left[32\tilde{B}^4(\tilde{B}^2 - J^2)^2 + 8\tilde{B}^2(11\tilde{B}^4 - 2\tilde{B}^2J^2 + 5J^4)\Gamma_{\text{off}}^2 + (96\tilde{B}^4 + 68\tilde{B}^2J^2 + 3J^4)\Gamma_{\text{off}}^4 \right. \\
& \left. \left. + 4(10\tilde{B}^2 + J^2)\Gamma_{\text{off}}^6 \right] \right\}
\end{aligned} \tag{E.18}$$

$$\begin{aligned}
P_{\uparrow\downarrow} = & \left\{ \Gamma_{\text{on}}^2 \left[32\tilde{B}^4(\tilde{B}^2 - J^2)^2 + 4\tilde{B}^2\Gamma_{\text{off}}^2(6\tilde{B}^4 - \tilde{B}^2J^2 + 4J^4) + 4\Gamma_{\text{off}}^4\tilde{B}^2(\tilde{B}^2 + 4J^2) \right] \right. \\
& + \Gamma_{\text{on}}\Gamma_{\text{off}} \left[12\tilde{B}^4(2\tilde{B}^2 - J^2)(\tilde{B}^2 - J^2) + \tilde{B}^4(22\tilde{B}^2 - J^2)\Gamma_{\text{off}}^2 \right] \\
& + 8\tilde{B}^6(\tilde{B}^2 - J^2)^2 + \Gamma_{\text{off}}^2\tilde{B}^4(14\tilde{B}^4 - 7\tilde{B}^2J^2 + 9J^4) + 2\Gamma_{\text{off}}^4\tilde{B}^4(5\tilde{B}^2 + 6J^2) \left. \right\} / \\
& \left\{ 8\Gamma_{\text{on}}^2(4\tilde{B}^2 + \Gamma_{\text{off}}^2) \left[4\tilde{B}^2(\tilde{B}^2 - J^2)^2 + (6\tilde{B}^4 + 2\tilde{B}^2J^2 + J^4)\Gamma_{\text{off}}^2 + (2\tilde{B}^2 + J^2)\Gamma_{\text{off}}^4 \right] \right. \\
& + 4\Gamma_{\text{on}}\tilde{B}^2\Gamma_{\text{off}} \left[12\tilde{B}^2(2\tilde{B}^2 - J^2)(\tilde{B}^2 - J^2) + (46\tilde{B}^2 + 7\tilde{B}^2J^2 - J^4)\Gamma_{\text{off}}^2 + (22\tilde{B}^2 - 4J^2)\Gamma_{\text{off}}^4 \right] \\
& + \tilde{B}^2 \left[32\tilde{B}^4(\tilde{B}^2 - J^2)^2 + 8\tilde{B}^2(11\tilde{B}^4 - 2\tilde{B}^2J^2 + 5J^4)\Gamma_{\text{off}}^2 + (96\tilde{B}^4 + 68\tilde{B}^2J^2 + 3J^4)\Gamma_{\text{off}}^4 \right. \\
& \left. \left. + 4(10\tilde{B}^2 + J^2)\Gamma_{\text{off}}^6 \right] \right\}
\end{aligned} \tag{E.19}$$

$$\begin{aligned}
P_{\downarrow\uparrow} = & \left\{ 4\tilde{B}^2\Gamma_{\text{on}}^2 \left[8\tilde{B}^2(\tilde{B}^2 - J^2)^2 + \Gamma_{\text{off}}^2(22\tilde{B}^4 + 7\tilde{B}^2 J^2 + 4J^4) + (13\tilde{B}^2 + 6J^2)\Gamma_{\text{off}}^4 + 2\Gamma_{\text{off}}^6 \right] \right. \\
& + \Gamma_{\text{on}}\Gamma_{\text{off}}\tilde{B}^4 \left[12(2\tilde{B}^2 - J^2)(\tilde{B}^2 - J^2) + (70\tilde{B}^2 + 27J^2)\Gamma_{\text{off}}^2 + 44\Gamma_{\text{off}}^4 \right] \\
& + \tilde{B}^4 \left[8\tilde{B}^2(\tilde{B}^2 - J^2)^2 + (30\tilde{B}^4 + 9\tilde{B}^2 J^2 + 9J^4)\Gamma_{\text{off}}^2 + (38\tilde{B}^2 + 27J^2)\Gamma_{\text{off}}^4 + 20\Gamma_{\text{off}}^6 \right] \Big\} / \\
& \left\{ 8\Gamma_{\text{on}}^2(4\tilde{B}^2 + \Gamma_{\text{off}}^2) \left[4\tilde{B}^2(\tilde{B}^2 - J^2)^2 + (6\tilde{B}^4 + 2\tilde{B}^2 J^2 + J^4)\Gamma_{\text{off}}^2 + (2\tilde{B}^2 + J^2)\Gamma_{\text{off}}^4 \right] \right. \\
& + 4\Gamma_{\text{on}}\tilde{B}^2\Gamma_{\text{off}} \left[12\tilde{B}^2(2\tilde{B}^2 - J^2)(\tilde{B}^2 - J^2) + (46\tilde{B}^2 + 7\tilde{B}^2 J^2 - J^4)\Gamma_{\text{off}}^2 + (22\tilde{B}^2 - 4J^2)\Gamma_{\text{off}}^4 \right] \\
& + \tilde{B}^2 \left[32\tilde{B}^4(\tilde{B}^2 - J^2)^2 + 8\tilde{B}^2(11\tilde{B}^4 - 2\tilde{B}^2 J^2 + 5J^4)\Gamma_{\text{off}}^2 + (96\tilde{B}^4 + 68\tilde{B}^2 J^2 + 3J^4)\Gamma_{\text{off}}^4 \right. \\
& \left. \left. + 4(10\tilde{B}^2 + J^2)\Gamma_{\text{off}}^6 \right] \right\}
\end{aligned} \tag{E.20}$$

$$\begin{aligned}
P_{\downarrow\downarrow} = & \left\{ \Gamma_{\text{on}}^2 \left[32\tilde{B}^4(\tilde{B}^2 - J^2)^2 + 4\Gamma_{\text{off}}^2\tilde{B}^2(22\tilde{B}^4 - 15\tilde{B}^2 J^2 + 8J^4) + \Gamma_{\text{off}}^4(52\tilde{B}^4 + 8\tilde{B}^2 J^2 + 8J^4) \right. \right. \\
& + 8(\tilde{B}^2 + J^2)\Gamma_{\text{off}}^6 \Big] + \Gamma_{\text{on}}\Gamma_{\text{off}}\tilde{B}^2 \left[12\tilde{B}^2(2\tilde{B}^2 - J^2)(\tilde{B}^2 - J^2) + (70\tilde{B}^4 - 13\tilde{B}^2 J^2 - 4J^4)\Gamma_{\text{off}}^2 \right. \\
& + 4(11\tilde{B}^2 - 4J^2)\Gamma_{\text{off}}^4 \Big] + 8\tilde{B}^6(\tilde{B}^2 - J^2)^2 + \tilde{B}^4\Gamma_{\text{off}}^2(30\tilde{B}^4 - 25\tilde{B}^2 J^2 + 19J^4) \\
& + \tilde{B}^2\Gamma_{\text{off}}^4(38\tilde{B}^4 + 25\tilde{B}^2 J^2 + 3J^4) + 4\tilde{B}^2\Gamma_{\text{off}}^6(5\tilde{B}^2 + J^2) \Big\} / \\
& \left\{ 8\Gamma_{\text{on}}^2(4\tilde{B}^2 + \Gamma_{\text{off}}^2) \left[4\tilde{B}^2(\tilde{B}^2 - J^2)^2 + (6\tilde{B}^4 + 2\tilde{B}^2 J^2 + J^4)\Gamma_{\text{off}}^2 + (2\tilde{B}^2 + J^2)\Gamma_{\text{off}}^4 \right] \right. \\
& + 4\Gamma_{\text{on}}\tilde{B}^2\Gamma_{\text{off}} \left[12\tilde{B}^2(2\tilde{B}^2 - J^2)(\tilde{B}^2 - J^2) + (46\tilde{B}^2 + 7\tilde{B}^2 J^2 - J^4)\Gamma_{\text{off}}^2 + (22\tilde{B}^2 - 4J^2)\Gamma_{\text{off}}^4 \right] \\
& + \tilde{B}^2 \left[32\tilde{B}^4(\tilde{B}^2 - J^2)^2 + 8\tilde{B}^2(11\tilde{B}^4 - 2\tilde{B}^2 J^2 + 5J^4)\Gamma_{\text{off}}^2 + (96\tilde{B}^4 + 68\tilde{B}^2 J^2 + 3J^4)\Gamma_{\text{off}}^4 \right. \\
& \left. \left. + 4(10\tilde{B}^2 + J^2)\Gamma_{\text{off}}^6 \right] \right\}
\end{aligned} \tag{E.21}$$

On resonance, $|\tilde{B}| = |J|$, and for $\Gamma_{\text{on}} \gg |J| \gg \Gamma_{\text{off}}$, this reduces to

$$P_{\uparrow\uparrow} = \frac{5}{24} \tag{E.22}$$

$$P_{\uparrow\downarrow} = \frac{1}{8} \tag{E.23}$$

$$P_{\downarrow\uparrow} = \frac{11}{24} \tag{E.24}$$

$$P_{\downarrow\downarrow} = \frac{5}{24} \tag{E.25}$$

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